

2nd International Conference on Engineering and Technology for Sustainable Development



ETSD 2026



ORGANIZED BY
FACULTY OF ENGINEERING AND TECHNOLOGY

Jashore University of Science and Technology
Jashore-7408, Bangladesh



ETSD 2026

2nd International Conference on Engineering and Technology for Sustainable Development

Organized by Faculty of Engineering and Technology, JUST

The abstracts of papers presented at ETSD 2026 are included in this book. All papers have been peer-reviewed by at least two independent reviewers. The short biographies and abstracts of keynote and invited speakers are also included; the contents herein do not necessarily reflect the opinion or endorsement of editors, organizers, or sponsors, and any liability cannot be imposed upon them.

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Publisher

Faculty of Engineering and Technology
Jashore University of Science and Technology
www.just.edu.bd



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2nd International Conference on Engineering and Technology for Sustainable Development



ETSD 2026

MESSAGES



2nd International Conference on Engineering and Technology for Sustainable Development

MESSAGE



Chief Patron

2nd International Conference on Engineering and
Technology for Sustainable Development

It is my great pleasure to welcome distinguished researchers, esteemed academics, industry professionals, and students to the 2nd International Conference on Engineering and Technology for Sustainable Development (ETSD 2026), hosted by Jashore University of Science and Technology.

This conference reflects our collective commitment to advancing knowledge, fostering innovation, and addressing the pressing challenges of sustainable development through engineering and technological excellence. In a world shaped by rapid transformation and global uncertainty, such platforms are essential for bringing together brilliant minds to exchange ideas, inspire collaboration, and create impactful solutions.

Jashore University of Science and Technology (JUST) has emerged as a leading public university in Bangladesh, with a strong focus on science, engineering, and applied research. Over the years, JUST has made notable progress in establishing modern laboratories, interdisciplinary research centers, and industry-oriented academic programs. Our Faculty of Engineering and Technology, along with other faculties, has been actively contributing to research in renewable energy systems, smart grid technologies, environmental sustainability, biomedical innovation, and data-driven solutions.

At JUST, we are deeply committed to promoting research that not only contributes to scientific knowledge but also addresses real-world challenges, particularly in developing countries like Bangladesh. We believe that locally grounded, scalable, and data-driven solutions can play a transformative role in sectors such as energy, agriculture, healthcare, and environmental management.

I am confident that the exchange of knowledge and ideas during this conference will foster innovation, encourage interdisciplinary collaborations, and contribute to a more sustainable and inclusive progress.

I extend my sincere appreciation to the organizing committee, reviewers, sponsors, and volunteers for their dedication and hard work in making this conference possible. I also thank all participants for their valuable contributions.

I wish the conference every success and hope that your experience at JUST will be both productive and memorable.

Professor Dr. Yearul Kabir

Vice Chancellor
Jashore University of Science and Technology



2nd International Conference on Engineering and Technology for Sustainable Development

MESSAGE



Convener

2nd International Conference on Engineering and
Technology for Sustainable Development

It gives me immense pleasure to welcome you to the *International Conference on Engineering and Technology for Sustainable Development (ETSD 2026)*, to be held at Jashore University of Science and Technology (JUST), Bangladesh. Organized by the Faculty of Engineering and Technology, this conference reflects our strong commitment to advancing research, innovation, and collaboration in addressing the pressing challenges of sustainable development.

In an era marked by rapid technological transformation and growing environmental concerns, the need for sustainable and inclusive solutions has never been more critical. ETSD 2026 aims to bring together researchers, engineers, academics, industry professionals, and policymakers from around the globe to share knowledge, exchange ideas, and foster interdisciplinary collaboration. The conference provides a vibrant platform to explore cutting-edge research and innovative practices across a wide range of disciplines, including electrical and electronic engineering, renewable energy, environmental science, computational intelligence, biomedical engineering, materials science, and emerging technologies.

The theme of this conference underscores the importance of integrating innovation with sustainability. Through technical sessions, discussions, and networking opportunities, ETSD 2026 seeks to inspire actionable solutions that contribute to a resilient and sustainable future. The diversity of topics and participants will undoubtedly enrich the dialogue and encourage collaborative approaches to global challenges.

At JUST, we take pride in fostering a research-driven academic environment that promotes excellence, creativity, and societal impact. Hosting this international conference is a reflection of our continued efforts to position the university as a hub of innovation and knowledge exchange. I am confident that ETSD 2026 will serve as a meaningful platform for participants to showcase their work, build professional networks, and initiate impactful collaborations.

I would like to express my sincere gratitude to the organizing committee, reviewers, sponsors, and volunteers for their dedicated efforts in making this conference possible. I also extend a warm welcome to all participants and wish you a productive and enriching experience at ETSD 2026.

Let us come together to “Innovate, Collaborate, and Educate” for a sustainable future.

Professor Dr. Engr. Imran Khan

Dean

Faculty of Engineering and Technology
Jashore University of Science and Technology



2nd International Conference on Engineering and Technology for Sustainable Development

MESSAGE



Conference Secretary

2nd International Conference on Engineering and
Technology for Sustainable Development

It is my great pleasure to welcome all distinguished guests, keynote speakers, session chairs, co-chairs, presenters, participants, and delegates to the **2nd International Conference on Engineering and Technology for Sustainable Development (ETSD 2026)**, organized by the **Faculty of Engineering and Technology, Jashore University of Science and Technology (JUST)**.

ETSD 2026 provides a valuable platform for researchers, academicians, professionals, industry experts, and students to share their latest research findings, innovative ideas, and practical experiences in the broad fields of engineering, technology, and sustainable development. This conference aims to encourage interdisciplinary collaboration and promote research that contributes to solving real-world challenges and achieving sustainable development goals.

As the Conference Secretary, I sincerely express my gratitude to the honorable Vice-Chancellor of JUST, the conference advisory committee, organizing committee, technical committee, keynote speakers, session chairs, reviewers, authors, sponsors, volunteers, and all concerned for their continuous support and cooperation in making this conference possible.

I would also like to thank all the authors and presenters for submitting their valuable research works. Their contributions have enriched the academic quality of this conference and made the proceedings meaningful. I hope that the discussions, presentations, and interactions during ETSD 2026 will create new opportunities for research collaboration and future innovation.

Finally, I wish all participants a successful, productive, and memorable conference.

Prof. Dr. Md. Jahangir Alam

Professor, Department of ChE
Jashore University of Science and Technology

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**KEYNOTE
SPEAKERS**



KEYNOTE SPEAKER



Prof. Dr. Salim Ahmed

Head of the Department of Process Engineering
Memorial University of Newfoundland (MUN)
St. John's, Canada

TALK TITLE

AI and ML for Process Safety: Prospects and Pitfalls

Abstract

Artificial Intelligence (AI) and Machine learning (ML) are influencing and shaping many aspects of our personal and professional life. For management of process safety, both personal traits and professional training play significant roles. This presentation will highlight some recent developments in the area of process safety based on AI and ML. More specifically, ML applications for process control, fault detection and diagnosis, process security and alarm system design will be discussed. A key to this development is the ability of ML tools to analyze large volumes of data far beyond human capacity. However, AI/ML models often suffer from issues of data quality, bias, and lack of interpretability, raising concerns about reliability and trust in safety-critical environments. Overreliance on automated systems may lead to complacency or erosion of human expertise. Furthermore, the integration of AI into legacy systems introduces cybersecurity risks and regulatory uncertainties, as existing safety standards are not always equipped to address algorithm-driven decision processes. Ethical considerations and accountability in case of system failure also remain unresolved. We will share the view that with emphasis on combining domain expertise with AI tools, ensuring transparency, and maintaining robust human oversight, the benefits of AI and ML tools can be harnessed in a sustainable way.

Biography

Dr. Salim Ahmed is currently serving as a Professor and Head of the Department of Process Engineering at Memorial University of Newfoundland (MUN), St. John's, Canada and the Director of the Centre for Risk, Integrity and Safety Engineering (C-RISE) at MUN. With expertise in process control, process modeling and process safety, Dr. Ahmed has contributed to research by supervising 30 PhD and Master students, publishing over 150 journal and conference articles, and book chapters and establishing new research programs. Dr. Ahmed received his B.Sc. and M.Sc. degrees in Chemical Engineering from Bangladesh University of Engineering and Technology (BUET), and Ph.D. in Process Control from University of Alberta, Canada. He has developed innovative approaches for warning system design using probabilistic modeling and fault detection methods using machine learning techniques. Dr. Ahmed serves as Associate Editor of Control Engineering Practice, Associate Editor of the Journal of the Franklin Institute, Editorial Board Member of Process Safety Progress and Regional Editor (North America) of International Journal of Reliability and Safety.



KEYNOTE SPEAKER



Prof. Dr. Muhammad Usman Ghani Khan

Sultan Qaboos IT Chair and Full Professor & COMSTECH Distinguished Scholar
Department of Computer Science
University of Engineering and Technology (UET)
Lahore, Pakistan

TALK TITLE

AI Powered Solutions for Real World Problems

Abstract

Artificial Intelligence is rapidly reshaping the global economy, with its projected contribution expected to reach nearly USD 15.7 trillion by 2030. Today, it stands as one of the most sought after technologies across industries worldwide. For developing nations, however, the real challenge lies not merely in adopting AI, but in building indigenous, practical, and scalable solutions that address local needs. This keynote presents the journey and impact of the National Center for Artificial Intelligence (NCAI), Pakistan, with a particular focus on contributions from UET Lahore. As a leading node within a nationwide network spanning six universities, NCAI at UET Lahore has been actively driving innovation through applied research and product-oriented development. Drawing upon over two decades of experience in artificial intelligence, machine learning, and interdisciplinary domains, this talk highlights how research at UET Lahore has translated into impactful real world systems. These include safe smart city solutions, healthcare and education technologies, intelligent surveillance, judiciary process automation, and media monitoring platforms. The focus is on moving beyond theoretical models toward deployable technologies that create measurable societal and economic impact. The keynote will further explore the development of sustainable research ecosystems, the strengthening of academia industry collaborations, and the mentoring of the next generation through advanced, high-impact training programs. Ultimately, it presents a vision in which AI is not just a technological advancement, but a catalyst for improving the socio economic landscape of Pakistan and other developing regions like Bangladesh through locally developed, globally relevant solutions.

Biography

Prof. Dr. Muhammad Usman Ghani Khan is a Sultan Qaboos IT Chair and Full Professor in the Department of Computer Science at UET Lahore. He completed his PhD from the University of Sheffield, UK, and is also an alumnus of UET.

With more than two decades of experience, his work spans across image processing, computer vision, bioinformatics, medical imaging, computational linguistics, and machine learning. He currently serves as



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the Director of the National Center for Artificial Intelligence (NCAI), where he is leading large-scale research and innovation efforts in Pakistan.

Over the years, he has established multiple specialized research labs, including Computer Vision & Machine Learning, Bioinformatics, Virtual Reality & Gaming, Data Science, Information Dissemination and Software Systems. His work is not just academic; he actively focuses on building practical, product oriented solutions and fostering industry collaborations.

He has been a visiting scholar at Stanford University in the Department of Psychology, completed a research internship at Toshiba in the UK, and has also mentored young innovators from South Asia at Google I/O in San Francisco.

As an educator, he teaches courses in Artificial Intelligence, Machine Learning, Deep Learning, and Generative AI. He leads several GenAI teaching initiatives for Judiciary, Government officers, Engineering faculty and doctors. He has authored several books, produced widely accessible video lectures, and published over 350 research papers with a very high cumulative impact. His research funding portfolio exceeds 1 billion PKR.

His broader vision is simple but powerful: to use technology and innovation to uplift the socio-economic landscape of Pakistan, while building strong global academic and research collaborations.

He is a COMSTECH Distinguished Scholar, recognizing his international standing within the OIC scientific community. COMSTECH (Committee on Scientific and Technological Cooperation) is an OIC body based in Islamabad, Pakistan, dedicated to strengthening science and technology collaboration among member states. It supports research, capacity building, and knowledge exchange through scholarships, fellowships, joint projects, and the Distinguished Scholar Program.

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2nd International Conference on Engineering and Technology for Sustainable Development

Organized by Faculty of Engineering and Technology, JUST

Conference Date: 15 May 2026

Venue: Sir Jagadish Chandra Bose Academic Building, JUST

Programme at a Glance

09:00 – 10:00	Registration & Kit Collection — Registration Desk, Ground Floor, Sir Jagadish Chandra Bose Academic Building, JUST
10:00 – 10:45	Inauguration Ceremony Professor Md. Shorif Hossain Gallery, Jatiyo Kobi Kazi Nazrul Islam Academic Building, JUST — Opening Remarks, Welcome Address
10:45 – 11:45	Keynote Sessions Professor Md. Shorif Hossain Gallery, Jatiyo Kobi Kazi Nazrul Islam Academic Building, JUST
11:45 – 12:00	Tea Break & Snacks — 2 nd Floor, Sir Jagadish Chandra Bose Academic Building, JUST
12:00 – 13:15	Morning Technical Sessions 5 Parallel Sessions — See Session Details Below
13:15 – 14:45	Prayer & Lunch Break — 1 st Floor, Sir Jagadish Chandra Bose Academic Building, JUST
14:45 – 16:00	Poster Presentation Session — 1 st Floor, Sir Jagadish Chandra Bose Academic Building, JUST
14:45 – 16:00	Afternoon Technical Sessions 4 Parallel Sessions — See Session Details Below
16:00 – 16:30	Tea & Refreshment — 2 nd Floor, Sir Jagadish Chandra Bose Academic Building, JUST
16:30 – 17:00	Award Ceremony Professor Md. Shorif Hossain Gallery, Jatiyo Kobi Kazi Nazrul Islam Academic Building, JUST
19:00 – 20:00	Gala Dinner — 3 rd & 4 th Floor, Teachers and Students Centre (TSC), JUST

Keynote Session | 10:45 – 11:45*Professor Md. Shorif Hossain Gallery, Jatiyo Kobi Kazi Nazrul Islam Academic Building, JUST***Keynote Session Chair: Professor Dr. Biplob Kumar Biswas, Department of ChE, JUST**

Talk Title	Keynote Speaker
AI Powered Solutions for Real World Problems	Prof. Dr. Muhammad Usman Ghani Khan Sultan Qaboos IT Chair and Full Professor & COMSTECH Distinguished Scholar Department of Computer Science University of Engineering and Technology (UET) Lahore, Pakistan
AI and ML for Process Safety: Prospects and Pitfalls	Prof. Dr. Salim Ahmed Head of the Department of Process Engineering Memorial University of Newfoundland (MUN) St. John's, Canada

Morning Technical Sessions | 12:00 – 13:15**Session T1SA — Morning Technical Session | 12:00 – 13:15**

Track 7: Petroleum, Mining, and Ocean Technology

Room: R#315, Sir Jagadish Chandra Bose Academic Building, JUST
Session Chair: Professor Dr. H. M. Zakir Hossain, Dept. of PME, JUST
Session Co-Chair: Dr. Mohammad Forrukh Hossain Khan, Dept. of PME, JUST
Session Moderator: Ibrahim Khalil, Mahmudul Hasan Rohan

#	Paper ID	Paper Title	Presenting Author(s)
1	38	Land Subsidence Prediction Due to Coal Extraction in Barapukuria Coal Mine	<i>Tonmoy Saha</i>
2	93	Wave Energy Assessment and Optimal Site Selection in the Bay of Bengal: A Multi-Criteria Analysis for Wave Energy Converter Deployment	<i>Animesh Roy</i>
3	104	Integrated Investigation of Self-heating and Combustion Risk in Coal Stockpiles at Barapukuria Coal Mine	<i>Labiba Akter</i>
4	110	Optimizing Mine Water Treatment Through Advanced Technologies: A Case Study of Barapukuria Coal Mine	<i>Md. Sohan Miah</i>
5	132	Petrophysical Evaluation of Habiganj Gas Field using Well Log Interpretation	<i>Md. Maharaj Mredul Soumic</i>

Session T1SB — Morning Technical Session | 12:00 – 13:15

Track 6: Mechanical Engineering, Production Technology, and Management

Room: R#316, Sir Jagadish Chandra Bose Academic Building, JUST
Session Chair: Professor Dr. A.S.M. Mojahidul Hoque, Dept. of IPE, JUST
Session Co-Chair: Dr. Md. Aminul Islam, Dept. of PME, JUST
Session Moderator: Md. Masud Rana, Md. Shafi

#	Paper ID	Paper Title	Presenting Author(s)
1	23	Assessment of Hardened Properties of Concrete Using VCCTL: A Simulation Based Approach	<i>Md. Shahadat Hossain</i>
2	108	Mechanical Shear Strengthening Techniques for RC Beams: A Critical Review	<i>Md. Farden Ahasan Sakib</i>

3	115	Experimental Investigation of Bio-Based PCM for Photovoltaic Panel Cooling	Nur Mohammad Robin
4	143	Thermal influence of flower shaped inner geometry on MHD natural convection flow of hybrid nanofluid inside a square cavity	Prianka Mallick
5	149	Effect of SMAW and TIG welding parameters on defects and Heat-affected Zone (HAZ) hardness in various steels	Pranta Dutt

Session T1SC — Morning Technical Session | 12:00 – 13:15

Track 8: Textile Engineering and Sustainable Technology

Room: R#304, Sir Jagadish Chandra Bose Academic Building, JUST		Session Chair: Professor Dr. Md. Moklesur Rahman, Dept. of PME, JUST Session Co-Chair: Dr. Chanchal Kumar Kundu, Dept. of TE, JUST Session Moderator: Saklain Mustak, Redwan Ahmed	
#	Paper ID	Paper Title	Presenting Author(s)
1	20	Sustainably Engineered PVA/CMC–Rice Husk Ash Bio-composites: Structure–Property Relationships and Barrier Enhancement toward Green Functional Materials	<i>Md. Mebtahul Jannat</i>
2	77	Environmentally Friendly Superabsorbent Bio-hydrogel for Agricultural Soil Conditioning and Plant Micronutrient Release	<i>Tarun Kumar Gayen</i>
3	78	Sustainable Fuel Production from Scrap Tires and Vegetable Waste via Electromagnetic Induction-Heated Co-Pyrolysis	<i>Fabiha Nawshin Satta</i>
4	81	A Comprehensive Review of Soil Reinforcement Using Natural and Synthetic Fibers: Mechanisms, Performance Comparison, and Emerging Sustainable Trends	<i>Sharifuzzaman Shafi</i>
5	85	Eco-Friendly Ceramic Tile Production Using Sugarcane Bagasse Ash and Eggshell Waste	<i>Md. Parvej Mosaraf Sohel</i>
6	148	Study on the Rockwell hardness behavior of coal fly ash incorporated injection molded thermoplastic composites	<i>M S Rabbi, Suman Bepari</i>

Session T1SD — Morning Technical Session | 12:00 – 13:15

Track 4: Electrical, Electronics, and Communication Engineering

Room: R#233, Sir Jagadish Chandra Bose Academic Building, JUST		Session Chair: Professor Dr. Mohammad Tofayal Ahmed, Dept. of PME, JUST Session Co-Chair: Dr. Md. Moznuzzaman, Dept. of EEE, JUST Session Moderator: Md. Al Muzahid Pranto, Aminul Hoque	
#	Paper ID	Paper Title	Presenting Author(s)
1	34	WSN-Based Solar Microgrid Health Monitoring for Rural Bangladesh	Tanjila Akter
2	46	Performance Enhancement of I-slot Microstrip Patch Antenna Using Different Substrate Materials	Anamika Mondol
3	56	Adoption of Floating Solar Photovoltaics (FSPV) for Wastewater Management: Technical, Environmental, and Economic Evaluation	BIPRO MONDAL
4	64	Unlocking the Potential of Agricultural Wastes for the Production of Next-Generation Capacitors: Rice Husk, Corncob, and Coconut Shell	Naimur Rahman

5	75	Design and Development of an Automated Soil Irrigation System using Arduino Uno	Ayon Kumar Bhowmick Ovi, Nabib Ahamed Sujon, Nafiul Islam Khan Nafi
6	76	Techno Economic Analysis of a Solar Powered High Density Fish Farm in Bangladesh	Nazmul Hasan

Session T1SE — Morning Technical Session | 12:00 – 13:15

Track 9: Nuclear, Plasma, Radiological Engineering and Processes

Room: R#101, Sir Jagadish Chandra Bose Academic Building, JUST		Session Chair: Professor Dr. Muhibul Haque Bhuyan, Dept. of EEE, AIUB Session Co-Chair: Dr. Md. Rafiul Hasan, Dept. of ChE, JUST Session Moderator: Abu Taher, Md. Naeem Howlader	
#	Paper ID	Paper Title	Presenting Author(s)
1	103	Integrated DFT–MD–ML Multiscale Modelling of Hydrogen Isotope Trapping and Oxide Evolution in Tungsten under Fusion-Relevant Plasma Exposure	Musavvir Haque
2	117	Structural and Optical Properties of Ni-Doped CuO Thin Films Prepared by CBD	Md Mosharof Hosen
3	121	Optical And Structural Properties of Ni-doped Lead Sulfide Thin Films Synthesized Via Chemical Bath Deposition Method	Md Mehedi Hasan Shuvo
4	124	Deposition Time Dependent Structural and Optical Properties of Cupric Sulfide Thin Films	Saifur Rahman Rakib
5	126	Preparation and Characterization of Lead Sulfide Thin Films Synthesized by Chemical Bath Deposition Technique	Sudip Sital Gosh
6	135	Structural, Optical and Electrical Properties of CuS Thin Films: Effect of Deposition Parameter	Parvez Mosarrof

Afternoon Technical Sessions | 14:45 – 16:00

Session T2SA — Afternoon Technical Session | 14:45 – 16:00

Track 1: Biomedical Engineering and Bioscience | Track 4: Electrical, Electronics, and Communication Engineering

Room: R#315, Sir Jagadish Chandra Bose Academic Building, JUST		Session Chair: Dr. Mehedi Hassan (Jewel), Dept. of EEE, JUST Session Co-Chair: Dr. Md. Kamrul Islam, Dept. of PME, JUST Session Moderator: Mohammad Basel, Milton Mistry	
#	Paper ID	Paper Title	Presenting Author(s)
1	30	Pulsatile Hemodynamic Analysis in Coronary Bifurcated Arteries with Aneurysm and Stenosis	Howlader Jannatul Naeem
2	157	Assessment of Syngas Quality from Peat and Biomass Co-Gasification in a Lab-Scale Bubbling Fluidized Bed Gasifier	Md. Raihan
3	181	Multi-Band Differential Entropy Feature Extraction with Ensemble Learning for EEG-Based Emotion Recognition	Rabita Hasan
4	79	Performance Enhancement of p-n CsGeI3-Based Solar Cell Through Device Architecture Optimization Using SCAPS	Apurbo Sharma
5	83	Next-Generation Electrical Systems for Efficient, Low-Emission Waste-to-Energy Conversion	Md. Ridwan Al Mustavy

6	174	Household Energy-Saving Behaviors and Effect On Campus Sustainability in AIUB	<i>Md. Jahidun Muntaka Farabi</i>
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Session T2SB — Afternoon Technical Session | 14:45 – 16:00

Track 2: Chemical Engineering and Material Science | Track 9: Nuclear, Plasma, Radiological Engineering and Processes

Room: R#316, Sir Jagadish Chandra Bose Academic Building, JUST		Session Chair: Professor Dr. Zaved Khan, Dept. of ChE, JUST Session Co-Chair: Dr. Most. Afroza Khatun, Dept. of ChE, JUST Session Moderator: Jannatun Nayeem, Rahannuma Orchy	
#	Paper ID	Paper Title	Presenting Author(s)
1	5	Soluble sulfide removal from tannery liming wastewater using Pontederia crassipes biosorbent	Prof. Dr. Md. Abul Hashem, Naima Khatun
2	43	Turbidity removal from composite tannery wastewater using natural adsorbent	Md. Shahriar Shahadat
3	89	Adsorption of soluble sulfide from tannery liming wastewater on thermally activated keratin biosorbent	S. M. Sany Isnain, Fariha Sumaiya
4	146	Green Synthesis of Bioactive ZnO Nanoparticles Using De-Bittered Moringa oleifera Leaves for Enhanced Antibacterial and Antioxidant Performance	Tamanna Rahman
5	136	A DFT Study on Penta-Boron Phosphorus (P-BP5) Nanosheet for Oxygen-Based Toxic Gases Sensing.	Nure Modina Sathi
6	140	A DFT Study on Pristine and Co-Doped Graphene Boron Nitride Nanosheets for PH3 and H2Se toxic gas Detection	Irin Akter Mim

Session T2SC — Afternoon Technical Session | 14:45 – 16:00

Track 3: Computational Intelligence and Applications

Room: R#304, Sir Jagadish Chandra Bose Academic Building, JUST		Session Chair: Professor Dr. Md. Alam Hossain, Dept. of CSE, JUST Session Co-Chair: Dr. Md. Kamrul Islam, Dept. of CSE, JUST Session Moderator: Al Shaharia Rafid, Masaba Rahman Suchi	
#	Paper ID	Paper Title	Presenting Author(s)
1	26	Machine-Learning-Based Energy Storage Optimization in Renewable-Powered Wastewater Treatment Plants for Heavy-Metal Removal	<i>MD. Estiak Ahmed</i>
2	98	A Hybrid Random Forest Surrogate Model for Accelerated Ion Implantation in Si, SiC, and GaAs Substrates	<i>Abdullah Shadek Fahim</i>
3	106	A Hybrid Empirical Machine Learning Approach for Ground Vibration Prediction in the Maddhapara Granite Mine	<i>Md. Manjurul Islam</i>
4	169	A Multi Model Machine Learning Approach for Long Term Rainfall Prediction in Dhaka	<i>Md Ashikur Rahman</i>
5	173	Beyond the Black Box: An Interpretable Machine Learning Framework for Quantifying Honey Adulteration with UV-Vis-NIR Spectroscopy	<i>Iqbal Hossain</i>

Session T2SD — Afternoon Technical Session | 14:45 – 16:00

Track 5: Environmental Science and Climate Change | Track 7: Petroleum, Mining, and Ocean Technology

Room: R#233, Sir Jagadish Chandra Bose Academic Building, JUST		Session Chair: Session Co-Chair: Session Moderator:	Professor Dr. Md Jahangir Alam, Dept. of ChE, JUST Dr. Shahanaz Parvin, Dept. of ChE, JUST Md.Foysal Hossen Shohag, Md. Iftekhar
#	Paper ID	Paper Title	Presenting Author(s)
1	1	Spatiotemporal Monitoring of Shoreline Dynamics along the Bay of Bengal Coast of Bangladesh Using GIS and Remote Sensing	<i>Md. Sharifuzzaman</i>
2	22	Investigation of heavy metal levels in soil Hazaribagh tannery area near Dhaka-Narayanganj-Demra dam, Dhaka, Bangladesh	<i>Jannatul Nime Tabassum</i>
3	61	Hydrochemical Assessment and Water Quality Index Evaluation of Industrial Canals in the National Special Economic Zone, NSEZ	<i>Partha Pratim Nath, Barkat Ullah Bhuiyan</i>
4	182	Assessing The Dominance of Survival-Based Adaptation Strategies Among Urban Climate Migrants In Khulna City, Bangladesh	<i>Khondoker Mahmud Parvez</i>
5	155	Integrated Desulfurization of Coal through Fenton Oxidation and Froth Flotation: A Hybrid Approach for Clean Coal Production	<i>Asif Akbar Siyam</i>
6	156	Energy Perspective on Barapukuria coal: Functional and Morphological Characterization Using FTIR and SEM-EDX.	<i>Md. Riyad</i>

Poster Presentation Session | 14:45 – 16:00

1 st Floor, Sir Jagadish Chandra Bose Academic Building, JUST	Session Chair: Professor Dr. Biplob Kumar Biswas, Dept. of ChE, JUST Session Moderator: Md. Mustakim Islam, Shyan Muhtasin Sazed
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#	Paper ID	Poster Title	Author(s)
1	2	A Blockchain-Enabled Framework for Secure and Transparent Smartphone Ownership Tracking	<i>Sadman Sakib</i>
2	3	Integrating Fog Computing and Content Caching: A Hybrid Framework for Intelligent Vehicular Networks	<i>Md. Manirul Islam</i>
3	6	Assessing Groundwater Dynamics and Its Effective Use in Agriculture for Climate-Resilient Groundwater Management	<i>Rial Ahmed, Mahmuda Siddika</i>
4	11	Silica supported Nickel coated Charcoal photocatalyst for enhancing Photocatalytic dye degradation of Rhodamine B	<i>Md Sumon Sarkar</i>
5	13	Design of Ni-TiO ₂ @Charcoal as Photocatalysts for Methylene Blue Degradation under UV-Visible Irradiation	<i>Mst. Refath Amena</i>
6	17	Urban Tree cover detection UAV, Remote Sensing and Deep Learning	<i>Md Sabbir Islam</i>
7	19	Comprehensive phytochemical profiling and pharmacological assessment of leaves of Adenosma caeruleum	<i>Raiyan Rahman Reon</i>
8	21	Estimating Soil Evaporation and Basal Crop Coefficients for Mulched vs. Non-mulched Maize: Experimental Measurements and Simulations	<i>Tasrif Rayhan</i>

#	Paper ID	Poster Title	Author(s)
9	24	Assessment of Noise Intensity and its Relationship with Land Use at Major Intersections in Khulna City.	<i>Sumiya Tasnim</i>
10	28	A Machine Learning-Driven IoT-Enabled Activity Monitoring System for Chronic Disease Prediction and Personalized Diet Planning	<i>Md Tarikul Islam</i>
11	29	Computational Analysis of Pulsatile Flow in Multiple Aneurysms: Influence of Aneurysm Size and Blood Rheology	<i>Sidratul Muntaha</i>
12	35	RUSLE Model-Based Soil Erosion Estimation for Prioritization of Mitigation Strategies: A Case Study in Gaibandha District, Bangladesh	<i>Akila Jamal</i>
13	42	Assessing Morphological Change and Future Dynamics of Three Major Transboundary Rivers in Sylhet District	<i>Arpita Hoom Chawdhury</i>
14	44	Predicting Evapotranspiration for Vegetable Crops by Simulating Root-Zone Soil Water with HYDRUS-1D	<i>Md. Farhan Galib</i>
15	47	Assessment of Salinity Related Hazard, Vulnerability and Risk in the South-West Coastal Region of Bangladesh Using Principal Component Analysis: A Case Study in Bagerhat District	<i>Maria Akter Pospo</i>
16	59	In Silico Design and Immunoinformatics Validation of a Multi-Epitope Vaccine Targeting Hemagglutinin and Fusion Proteins of Morbillivirus	<i>Md. Morshed Zaman</i>
17	63	Computational Hemodynamics of Pulsatile Flow in Saccular and Fusiform Aneurysms: The Role of Blood Rheology	<i>Farjana Mitu</i>
18	70	Alternate Wetting and Drying for Climate-Resilient Boro Rice: Impacts on Yield, Water Use Efficiency, and Water Quality	<i>Md. Ashrafuzzaman</i>
19	71	Ecological health of the Kushiya River in the Meghna Basin as indicated by hydrological assessment	<i>Md. Abdur Rouf</i>
20	84	Assessment of Land Deformation at Barapukuria Coal Mine Area and its Impacts with Geospatial Techniques	<i>Md. Rubel Islam</i>
21	91	Household waste management behavior pattern among urban and rural people of Jashore district	<i>Ananna Krishna Roy</i>
22	97	Geo-Spatial and Machine Learning-Based Assessment Of Urban Groundwater Sustainability in The Rangpur Region, Bangladesh	<i>Md. Mosharaf Hosen</i>
23	101	Sustainable Ventilation System: Selection Air-Distribution Design for Maddhapara Granite Mine	<i>Farhana Yesmin</i>
24	105	Assessing Urban Density Transformation in Chattogram City Corporation, Bangladesh (2000-2024): A Remote Sensing and GIS Approach with LULC and NDBI Analysis	<i>Md. Omar Faruk</i>
25	107	Interface Engineering of PMMA/g-C ₃ N ₄ Nanocomposites: Influence on Mechanical, Thermal and Efficient UV Shielding Properties	<i>Md. Lalon Mia, Md. Al Shahariar</i>
26	118	Effect of non-Hermitian modulation for topological acoustic transmission in a 2D periodic system based on phononic crystal	<i>Md. Shuzon Ali</i>
27	123	FT-Conv-Net: A Deep Learning Framework For Ultrasound-Based PCOS Diagnosis	<i>Md Mahe Alam</i>

#	Paper ID	Poster Title	Author(s)
28	125	A Probabilistic Prediction of Bangladesh's Earthquake Activity and Vulnerable Regions.	<i>Md. Abir hossain</i>
29	129	Modelling Urban Surface Temperature Variability in Gazipur: A Multi-Temporal Remote Sensing and Machine Learning Approach	<i>Md. Taofikul Islam</i>
30	133	Vertical Distribution and Physico-Chemical Controls on Microplastic Deposition in Flood-Affected Soils.	<i>Md. Nazimul Islam Methon</i>
31	141	Ethical Challenges in AI and Gene Editing: Comparative Analysis of Privacy Risks and Regulatory Implications	<i>Ritu Dey</i>
32	151	Remote Sensing-Based Analysis Of Land Impacts On PM2.5 And Gaseous Pollutants: A Study Of Dhaka, Bangladesh	<i>Jubayad Bin Showkat</i>
33	158	Calculation of coalbed methane resource potential of Barapukuria coal mine using Meisner and Kim equation	<i>Sheikh Ayman Ahmad</i>
34	167	Genomic Insights into Multidrug-Resistant Enteric Escherichia coli in Adult Enteritis Patients in Bangladesh	<i>Babry Fatema</i>
35	171	Smart Fog Harvesting for Urban Air Quality and Water Augmentation: An Integrated Green Technology Approach for Climate Resilience in Bangladesh	<i>Nawer Islam Sattik</i>
36	172	Microwave-Assisted Extraction Optimization: Comparative Analysis of Gelatin Quality from Indigenous versus Crossbred Cattle Skins	<i>Iqbal Hossain</i>
37	178	Next-Generation Energy Dispatch Optimization Using an Automated Priority-Driven Control System for Bangladesh	<i>MD. Jahidul Islam Shanto</i>

2nd International Conference on Engineering and Technology for Sustainable Development



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ABSTRACTS

Track 01: Biomedical Engineering and Bioscience**Paper ID: 30****Presentation Mode: Oral****Pulsatile Hemodynamic Analysis in Coronary Bifurcated Arteries with Combined Aneurysm and Stenosis****Howlader Jannatul Nieem *, Chinmayee Podder and Sidratul Muntaha***Department of Mathematics, University of Barishal, Kornokathi, Barishal-8254, Bangladesh***Corresponding author: nieemislam53@gmail.com***Abstract**

Coronary bifurcations are common sites for complex vascular pathologies due to the complex geometry and highly pulsatile nature of coronary blood flow. The coexistence of aneurysm and stenosis in these regions can profoundly disturb local hemodynamics, creating heterogeneous wall shear stress (WSS) distributions, oscillatory shear index (OSI) variations, and flow recirculation zones that may contribute to disease progression, plaque instability, or aneurysm growth. Such hemodynamic disturbances are critical because they directly influence endothelial cell function and may exacerbate localized vascular remodeling or thrombosis. Despite the clinical significance, the interactive effects of coexisting aneurysms and stenoses on coronary hemodynamics remain poorly understood. The objective of this study is to computationally investigate pulsatile blood flow within a coronary bifurcated artery affected by both aneurysmal dilation and stenotic narrowing, and to assess the interactive hemodynamic consequences of these coexisting pathologies. An idealized three dimensional coronary bifurcation geometry was constructed in SolidWorks. Time dependent incompressible Navier-Stokes equations were solved using the finite element method in COMSOL Multiphysics. Physiologically realistic pulsatile coronary waveforms were applied at the inlet, while traction free boundary conditions were imposed at the outlets. Time resolved velocity fields, pressure distribution, WSS, and OSI were analyzed over a full cardiac cycle to quantify transient flow behavior. Results demonstrate that the stenotic segment induces accelerated flow and elevated peak WSS, whereas the aneurysmal region develops recirculating vortices, low-shear zones, and delayed diastolic flow reversal. OSI is markedly elevated along the aneurysm wall and downstream of the stenosis, indicating regions susceptible to endothelial dysfunction and potential aneurysm progression. These findings demonstrate that the combined presence of aneurysm and stenosis produces highly unsteady hemodynamic patterns with marked spatial variation, emphasizing the value of pulsatile simulations for understanding disease progression and improving predictive vascular modeling.

Keywords: *Stenosis, aneurysm, finite element method, pulsatile blood flow, bifurcated artery, hemodynamic behavior, wall shear stress (WSS), oscillatory shear index (OSI).*

Track 01: Biomedical Engineering and Bioscience**Paper ID: 157****Presentation Mode: Oral****Assessment of Syngas Quality from Peat and Biomass Co-Gasification in a Lab-Scale Bubbling Fluidized Bed Gasifier****MD. Raihan^{1,2}, Md. Riyad^{1,2}, Asif Akbar Siyam^{1,2}, Sheikh Ayman Ahmad^{1,2}, Minhaj Uddin Monir^{1,2 *}**¹*Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh*²*Energy Conversion Laboratory, Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh***Corresponding Author: monir_pme@just.edu.bd***Abstract**

The combined gasification of peat and agricultural waste has emerged as a practical way to generate cleaner syngas while lowering reliance on fully fossil-based fuels. This study focuses on the behaviour of peat and biomass when they are gasified together in a laboratory-scale bubbling fluidized bed (BFB) reactor operated with oxygen as the gasifying agent. The gasifier is built from stainless steel and includes a plenum chamber, a distributor plate, a sand-filled bed, and a freeboard zone, supported by a shell furnace for heating. A cyclone, tar collection system, and online gas analyzer are connected to the unit to monitor product gases and condensable compounds. Experiments will be performed between 600 °C and 900 °C using several peat–biomass blend ratios (1:0, 75:25, 50:50, 25:75, and 0:1). The goal is to observe how temperature and feed composition affect the amount and quality of the syngas. Tar released during co-gasification will also be examined. The results are expected to offer a clearer understanding of whether mixtures of peat and biomass can serve as a reliable feedstock for producing syngas and if this approach has potential for future scale-up to industrial systems.

Keywords: *Co-gasification, Bubbling fluidized bed, Syngas, Renewable energy, Peat*

Track 01: Biomedical Engineering and Bioscience**Paper ID: 181****Presentation Mode: Oral****Multi-Band Differential Entropy Feature Extraction with Ensemble Learning for EEG-Based Emotion Recognition****Rabita Hasan****Department of Electronics and Communication Engineering (ECE), Khulna University of Engineering and Technology, Khulna.***Corresponding Author: rabitahasan32@gmail.com***Abstract**

Emotion recognition from electroencephalogram (EEG) signals has become a major research focus in affective computing, with important applications in healthcare, human– computer interaction and brain– computer interfaces. Reliable identification of emotional states from neural activity enables adaptive systems that respond intelligently to user affect, supporting mental health monitoring, personalized learning and emotion-aware technologies. Among existing feature extraction techniques, Differential Entropy (DE) has shown strong potential because it effectively quantifies frequency-specific activity while capturing the uncertainty and distributional complexity of EEG signals. These properties provide stable and discriminative representations across frequency sub-bands, making DE suitable for emotion recognition. This study aims to classify emotional dimensions, specifically valence and arousal, using DE-based features combined with ensemble machine learning models. EEG recordings from 32 participants in the DEAP dataset were pre-processed using a fourth order Butterworth band-pass filter (0.5–45 Hz) to suppress noise and retain relevant neural information. The signals were segmented into non-overlapping three-second windows, and DE features were extracted from five canonical sub-bands: delta, theta, alpha, beta and gamma. These features were used to train three ensemble classifiers: Random Forest, Gradient Boosting and Extreme Gradient Boosting. Model performance was evaluated using a ten-fold cross validation strategy to ensure robustness and reduce bias. Experimental results indicate that Extreme Gradient Boosting achieved the highest performance, obtaining average accuracies of 87.56% for valence and 84.23% for arousal, while Random Forest and Gradient Boosting produced slightly lower yet competitive results. Additional evaluation metrics, including precision, recall and F1-score further confirmed the reliability of the proposed framework. Feature importance analysis indicated that DE features from the alpha, beta, and gamma bands played a dominant role in emotion classification, with beta and gamma bands exhibiting higher discriminative power due to their strong links to cognitive activity and affective arousal. Overall, DE features with ensemble models enable generalizable EEG-based emotion recognition.

Keywords: *DEAP Dataset, Ensemble Models, EEG Signals, Emotion Classification, Differential Entropy.*

Track 01: Biomedical Engineering and Bioscience

Paper ID: 19

Presentation Mode: Poster

Comprehensive phytochemical profiling and pharmacological assessment of leaves of *Adenosma caeruleum***Raiyan Rahman Reon, Mahjabin Sara, Prof. Dr. Samir Kumar Sadhu****Pharmacy Discipline, Khulna University***Corresponding Author: sksadhu1969@yahoo.com***Abstract**

This study explores the phytochemical composition and pharmacological activities of *A. caeruleum* leaf extract, a medicinal plant of the Plantaginaceae family native to the Sundarbans mangrove forest. Traditionally, the species has been employed to treat postpartum ailments, hepatic and biliary disorders, jaundice, fever, and pain. Phytochemical screening confirmed the presence of alkaloids, flavonoids, phenolics, steroids, saponins, terpenoids, and glycosides, while chromatographic separation yielded phenylethanoid glycosides (crenatoside, verbascoside, cistanoside F, campneoside I, apigenin 7-O- β -d-glucuronopyranoside, apigenin 7-O- β -d-glucopyranoside) and monoterpenoids (β -pinene, 1,8-cineole, β -terpinene) along with monoterpenoid peroxides as reported in previous literature. Antioxidant activity was notable, with significant free radical scavenging capacity (IC_{50} : 37.72 μ g/mL) comparing with standard ascorbic acid (IC_{50} : 19.51 μ g/mL). Quantitative analysis revealed high phenolic (79.07 mg GAE/g), flavonoid (393.88 mg QE/g), tannin (90.6 mg GAE/g), and total antioxidant capacity (244.21 mg AAE/g). Acute toxicity studies confirmed safety in murine models. Antimicrobial activity showed moderate inhibition zones (1–10 mm at 250–500 μ g/disc), while analgesic evaluation demonstrated significant inhibition of writhing (66.87% and 71.27% at 250 and 500 mg/kg, respectively), comparable to Diclofenac Na (81.92%). The extract also exhibited dose-dependent anti-inflammatory activity and mild antipyretic effects. Importantly, *A. caeruleum* demonstrated significant hepatoprotective potential in paracetamol-induced liver injury, with 500 mg/kg reducing ALT (57.36 U/L), AST (111.34 U/L), and bilirubin (0.704 mg/dL) to near-normal levels, comparable to Silymarin. Furthermore, molecular docking studies revealed strong binding affinities of key phytoconstituents with antioxidant and hepatoprotective target proteins, supporting the experimental findings. Overall, the results highlight *A. caeruleum* as a promising hepatoprotective plant rich in bioactive compounds, reinforcing its ethnomedicinal relevance and justifying further pharmacological and mechanistic investigations.

Keywords: *A. caeruleum*, Phytochemical screening, Mangrove medicinal plants, Antioxidant activity, Hepatoprotective potential, Paracetamol-induced liver injury, Molecular docking,

Track 01: Biomedical Engineering and Bioscience**Paper ID: 29****Presentation Mode: Poster****Computational Analysis of Pulsatile Flow in Multiple Aneurysms: Influence of Aneurysm Size and Blood Rheology****Sidratul Muntaha^{*}, Chinmayee Podder and Howlader Jannatul Nieem***Department of Mathematics, University of Barishal, Kornokathi, Barishal-8254, Bangladesh**^{*} Corresponding author: smuntaha430@gmail.com***Abstract**

Abdominal aortic aneurysm (AAA) is a life-threatening vascular dilation that may extend into the iliac arteries, forming multiple interconnected aneurysms. Such complex geometries significantly affect local blood flow patterns, wall shear stress (WSS) distribution, which are crucial to the development of endothelial damage, thrombosis, possible rupture. In this study, transient blood-flow behavior is examined in a computational model containing multiple aneurysms in the infrarenal aorta and the left and right iliac arteries, emphasizing the effects of aneurysm dimensions and blood rheology. An idealized 3D model of the infrarenal aorta with iliac bifurcation and aneurysmal bulges was developed. Time-dependent simulations were performed in COMSOL Multiphysics under laminar flow conditions using a pulsatile inlet velocity and zero pressure outlet boundary. The arterial wall was assumed rigid, and blood was modeled as both a Newtonian and a non-Newtonian Carreau fluid. An enlarged abdominal aneurysm model was also analyzed to evaluate the impact of geometric variation. Flow separation and vortex patterns were seen inside all the aneurysm bulges, especially when the flow reached peak systole. Areas with low wall shear stress (WSS) and high oscillatory shear index (OSI) were mainly found along the aneurysm walls. The bigger abdominal aneurysm showed stronger flow recirculation and more variation in WSS, particularly near the iliac bifurcation. The Carreau model gave slightly lower shear stress than the Newtonian model, showing that blood rheology has an important effect on the results. The results show that larger aneurysms and changes in blood rheology strongly affect local blood flow, increasing endothelial damage and thrombus risk in low-shear areas. These effects may weaken the aneurysm wall and raise rupture risk. The findings improve understanding of multiple-aneurysm mechanics and can help guide better diagnosis and treatment.

Keywords: *Infrarenal aorta, Iliac arteries, Multiple aneurysms, Hemodynamics, Wall shear stress, Oscillatory shear index, Blood rheology, Computational fluid dynamics*

Track 01: Biomedical Engineering and Bioscience**Paper ID: 59****Presentation Mode: Poster****In Silico Design and Immunoinformatics Validation of a Multi-Epitope Vaccine Targeting Hemagglutinin and Fusion Proteins of Morbillivirus****Md. Morshed Zaman*, Trina Nishita***Faculty of Fisheries, Sylhet Agricultural University, Sylhet-3100, Bangladesh,***Corresponding author: morshed.fisheries@student.sau.ac.bd***Abstract**

Morbillivirus, particularly the measles virus, remains a highly contagious pathogen responsible for significant global morbidity and mortality. Despite the availability of vaccines, emerging viral variants and incomplete immunization coverage necessitate the development of novel vaccine candidates. This study aimed to design a multi-epitope subunit vaccine targeting conserved regions of Morbillivirus hemagglutinin (H) and fusion (F) glycoproteins using computational immunoinformatics approaches. Antigenic protein sequences were retrieved from NCBI and analyzed using VaxiJen and ProtParam to assess antigenicity, stability, and physicochemical properties. T- and B-cell epitopes were predicted using IEDB tools for MHC class I, class II, and B-cell binding affinity. Non-allergenic and non-toxic epitopes with high antigenicity and full conservancy were selected. Linkers (AAY and GPGPG) and an adjuvant sequence (β -defensin) were incorporated to construct the final vaccine. Structural modeling, refinement, and validation were performed to evaluate conformational stability. Molecular docking analyzed vaccine-TLR interactions, and immune simulation was conducted to predict humoral and cellular immune responses. The final construct demonstrated high antigenicity (VaxiJen score 1.00), stability (instability index 20.28), and favorable hydropathicity (GRAVY -0.131). Docking studies revealed strong binding affinity with immune receptors, supporting its potential to elicit robust immune responses. Immune simulations predicted elevated IgM, IgG1, and IgG2 levels, along with increased memory B and T cell populations, indicating long-term immunity. The designed multi-epitope vaccine candidate targeting the hemagglutinin and fusion proteins of Morbillivirus exhibits promising immunogenicity, stability, and interaction potential. These findings support its further in vitro and in vivo validation for future vaccine development.

Keywords: *Morbillivirus, Measles virus, Multi-epitope vaccine, Immunoinformatics, Reverse vaccinology, Hemagglutinin protein, Fusion protein, Epitope prediction*

Track 01: Biomedical Engineering and Bioscience**Paper ID: 63****Presentation Mode: Poster****Computational Hemodynamics of Pulsatile Flow in Saccular and Fusiform Aneurysms: The Role of Blood Rheology****Farjana Mitu* and Chinmayee Podder****Department of Mathematics, University of Barishal, Barishal-8254, Bangladesh***Corresponding author: farjanamitu055@gmail.com, cpodder@bu.ac.bd***Abstract**

Abnormal dilations of blood vessels, known as arterial aneurysms, represent a significant clinical concern because they can alter normal hemodynamics and increase the risk of rupture. The geometric diversity of aneurysms including saccular and fusiform contributes to complex flow behavior and necessitates detailed investigation to understand their biomechanical environment. This study employs computational fluid dynamics (CFD) to analyze pulsatile blood flow in three dimensional arterial models with varying aneurysmal morphologies, developed in SolidWorks and simulated using COMSOL Multiphysics. The numerical simulations are performed using the finite element method (FEM), enabling accurate spatial and temporal resolution of flow patterns. Unsteady, time-dependent simulations are performed under both Newtonian and Carreau non Newtonian assumptions to capture the rheological behavior of blood more realistically. Key hemodynamic parameters including velocity field, pressure distribution, wall shear stress (WSS), oscillatory shear index (OSI), time-averaged wall shear stress (TAWSS), and relative residence time (RRT) are computed and compared across different models. The results highlight notable differences between Newtonian and non-Newtonian simulations, particularly in regions of flow separation and recirculation within and downstream of the aneurysm. Elevated OSI and RRT values are observed in non-Newtonian cases, suggesting a higher tendency for disturbed flow and thrombus formation. The study provides a comparative insight into the influence of aneurysm geometry and blood rheology on vascular hemodynamics. The findings emphasize the necessity of incorporating non Newtonian properties in CFD simulations and offer a refined understanding of risk factors associated with aneurysm progression and potential rupture.

Keywords: *Arterial aneurysm, Computational fluid dynamics (CFD), Hemodynamics, Pulsatile blood flow, Finite element method (FEM), Non-Newtonian fluid*

Track 01: Biomedical Engineering and Bioscience**Paper ID: 123****Presentation Mode: Poster****FT-Conv-Net: A Deep Learning Framework for Ultrasound-Based PCOS Diagnosis****Md Rakibul Islam Midul¹, Md Mahe Alam², Asad Ahmed³, Rabul Hussain Iaskar**¹*Department of Electronics and Communication Engineering, National Institute of Technology Silchar Assam, India*²*Department of Electronics & Telecommunication Engineering Rajshahi University of Engineering & Technology, Bangladesh*³*Department of Computer Science and Engineering, National Institute of Technology Silchar Assam, India***Abstract**

Polycystic Ovary Syndrome (PCOS) is a common female hormonal disorder that is revealed during the reproductive years with symptoms of irregular menstrual cycle and elevated levels of androgen. It creates a great hindrance to female pregnancy. PCOS also causes other syndromes, such as alopecia, hirsutism, acne, hyperandrogenemia, and oligoovulation. PCOS is detected by gynecologists primarily based on the number and size of follicles present in the ovaries. This method is subject to human error. In recent times, PCOS detection using computer-aided diagnosis (CAD) has gained popularity for providing more accurate diagnosis and reducing health risks for women by enabling early prediction. Several Machine Learning (ML) and Deep Learning (DL) models for PCOS detection have been proposed in different studies; however, these models did not adequately address the preprocessing requirements essential for optimal performance. Effective preprocessing is crucial for the design of robust and reliable systems. In this article, the FTConv-Net model has been developed to analyze the effect of data preprocessing in PCOS prediction using deep learning. This research article classifies PCOS and non-PCOS from ultrasound ovary images by using fine-tuning pre-trained CNN models such as VGG16, VGG19, ResNet-50, Inception V3, and DenseNet121 with image preprocessing techniques such as CLAHE for image enhancement and Watershed for image segmentation. Inception V3 with preprocessing shows the highest accuracy of 98.56%, which indicates that this model can be implemented to diagnose PCOS more effectively and accurately.

Keywords: *Polycystic Ovary Syndrome (PCOS), Computer-aided diagnosis (CAD), Machine learning, Deep learning, Ultrasound imaging, Image preprocessing, CLAHE, Watershed segmentation*

Track 01: Biomedical Engineering and Bioscience**Paper ID: 167****Presentation Mode: Poster****Genomic Insights into Multidrug-Resistant Enteric *Escherichia coli* in Adult Enteritis Patients in Bangladesh****Babry Fatema, Asaduzzaman Asad, Mosabbir Ahmed, Zhahirul Islam****Laboratory of Gut-Brain Axis, Infectious Disease Division (IDD), icddr,b, Dhaka-1212, Bangladesh***Abstract**

AMR in enteric *E. coli* poses a growing global health challenge, particularly in low-resource settings such as Bangladesh, where rapid resistance dissemination is largely mediated by mobile genetic elements (MGEs), including plasmids and integrons; integrating genomic surveillance with computational analyses offers a powerful approach to understanding resistance dynamics and informing public-health strategies. To characterize the phenotypic resistance and genomic determinants of MDR *E. coli* from adult enteritis patients, twenty-six isolates obtained from 40 stool specimens at icddr,b, Dhaka Hospital underwent antimicrobial susceptibility testing against twelve clinically relevant antibiotics, while six representative MDR isolates underwent whole-genome sequencing (WGS) to profile resistance genes, virulence factors, plasmid replicons, plasmid mobility, integrons, and sequence types, with computational analyses further identifying high-risk plasmids and resistance clusters. High resistance was observed to β -lactams (90%), cephalosporins (85%), and carbapenems (75%); WGS revealed diverse high-risk lineages including ST44, ST648, ST354, and ST131, along with 143 virulence-associated genes, diverse integron cassette arrays such as *dfrA12*, *ant(3'')-Ia*, and CALIN structures, and plasmid replicons comprising 15% conjugative, 38% mobilizable, and 47% non-mobilizable elements, with a ~170 kbp conjugative IncF plasmid (AA324) carrying *bla*CTX-M-15, *bla*TEM-1B, and *qnrS* detected in two isolates, indicating high horizontal gene transfer potential. MDR *E. coli* circulating in Dhaka harbor potent MGEs facilitating resistance spread, and this study demonstrates how genomic and computational insights can enable data-driven AMR monitoring and precision antimicrobial stewardship, providing actionable intelligence to improve public health outcomes in high-burden settings.

Keywords: *Multidrug-Resistant *E. coli*; Genomic Surveillance; Plasmid-Mediated Resistance*

Track 2: Chemical Engineering and Material Science**Paper ID: 05****Presentation Mode: Oral****Soluble sulfide removal from tannery liming wastewater using *Pontederia crassipes* biosorbent****Md. Abul Hashem*, Shihab Uddin, Tarik Anam Tonmoy, Naima Khatun, Nusrat Ara Bhuyan***Khulna University of Engineering & Technology (KUET), Faculty of Mechanical Engineering, Department of Leather Engineering, Khulna-9203, Bangladesh***Corresponding author: hashem_518@yahoo.com***Abstract**

In tannery, liming is an alkaline hydrolysis of raw animal skin to remove epidermis, hair/wool, non-structural protein, fat, grease, and swelling of collagen protein. The chemicals (sodium sulfide, lime) used in the liming process generate a significant amount of wastewater containing hazardous pollutants, e.g., soluble sulfide, high chemical oxygen demand (COD), biochemical oxygen demand (BOD), and total dissolved solids (TDS). In this research, a thermally electrified biosorbent derived from *Pontederia crassipes* is used in batch experiments to remove soluble sulfide from liming wastewater. The collected *Pontederia crassipes* is sun-dried to remove moisture and subsequently subjected to thermal electrification at 350°C for 2 hours. The liming wastewater was pretreated with alum to facilitate coagulation and sedimentation, and the resulting supernatant was employed for subsequent batch adsorption experiments. The physicochemical parameters of the liming wastewater: pH, sulfide, salinity, TDS, COD, and BOD were 5889.15±5.95 mg/L, 27.63±0.11 mg/L, 24.02±0.3 g/L, 2883.21±6.2 mg/L, and 89658.21±7 mg/L, respectively. The batch experiments were conducted under optimized conditions for treating the liming wastewater, and the physicochemical parameters were assessed. At optimized conditions, the maximum sulfide removal efficiency was 93.34%, with the reductions of BOD (26.47%), COD (28.62%), and salinity (17.59%). SEM images, EDS spectra, and FTIR ensured the adsorption of sulfide onto the equipped biosorbent. The experimental outcomes indicate that water hyacinth can be effectively utilized as a sustainable and low-cost biosorbent for the efficient removal of sulfides from the tannery liming wastewater.

Keywords: *Sodium sulfide, Hydrogen sulfide, Batch experiment, Wastewater, Thermal activation*

Track 2: Chemical Engineering and Material Science**Paper ID: 43****Presentation Mode: Oral****Turbidity Removal From Composite Tannery Wastewater Using Natural Adsorbent****Md. Abul Hashem*, Md. Shahriar Shahadat, Syed Ehtashamul Hoque, Md. Fakhurul Hassan***Khulna University of Engineering & Technology (KUET), Faculty of Mechanical Engineering, Department of Leather Engineering, Khulna-9203, Bangladesh***Corresponding author: hashem_518@yahoo.com***Abstract**

Anthropogenic actions, such as industrial activities, continuously pollute the water. The tanning industry discharged the generated wastewater with high turbidity into the environment. Turbidity affects the ecosystem, as it is responsible for the malfunctioning of the photosynthesis process in aquatic organisms by interrupting the light penetration through the water body. In this study, an attempt has been made to investigate the removal of turbidity from tannery composite wastewater with different adsorbents. The ground coconut (*C. nucifera*) shell and its activated charcoal, ground mahogany (*S. macrophylla*) fruit shell and its activated charcoal, activated sugarcane (*S. officinarum*) bagasse charcoal were used to remove turbidity from the composite tannery wastewater. Results reveal that the turbidity removal for ground *C. nucifera* shell, activated *C. nucifera* shell charcoal, ground *S. macrophylla* fruit shell, activated *S. macrophylla* fruit shell charcoal, and activated *S. officinarum* bagasse charcoal was 65.34% and 65.03%, 68.41% and 66.26%, 71.14%, respectively. Among the adsorbents, thermally activated *S. officinarum* bagasse charcoal had the highest removal capacity. In a batch test, various parameters were optimized to obtain the maximum turbidity removal from composite tannery wastewater using the thermally activated *S. officinarum* bagasse charcoal. The turbidity removal was achieved at 73.66% with optimum conditions (dose 0.05g/50 mL wastewater, stirring time 30 min and with a relative solution pH 9.0), which revealed a reduction of chemical oxygen demand (COD), biochemical oxygen demand (BOD), total dissolved solids (TDS), total suspended solids (TSS), and total solids (TS) by 23.71%, 19.91%, 27.99%, 53.67% and 37.68%, respectively. The Pseudo-Second-Order (PSO) kinetics and the Freundlich isotherm were well fitted. Thus, the study revealed an innovative way of utilizing thermally activated *S. officinarum* bagasse charcoal could be used to remove turbidity from tannery wastewater.

Keywords: *Composite tannery wastewater, Sugarcane bagasse, Turbidity, Adsorption, Removal efficacy.*

Track 2: Chemical Engineering and Material Science**Paper ID: 89****Presentation Mode: Oral****Adsorption of soluble sulfide from tannery liming wastewater on thermally activated keratin biosorbent****Md. Abul Hashem*, S. M. Sany Isnain, Fariha Sumaiya, Md. Shahriar Shahadat, Md. Mukimujjaman Miem***Department of Leather Engineering, Khulna University of Engineering & Technology (KUET), Khulna-9203, Bangladesh***Corresponding author: hashem_518@yahoo.com***Abstract**

In tannery, liming is the first chemical treatment to remove hair, wool, epidermis, soluble proteins, fat, and grease. In the conventional liming, sodium sulfide (Na_2S) and lime (CaO) are used. The generated liming wastewater contains soluble sulfide, total dissolved solids (TDS), salinity, and electrical conductivity (EC) with a high content of biochemical oxygen demand (BOD) and chemical oxygen demand (COD). Upon acidification, sulfide in wastewater generates toxic hydrogen sulfide (H_2S) gas, which is harmful to humans and the environment. This study is focused on the adsorption of soluble sulfide from tannery liming wastewater on thermally activated biosorbent. The chemically dehaired goat hair waste was collected from the tannery, sun-dried, and oven-dried for 12 hours. The oven-dried goat hair was thermally activated at 450°C for 150 minutes. Batch-wise, sulfide adsorption was conducted considering varying parameters- biosorbent dose, contact time, and settling time. Before and after adsorption, the biosorbent was characterized through Scanning Electron Microscope (SEM), Energy Dispersive Spectroscopy (EDS), and Fourier Transform Infrared (FTIR). The sulfide, TDS, salinity, EC, and pH of the raw liming wastewater were 7176 mg/L, 11.28 g/L, 10.34 ppt, 18.38 mS, and 11.28, respectively. Results reveal the noticeable reduction of sulfide (92.9%) at optimized conditions, biosorbent dose at 0.2 g/30 mL wastewater with 10 minutes contact time. SEM images and EDS spectra ensured the absorption of sulfide and other pollutants from the liming wastewater. Thus, simple and adaptable techniques can be used to remove sulfide from liming wastewater using thermally activated biosorbent derived from chemically dehaired goat hair waste of tannery.

Keywords: *Hazardous, Environment, Waste management, Thermal activation, Toxic.*

Track 2: Chemical Engineering and Material Science**Paper ID: 146****Presentation Mode: Oral****Green Synthesis of Bioactive ZnO Nanoparticles Using De-Bittered Moringa oleifera Leaves for Enhanced Antibacterial and Antioxidant Performance****Tamanna Rahman, Fahima Siddikey, Md. Abu Rayhan, Wahidu Zzaman****Department of Food Engineering and Tea Technology, Shahjalal University of Science and Technology, Sylhet 3114, Bangladesh***Corresponding Author: wahid-ttc@sust.edu***Abstract**

This study reports the green synthesis of zinc oxide nanoparticles (ZnO-NPs) using de-bittered Moringa oleifera leaf extract as a sustainable and eco-friendly alternative to conventional chemical methods. The de-bittered extract was then used for ZnO-NPs synthesis, and the resulting nanoparticles were characterized using UV–Vis spectroscopy, FTIR, SEM, and EDS, confirming their nanoscale dimensions and biofunctional surfaces. SEM revealed a distinctive flower-like morphology with particle sizes ranging from approximately 45 to 82 nm. UV–Visible spectroscopy showed a high bandgap energy of 3.40 eV. Elemental composition analysis via Energy Dispersive X-ray Spectroscopy (EDS) demonstrated the purity of the ZnO nanoparticles. Fourier-transform infrared (FTIR) spectroscopy identified multiple functional groups on the nanoparticle surface, including O–H, C–H, C=O, C=C, and C–H bonds, indicating the presence of phytochemical capping agents such as flavonoids and phenolic acids. Antioxidant activity, evaluated through the DPPH assay, showed superior radical scavenging capacity. The antibacterial efficacy of green-synthesized ZnO nanoparticles (GsZnO-NPs) was evaluated at varying concentrations (50, 100, and 200 µg/mL) against Gram-positive and Gram-negative bacteria using the zone-of-inhibition (ZOI) method, where GsZnO-NPs at 200 µg/mL exhibited a high ZOI of 26.75 mm against Escherichia coli 30.00 mm against Staphylococcus aureus, and 24.50 mm against Bacillus subtilis, all of which were comparatively higher than those of the standard drug Cefpodoxime. These results highlight the potential of de-bittered M. oleifera extract in producing bioactive ZnO-NPs for food, pharmaceutical, and cosmeceutical applications.

Keywords: *Green synthesis, Zinc oxide nanoparticles, De-bittered Moringa oleifera, Antibacterial activity and Antioxidant properties*

Track 2: Chemical Engineering and Material Science

Paper ID: 11

Presentation Mode: Poster

Silica supported Nickel coated Charcoal photocatalyst for enhancing Photocatalytic dye degradation of Rhodamine B

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Abstract

Recently, extensive research has been conducted to improve the dye degradation activity of titanium dioxide (TiO₂) photocatalysts under UV-Vis irradiation. The study examines the Ni-TiO₂@Charcoal photocatalyst, aiming to enhance the degradation of methylene blue dye under UV-Vis irradiation significantly. Ni-TiO₂@charcoal photocatalysts were synthesized using wet impregnation methods and tested under various conditions, including pH, dye concentration, time, and catalyst dosage. Characterization of the prepared samples was conducted using X-ray diffraction (XRD), Fourier Transform Infrared (FTIR) spectroscopy, scanning electron microscopy (SEM), Energy-Dispersive X-ray spectroscopy (EDX), and Diffuse Reflectance Spectroscopy (DRS). The XRD results indicated that the anatase phase of TiO₂ was predominant. FTIR analysis confirmed the successful incorporation of nickel ions and charcoal into the TiO₂ matrix. Additionally, DRS results showed that the inclusion of nickel ions and charcoal reduced the band gap energy of TiO₂ to 3.13 eV. The photocatalyst achieved a degradation rate of 95.88% for methylene blue and demonstrated pseudo-first-order kinetics ($R^2 = 0.9969$). The catalyst's activity decreased at higher dye concentrations but improved with longer contact times, underscoring the essential role of charcoal in the degradation process. The synthesized Ni-TiO₂@charcoal photocatalyst proved to be highly effective in treating textile effluents under UV-Vis irradiation.

Keywords: Charcoal, Rhodamine-B, Photocatalysis, Adsorption, SiO₂-NiC composite, wastewater

Track 2: Chemical Engineering and Material Science

Paper ID: 13

Presentation Mode: Poster

Design of Ni-TiO₂@Charcoal as Photocatalysts for Methylene Blue Degradation under UV-Visible Irradiation

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Abstract

This study reports the green synthesis of zinc oxide nanoparticles (ZnO-NPs) using de-bittered *Moringa oleifera* leaf extract as a sustainable and eco-friendly alternative to conventional chemical methods. The de-bittered extract was then used for ZnO-NPs synthesis, and the resulting nanoparticles were characterized using UV-Vis spectroscopy, FTIR, SEM, and EDS, confirming their nanoscale dimensions and biofunctional surfaces. SEM revealed a distinctive flower-like morphology with particle sizes ranging from approximately 45 to 82 nm. UV-Visible spectroscopy showed a high bandgap energy of 3.40 eV. Elemental composition analysis via Energy Dispersive X-ray Spectroscopy (EDS) demonstrated the purity of the ZnO nanoparticles. Fourier-transform infrared (FTIR) spectroscopy identified multiple functional groups on the nanoparticle surface, including O-H, C-H, C=O, C=C, and C-H bonds, indicating the presence of phytochemical capping agents such as flavonoids and phenolic acids. Antioxidant activity, evaluated through the DPPH assay, showed superior radical scavenging capacity. The antibacterial efficacy of green-synthesized ZnO nanoparticles (GsZnO-NPs) was evaluated at varying concentrations (50, 100, and 200 µg/mL) against Gram-positive and Gram-negative bacteria using the zone-of-inhibition (ZOI) method, where GsZnO-NPs at 200 µg/mL exhibited a high ZOI of 26.75 mm against *Escherichia coli* 30.00 mm against *Staphylococcus aureus*, and 24.50 mm against *Bacillus subtilis*, all of which were comparatively higher than those of the standard drug Cefpodoxime. These results highlight the potential of de-bittered *M. oleifera* extract in producing bioactive ZnO-NPs for food, pharmaceutical, and cosmeceutical applications.

Keywords: Green synthesis, Zinc oxide nanoparticles (ZnO-NPs), *Moringa oleifera*, De-bittered extract, Nanobiotechnology, Flower-like morphology, Bandgap energy

Track 3: Computational Intelligence and Applications**Paper ID: 26****Presentation Mode: Oral****Machine-Learning-Based Energy Storage Optimization in Renewable-Powered Wastewater Treatment Plants for Heavy-Metal Removal****Airin Akter, Md. Estiak Ahmed, Shafiun Nahin Shourav, Zabin Tasmin***American International University Bangladesh, Dhaka, Bangladesh***Abstract**

Such treatment of heavy metals-laden industrial wastewater is extremely energy-intensive, while the integration of renewable energy sources introduces intermittent and operational uncertainty. This paper presents a unified machine-learning-based energy-storage optimization framework for a renewable-powered WWTP specifically designed for heavy-metal removal. The proposed system integrates a PV generation unit, a BESS, and an adsorption/ion-exchange removal process through a hierarchical, AI-enabled supervisory controller. A multi-model learning architecture comprises LSTM networks for dynamic load forecasting, Random Forest regressors for solar-generation prediction, and an ANN meta-model for adsorption-unit effluent estimation that provides real-time predictions to an MPC optimizer. The MPC jointly optimizes BESS charge-discharge cycles, power-electronic converter setpoints, and adsorption/regeneration timing in a manner that minimizes energy consumption while maintaining target pollutant-removal efficiency. Simulation results for a 400-kW PV + 1-MWh Li-ion BESS industrial WWTP have demonstrated the capability of the proposed framework to reduce grid-energy consumption up to 27%, lower peak-demand power by 30-35%, and decrease battery degradation by 18% through intelligent cycle management. Besides, heavy-metal removal efficiency increases from 98.1% to 99.6%, keeping the effluent concentration below 0.05 mg/L under widely varying influence conditions. This work establishes, for the first time, a cross-disciplinary connection between power-electronic-driven energy optimization and advanced heavy-metal remediation, demonstrating how electrical and electronic engineering methods can significantly enhance environmental process performance while reducing operating costs. The proposed framework provides a scalable foundation for next-generation sustainable wastewater treatment infrastructure.

Keywords: *Wastewater treatment plant (WWTP), heavy-metal removal, battery energy storage system (BESS), machine learning, predictive control, renewable energy integration, power electronics.*

Track 3: Computational Intelligence and Applications**Paper ID: 98****Presentation Mode: Oral****A Hybrid Random Forest Surrogate Model for Accelerated Ion Implantation in Si, SiC, and GaAs Substrates****Abdullah Shadek Fahim*, Abu Taher, Md. Mushfiqur Rahman, Musavvir Haque, Md. Shakil Hossain, Nabila Rahaman Ritu, Irin Akter Mim, Nure Modina Sathi, Md. Amzad Hossain***Department of Electrical and Electronic Engineering, Jashore University of Science and Technology***Corresponding Author: fahim.just.19@gmail.com***Abstract**

Computationally intensive Stopping and Range of Ions in Matter (SRIM) simulations were used to design and optimize advanced semiconductor materials, including silicon (Si), silicon carbide (SiC), and gallium arsenide (GaAs), to be used in photovoltaic (PV) solar cells and electric vehicle (EV) power devices. These simulations are physically accurate but require substantial computational resources and time, thus forming a key bottleneck in the fast development of next-generation energy materials. To significantly address this shortcoming, a hybrid Random Forest-based surrogate model has been developed to effectively replicate the results of SRIM ion implantation across a wide range of conditions. The scheme works in two consecutive stages: firstly, a classifier separates zero-range and profile events with a total accuracy of 91%; secondly, a multi-regressor model predicts nine critical physical parameters, such as projected range, straggle, lateral spread, vacancy concentration, backscattered ions, and transmitted ions. Approximately 5,600 validated SRIM simulations, including a wide range of ion species (Ar, As, B, Mg, P) and a variety of implantation energies, angles, and substrate thicknesses of Si, SiC, and GaAs targets, were used to train the surrogate model. The coefficient of determination (R^2) was obtained at an average of 0.97, demonstrating a high level of predictive power. Root Mean Square Error (RMSE) analysis confirmed that the Random Forest model achieved an error of only $\sim 275\text{\AA}$ for a projected range prediction, ensuring physical consistency with SRIM-simulated outputs. By providing real-time and accurate predictions, the surrogate model reduces the need for extensive SRIM simulations, which impose a significant computational and energy cost. The development enables rapid optimization of implantation parameters to achieve accurate doping, defect minimization, and junction optimization during solar cell fabrication, as well as the development of high-performance SiC-based EV power electronics.

Keywords: *SRIM simulation, Random Forest, Surrogate model, Ion implantation, Semiconductor materials, Photovoltaic cells, Electric vehicle (EV) electronics, Machine learning*

Track 3: Computational Intelligence and Applications**Paper ID: 106****Presentation Mode: Oral****A Hybrid Empirical Machine Learning Approach for Ground Vibration Prediction in the Maddhapara Granite Mine****Md. Manjurul Islam^{1,2}, Md. Sohan Miah^{1,2}, Istiak Ahammod^{1,2}, Mohammad Forrukh Hossain Khan^{1,2*}**¹Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh²Mine Planning and Design Laboratory, Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh*Corresponding Author: Mohammad Forrukh Hossain Khan (dr.forrukh@just.edu.bd)**Abstract**

Blast-induced ground vibration is a critical environmental and operational concern in hard-rock mines, particularly in densely populated regions such as the Maddhapara Granite Mine (MGM) in Bangladesh. This study integrates field-recorded vibration data with predictive modelling approaches to improve the accuracy of Peak Particle Velocity (PPV) estimation and ensure safe blasting practices. A total of 46 vibration events were measured using the Instantel Micromate system at six vulnerable surface locations around the mine. The measured PPV values ranged from 0.315 to 6.636 mm/s, with frequencies between 2.1 and 9.5 Hz, all remaining within the safety limits specified by USBM RI-8507 and DIN 4150 standards. Using this data, a site-specific USBM scaled-distance relationship, $PPV = 3.290(SD)^{-0.113}$, was derived through regression analysis ($R^2 = 0.521$), forming the foundation for reliable vibration prediction at the mine. To enhance prediction accuracy beyond the empirical USBM approach, the study evaluates the feasibility of integrating machine-learning frameworks- specifically ANN and hybrid PSO-ANN models- commonly applied in similar hard-rock PPV prediction studies. Given that the MGM dataset includes well-defined inputs such as maximum charge per delay, scaled distance, and measured PPV, these models are fully compatible and can be trained without methodological conflict. Their use is justified because the USBM model alone explains only a moderate share of the variance, indicating non-linear interactions within the geological medium. The combined empirical-and-AI approach strengthens predictive capability and supports safer, more optimized blasting practices at MGM.

Keywords: *Ground vibration, PSO-ANN, Peak Particle Velocity (PPV), Blasting optimization*

Track 3: Computational Intelligence and Applications**Paper ID: 169****Presentation Mode: Oral****A Multi Model Machine Learning Approach for Long Term Rainfall Prediction in Dhaka****Md Ashikur Rahman* Prof. A. K. Barbhuiya***Department of Civil Engineering, National Institute of Technology Silchar, Assam, India.***Corresponding Author: mdashikur_ug_22@civil.nits.ac.in***Abstract**

Urgent rain forecasting will be necessary to support climate-resilient planning in Dhaka, where rapid urbanization can quickly pose a threat to infrastructure and human safety due to frequent flooding. This paper aims to develop a predictive model for future rainfall predictions using 25 years of rainfall data from three different machine learning models, sourced from the NASA POWER database. Three different models were used to test their ability to depict the complex temporal variability of rainfall in the area: these include Support Vector Machine (SVM), Random Forest (RF), and Artificial Neural Network (ANN). The data was split into a training set (70%) and a testing set (30%) to ensure the evaluation and reduce overfitting. All these models were trained on critical meteorological parameters, such as past precipitation and other climatically related dynamics known to affect rainfall dynamics. The comparative analysis of the models reveals that the Random Forest model exhibits the most predictive reliability among the others, being more predictive and stable in generalization when applied to previously unknown data. The ANN model achieves a fair performance, while the SVM exhibits relatively lower adaptability to the nonlinear nature of the rainfall data. Besides model evaluation, the model was used to create spatial prediction maps that depicted the projected rainfall distribution in Dhaka. These maps indicate new trends that can serve as early warnings of potential zones of strong-intensity rainfall, helping to evaluate flood threats and manage city water resources. The general results suggest that machine learning, specifically the Random Forest approach, is a promising channel for improving rainfall prediction quality in monsoon-controlled regions. The research paper has provided a framework that can serve the purpose of future climate analysis, urban planning, and decision-making in other data-limited areas with ease.

Keywords: *Rainfall forecasting, Machine learning, Random Forest (RF), Support Vector Machine (SVM), Artificial Neural Network (ANN), NASA POWER database, Dhaka, Climate resilience*

Track 3: Computational Intelligence and Applications**Paper ID: 173****Presentation Mode: Oral****Beyond the Black Box: An Interpretable Machine Learning Framework for Quantifying Honey Adulteration with UV-Vis-NIR Spectroscopy****Md. Abdullah Al Noman*, Anannya Barua Nijhum, Iqbal Hossain, Mohammad Gulzarul Aziz, Afzal Rahman***Department of Food Engineering and Technology, Faculty of Agricultural Engineering and Technology, Bangladesh Agricultural University, Mymensingh 2202, Bangladesh.***Corresponding Author: noman.1707002@bau.edu.bd***Abstract**

Honey adulteration is a persistent global issue demanding rapid, accurate, and transparent analytical solutions. This study demonstrates a framework for quantifying honey adulteration using UV-Vis-NIR spectroscopy (200–900 nm) coupled with interpretable machine learning. Four types of honey (Wild, Blackseed, Mustard, Rubber) were adulterated with Corn Syrup, Glucose Syrup and Caramel Color at varying concentrations (1%, 10%, 20%, 30%). Five regression models (LightGBM, CatBoost, XGBoost, Random Forest, K-Nearest Neighbors) were evaluated on a reduced feature set from Random Forest (RF), Recursive Feature Elimination with Cross-Validation (RFECV), and Shapley Additive Explanations (SHAP). LightGBM model demonstrated the highest predictive accuracy, achieving $R^2 = 0.938$ and $RMSE = 0.0258$. This optimized model significantly outperformed other tested ensemble methods (CatBoost $R^2 = 0.934$, XGBoost $R^2 = 0.923$, RF $R^2 = 0.903$, KNN $R^2 = 0.875$). Eight chemically significant wavelengths (243, 373, 380, 382, 394, 576, 578, and 899 nm) were identified as critical predictors. SHAP analysis was employed to provide a transparent interpretation of the final model. This analysis elucidates the specific impact of each wavelength on the model's predictions, confirming their correlation with known chemical markers. Combining advanced ensemble learning with explainable AI, this integrated approach offers a robust, rapid, and scientifically grounded methodology for honey authentication, suitable for high-throughput quality control.

Keywords: *Honey Adulteration, UV-Vis-NIR Spectroscopy, Ensemble Learning, Explainable AI, SHAP*

Track 3: Computational Intelligence and Applications**Paper ID: 02****Presentation Mode: Poster****A Blockchain-Enabled Framework for Secure and Transparent Smartphone Ownership Tracking****Fahim Haider, Sadman Sakib, Mirza Sadman Mehrab, Mahmud Iftekhhar Asef, Md Manirul Islam, Samiur Rahman Khan****Abstract**

The global surge in smartphone adoption expected to surpass 7.7 billion active devices by 2025—has amplified issues of theft, identity misuse, and untraceable resale. Centralized systems for International Mobile Equipment Identity (IMEI) management have proved vulnerable to tampering, duplication, and jurisdictional inconsistency. This paper presents a blockchain-enabled framework that provides a decentralized, tamper-proof mechanism for smartphone ownership registration and transfer. Each device's IMEI is cryptographically bound to verified digital identities, ensuring authenticity and traceability across borders. Smart contracts automate ownership verification, transfer, and theft reporting, while permissioned blockchain architecture enforces regulatory compliance and privacy. A prototype implemented on Ethereum and Hyperledger Fabric demonstrated superior performance, achieving subsecond transaction latency and high throughput with minimal operational cost. The framework addresses challenges in privacy preservation, interoperability, and governance, contributing a scalable model for national and international deployment. Results indicate that the system can reduce fraudulent ownership claims by up to 92% and streamline blacklisting operations by integrating with CEIR and GSMA IMEIDB..

Keywords: *Blockchain, Smartphone Ownership Tracking, Smart Contracts, IMEI Registration, Decentralized Identity.*

Track 3: Computational Intelligence and Applications**Paper ID: 03****Presentation Mode: Poster****Integrating Fog Computing and Content Caching: A Hybrid Framework for Intelligent Vehicular Networks****Md Didarul Alam Alif, Imtiaz Ahmed, Fahad Bin Ahmed, Md Manirul Islam, Samiur Rahman Khan, Umme Salsabil***American International University-Bangladesh***Corresponding Author: manirul@aiub.edu***Abstract**

This research paper discusses the hybrid approach towards modernizing efficient Vehicular Networks with a focus on content caching techniques integrated at the fog computing layer. The main goal is to enhance the vehicular network system's performance by resolving latency, bandwidth, and general efficiency-related problems. Through the use of simulated elements including vehicles, On-Board Units (OBUs), fog nodes, content caching servers, and cloud computing servers, the research methodically assesses the current vehicular network system design. To capture and preprocess data on system configuration, data generation, fog computing, communication to the cloud, cloud computing, data delivery, and efficient retrieval, data collection uses OBUs, content caching servers, and fog computing nodes. Dependent factors include latency, bandwidth utilization, cache hit rates, and system efficiency, whereas independent variables include vehicle density, communication range, fog node dispersion, and cache size. When analyzing data, statistical approaches are used, giving ethical issues for data protection and privacy first priority. Using repeated runs and well-established modeling tools, the study improves validity while acknowledging its limits. The architecture of the hybrid system includes direct car-to-cloud connectivity, vehicle-to-fog node communication, and a hybrid combination of the two. The framework and visual design show how data moves from OBU to cloud computing via fog computing layers. The literature study highlights Vehicular Networks in the perspective of larger transportation technologies, highlighting the difficulties, developments, and significance of fog computing, routing protocols, and big data solutions. The study's conclusion highlights its shortcomings and suggests future lines of inquiry for fog computing for Vehicular Networks. This provides some insight into how vehicular network technology is evolving for intelligent transportation systems that are safer, more efficient, and data driven.

Keywords: *Vehicular Networks, fog computing, content caching, intelligent transportation systems, hybrid communication, latency, bandwidth, simulation, data privacy, Big Data solutions.*

Track 3: Computational Intelligence and Applications**Paper ID: 17****Presentation Mode: Poster****Urban Tree cover detection UAV, Remote Sensing and Deep Learning****Md Sabbir Islam* and Dr. Md. Rahedul Islam***Department of geography and environment, Pabna University of Science and Technology, Pabna-6000, Bangladesh.***Corresponding Author: sabbir.210908@s.pust.ac.bd***Abstract**

Trees are the essential elements of the environment to conserve ecology, air quality, climate, biodiversity and sustainability. Correct estimation of trees is very essential for managing sustainable development, protection of biodiversity and monitoring climate. Traditional field based initiatives are costly, time consuming and restricted to local region. So main objective of this study is to identify urban tree coverage area with drone-satellite hybrid model and assessing green space. Our study gives a workflow to detect tree with high resolution UAV image integrated with satellite (Sentinel-1, Sentinel-2) data with Mask R-CNN and Python based model. We took drone image less than 1 cm spatial resolution to train model on R-CNN and Python based model Detectron2 which will be interpolated to Sentinel 10 m image. It detects of per pixel by using Sentinel-1 (SAR) and Sentinel-2 (MSI). This will help to scale up monitoring trees, urban forest, environment planning and sustainable development. Result shows tree coverage on Python based model. It will offer promising application for urban environment and climate adaption in Bangladesh.

Keywords: *UAV, Remote Sensing, Deep Learning, Urban Forest, Tree Detection*

Track 3: Computational Intelligence and Applications**Paper ID: 28****Presentation Mode: Poster****A Machine Learning-Driven IoT-Enabled Activity Monitoring System for Chronic Disease Prediction and Personalized Diet Planning****Md Yasir Arafat¹, Md Tarikul Islam¹, and Tanvir Ahmad²**¹Department of Computer Science and Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh² Department of Nutrition and Food Technology, Jashore University of Science and Technology, Jashore-7408, Bangladesh**Abstract**

The healthcare sector has undergone significant transformation with the advent of the *Internet of Things (IoT)*, which enables the collection of large-scale patient data and facilitates novel approaches to disease prediction and prevention. Predictive analytics increasingly leverage such data to develop models capable of forecasting chronic diseases. Contemporary wearable devices, integrated with intelligent internet-based systems, continuously monitor key health indicators, including heart rate, blood pressure, blood glucose, and oxygen saturation (SpO₂). In this study, we propose a *Machine Learning-Driven IoT-Enabled Activity Monitoring System* for predicting long-term conditions, such as cardiovascular diseases and diabetes, using comprehensive patient datasets. The system employs a rigorously designed machine learning framework capable of multi-class classification, enabling early disease detection, timely risk notification, and personalized diet recommendations based on user health findings, activity and calorie expenditure. The framework continuously adapts predictive models in real-time using data from multiple sources, providing a scalable and cost-effective solution for ongoing health monitoring and recommending healthy diet plan. By integrating IoT-enabled smart wearables and machine learning analytics, the system enhances chronic disease management, improves quality of life, and delivers personalized dietary planning. This forward-looking approach demonstrates the potential to transform healthcare delivery in an increasingly connected, data-driven environment.

Keywords: *Internet of Things (IoT), Healthcare analytics, Predictive modeling, Machine learning, Chronic disease management, Cardiovascular disease prediction, Diabetes detection*

Track 3: Computational Intelligence and Applications**Paper ID: 97****Presentation Mode: Poster****Geo-Spatial and Machine Learning-Based Assessment of Urban Groundwater Sustainability in The Rangpur Region, Bangladesh.****Md. Abir Hasan^{1*}, Md. Mosharaf Hosen², Md. Nayem Hossain³, Md. Abir Hossain⁴**¹Assistant Professor, ²³Department of Civil Engineering,⁴Department of Geography & Environment, Pabna University of Science and Technology, Bangladesh.^{*}Corresponding Author: hasanabir.ce@pust.ac.bd**Abstract**

Ensuring the long-term sustainability of groundwater is one of the major challenges for the northern region of Bangladesh. This study represents an integrated geo-spatial analysis and machine learning-based assessment, utilizing long-term hydrological data from eight observation wells in the Rangpur region for the period 1995-2024. The main goals of this study are to examine spatio-temporal variations in water level, delimit potential groundwater recharge, and predict future risk zones under urban expansion and climate change. Different thematic layers, including land use and land cover (LULC), rainfall, digital elevation model (DEM), soil, geology, lineament density, and slope, were overlaid in a GIS environment to delineate groundwater prospect zones. Moreover, a deep learning model for the estimation of water table depths (WTD) using spatial and temporal features is also developed. The overall accuracy for the predictive rate of the model was 66.7%, suggesting high prediction capability. This also demonstrates the machine learning's ability to estimate influential environmental conditions, such as water table levels. In conclusion, the hybrid GIS and ML approach would be a useful decision-support tool for managing groundwater sustainability for the urban areas of Bangladesh.

Keywords: *Geo-Spatial Analysis, Machine Learning, Urban Areas, Groundwater Sustainability, Rangpur Region.*

Track 3: Computational Intelligence and Applications**Paper ID: 129****Presentation Mode: Poster****Modelling Urban Surface Temperature Variability in Gazipur: A Multi-Temporal Remote Sensing and Machine Learning Approach****Md.Taofikul Islama, Md.Iftekher Hasan Imona*, Fahmida Sultanaa, Md.Sofiul Islama***Urban and Rural Planning Discipline, Khulna University, Khulna 9208, Bangladesh***Corresponding Author: iftekherimon@gmail.com***Abstract**

Urbanization-induced micro-climatic shifts have emerged as critical challenges for sustainable urban development, particularly in rapidly transforming peri-urban areas of Bangladesh as it is one of the most densely populated country in the world. This study aims to model and analyse the spatio-temporal dynamics of the urban micro-climate across Gazipur district using multi-index statistical and machine learning approaches. Multi-temporal satellite datasets from 2015, 2020, and 2025 will be employed to derive key remote sensing indices, including the Normalized Difference Vegetation Index (NDVI), Normalized Difference Built-up Index (NDBI), Normalized Difference Water Index (NDWI), Atmospherically Resistant Vegetation Index (ARVI), Soil-Adjusted Vegetation Index (SAVI), Modified Normalized Difference Water Index (MNDWI), Digital Elevation Model (DEM), and Land Surface Temperature (LST). Multi Linear Regression will be utilized to quantify the relationships between LST and the derived biophysical and anthropogenic parameters, identifying the most significant environmental drivers of surface thermal variation (expected $R^2 > 0.70$). To capture the combined influence of these factors, Principal Component Analysis (PCA) will be applied (excluding LST) to develop an integrated Atmospheric Risk Index (ARI), which will subsequently be correlated with spatio-temporal LST trends to assess its explanatory power (anticipated correlation coefficient $r > 0.65$). A Random Forest (RF) regression model will then predict LST under the evolving land surface conditions, enabling high-accuracy identification of urban heat hotspot zones (expected model accuracy $> 85\%$). Change detection analyses between the three temporal datasets will illustrate progressive thermal intensification and vegetation-bare surface transformations over the decade, highlighting zones with potential surface temperature increases exceeding $3-5^{\circ}\text{C}$. The findings will provide a comprehensive understanding of urban micro-climate evolution in Gazipur, supporting local authorities in adaptive urban design, ecological zoning, and sustainable climate resilience planning.

Keywords: *Urbanization, Micro-climate, Land Surface Temperature (LST), Remote sensing, Spatio-temporal dynamics, Gazipur, Bangladesh, NDVI, NDBI, Land cover change*

Track 3: Computational Intelligence and Applications**Paper ID: 141****Presentation Mode: Poster****Ethical Challenges in AI and Gene Editing: Comparative Analysis of Privacy Risks and Regulatory Implications****Ritu Dey, Anim Chandra Debnath, Irtiza Yesmeen, Ahnaf Ahmad, Srabon D Costa, Jannat Ferdous****American International University – Bangladesh, Dhaka 1229, Bangladesh***Corresponding Author: jannatul@aiub.edu***Abstract**

Rapid advancements in artificial intelligence (AI) and gene editing have introduced complex ethical challenges and privacy concerns in healthcare and biotechnology. Evaluating their multidimensional impact is essential for effective regulation. This study examines 200 documented ethical challenges (2019–2024) in AI and gene editing to identify key risk factors, regional patterns and predictors of ethical impact. Quantitative analysis was performed using datasets from seven international sources (MIMIC-IV, Open Neuro, WHO GHO, IHME GBD, EGA, INDEPTH/Chakaria HDSS, Bangladesh Open Data). Statistical techniques included descriptive statistics, chi-square tests, ANOVA, Pearson correlation and multiple regression. Variables analyzed were impact scores, public awareness, regulatory complexity, severity, geography and stakeholder involvement. AI accounted for 37.5% of challenges (mean impact score 59.2) and gene editing 34.0% (58.9). South Asia and Africa had the highest concentration (33.5%). Severity emerged as the strongest predictor (R^2 increased from 0.019 to 0.35–0.45 with inclusion of severity and contextual variables). Public awareness averaged 5.3/10, highlighting gaps in understanding. Severity level is a critical determinant of ethical impact. Regional disparities suggest the need for tiered regulatory approaches immediate measures for AI transparency and genetic discrimination, intermediate stakeholder engagement and long term global ethical standards. Findings provide empirical guidance for policymakers addressing emerging technological ethics.

Keywords: *Artificial Intelligence, Gene Editing, Ethical Challenges, Privacy Risks, Regression Analysis.*

Track 4: Electrical, Electronics, and Communication Engineering**Paper ID: 34****Presentation Mode: Oral****WSN-Based Solar Microgrid Health Monitoring for Rural Bangladesh****Md. Estiak Ahmed, Tanjila Akter***American International University-Bangladesh (AIUB), Dhaka Bangladesh**Eden Mohila College, Dhaka, Bangladesh***Abstract**

Small solar microgrids have ensured reliable electricity supply in most rural areas of Bangladesh. However, these systems often suffer from performance issues originating from dust on panels, battery degradation, high temperatures, irregular maintenance, and the absence of real-time monitoring. When a component fails, power outages affect the whole community due to technicians arriving late, and faults remain undetected for long periods. In this paper, a WSN-based health monitoring system is proposed that continuously observes the condition of each part of a rural solar microgrid and automatically detects early signs of failure. In the new approach, several low-power wireless sensor nodes have been installed at key points of the microgrid that include solar panels, batteries, charge controller, and load distribution line. Each sensor node measures parameters such as temperature, solar irradiance, dust level, battery voltage, State of Charge (SoC), current flow, and load consumption. These nodes will communicate among themselves in a multi-hop WSN to enable the system to cover large areas in rural settings without relying on mobile networks. All data collected will be forwarded to a local gateway, which will process the information using a simple rule-based and machine-learning-assisted diagnostic algorithm. The algorithm will detect abnormal conditions relating to overheating, excess dust accumulation, low battery health, wiring faults, and/or load imbalance. If a problem is identified, an alert will be raised that can be viewed by operators via a mobile app or through a local display. Simulation and prototype testing of the proposed WSN monitoring system for microgrids can reduce unexpected shutdowns, improve battery lifetime, and help microgrid operators take corrective actions against such failures before they happen. This work represents a practical, low-cost, scalable solution to make rural solar microgrids in Bangladesh more reliable, efficient, and easier to maintain.

Keywords: *Wireless Sensor Network (WSN), Solar Microgrid, Health Monitoring, Predictive Maintenance, Rural Electrification.*

Track 4: Electrical, Electronics, and Communication Engineering**Paper ID: 46****Presentation Mode: Oral****Performance Enhancement of I-Slots Microstrip Patch Antenna Using Different Substrate Materials****Anamika Mondol****Khawaja Yunus Ali University***Corresponding Author: anuanamika387@gmail.com***Abstract**

Microstrip Antenna is a type of antenna that consists of a flat conductive patch mounted on a grounded dielectric substrate. It is widely used in advanced wireless communication systems due to its compact size, low profile, and easy fabrication. IN recent years microstrip antenna has attracted more attention. It is an important device, for communication above 1GHz.It is used in satellite communication, mobile communication and aircraft communication application. In this paper, the performance of square shape Microstrip patch Antenna is analyzed by changing different substrate materials with “I” slot over the patch and this is the proposed antenna. Here we discussed about the comparison of different substrate materials namely Lead glass, Roger (R03003), Silicon (lossy), FR-4(lossy), Quartz (Fused) (lossy) and Rubber are used to achieve better gain and directivity. Among these materials lead glass provides the better results. The enhanced performances of proposed antenna with lead glass materials are compared with the conventional Microstrip patch antenna. The enhanced directivity and gain of the proposed antenna is 8.64dBi and 7.87dBi respectively.

Keywords: *Substrate material, MPA, Gain, Directivity, Return loss, VSWR, Enhance Performance, High frequency, Wireless Application, CST (2018)*

Track 4: Electrical, Electronics, and Communication Engineering**Paper ID: 56****Presentation Mode: Oral****Adoption of Floating Solar Photovoltaics (FSPV) for Wastewater Management: Technical, Environmental, and Economic Evaluation****Bipro Mondal, Dr. Biplab Satpati****Department of Electrical Engineering, The University of Burdwan, India***Corresponding Author: biplabsatpati@yahoo.co.in***Abstract**

Floating Solar Photovoltaics (FSPV) have emerged as an innovative alternative to land-based PV systems, addressing concerns such as land scarcity, higher operating temperatures, and excessive water evaporation. By utilizing reservoirs and wastewater ponds, FSPV enhances energy yield through natural cooling while conserving water and reducing land-use pressure. Objective : This study aims to evaluate the technical performance, environmental benefits, and economic feasibility of FSPV compared to conventional land-based PV (LBPV) installations. The primary goal is to determine whether FSPV can deliver higher efficiency, sustainability, and cost-effectiveness, particularly for wastewater management applications. Methods : A 12 W FSPV system was simulated using PVsyst 7.4, modeling solar irradiance, water temperature, wind effects, cabling configuration, inverter placement, and anchoring systems. Field data such as voltage, current, ambient temperature, and plate temperature were collected. Performance was assessed through I–V curves, performance ratio (PR), capacity utilization factor (CUF), and system loss evaluations. Economic feasibility was analyzed using CERC guidelines to compute LCOE, NPV, IRR, and payback period. Sensitivity analysis examined variations in discount rate, loan interest, and electricity selling rate. Results : The FSPV system demonstrated superior performance due to evaporative cooling, achieving a PR of 86% and CUF of 20%, outperforming LBPV values of 80% and 15%, respectively. Lower module temperatures resulted in higher electrical efficiency. Environmentally, FSPV reduced evaporation losses and required no land acquisition. Economically, FSPV showed lower LCOE and faster payback when land cost was excluded. Sensitivity analysis indicated strong influence of discount rate and electricity selling rate on project viability. Overall, FSPV presents a high-potential solution for sustainable energy generation and water conservation.

Keywords: *Floating Solar Photovoltaics, Wastewater Management, Performance Ratio, Economic Analysis, Renewable Energy.*

Track 4: Electrical, Electronics, and Communication Engineering**Paper ID: 64****Presentation Mode: Oral****Unlocking the Potential of Agricultural Wastes for the Production of Next-Generation Capacitors: Rice Husk, Corncob, and Coconut Shell.****Naimur Rahman*, Md. Shakil Ahamed, Md. Abdul Kader, Khondoker Mushfikur Rahman, Md. Jahidul Haque***Department of Ceramic & Metallurgical Engineering, Rajshahi University of Engineering & Technology (RUET), Rajshahi-6204, Bangladesh***Corresponding Author: 2106036@student.ruet.ac.bd***Abstract**

Agricultural waste resources such as rice husk, corn cob, and coconut shell are readily available but not well explored materials that could be converted into high-surface-area activated carbon for supercapacitors. This conversion is important for the storage of renewable energy. The challenge of waste management could therefore be addressed through the production of storage devices using these wastes. This work therefore provides a low-cost technique for the production of activated carbon using these biomasses for the storage of energy. The biomasses were pyrolyzed using a pressure cooker that has been modified to ensure that the biomasses undergo low-oxygen conditions achieved through volatile control. The resultant biomasses were then activated using potassium hydroxide and a temperature of 700°C with a temperature rise of 3°C/min to obtain porous activated carbon. Electrodes were produced through the deposition of a material composed of 85% activated carbon and pyrolyzed biomasses (conducting additives of 10%) and 5% binder and current collectors. Supercapacitors were then assembled by separating the material with cotton and using aqueous electrolytes. Initial tests were done and revealed the capability of the work to convert raw biomasses into both biochar and activated carbon, thus the practicality of the work. The biomasses used are composed of organic materials that readily pyrolyze and could be transformed into activated carbon through the reported low-cost method. So the study may offer a cost-effective and environmentally favorable way to electrode materials for sustainable energy storage.

Keywords: *Agricultural waste, Pyrolysis, Activated carbon, Electrochemical, Energy storage.*

Track 4: Electrical, Electronics, and Communication Engineering**Paper ID: 75****Presentation Mode: Oral****Design and Development of an Automated Soil Irrigation System using Arduino Uno****Ayon Kumar Bhowmick Ovi¹, Nabib Ahamed Sujan¹, Nafiul Islam Khan Nafi¹, and Muhibul Haque Bhuyan^{2*}**¹*Department of Computer Science and Engineering, American International University-Bangladesh (AIUB)*²*Department of Electrical and Electronic Engineering, American International University-Bangladesh (AIUB)***Corresponding Author: muhibulhb@aiub.edu***Abstract**

Effective water resource management is a critical challenge in current agricultural sector, especially in water-starved areas and growing demand for food production. Currently, manual irrigation system causes huge amount of water wastage, which yields less amount of crop production and higher manpower requirements. This research work aims to design and development of an automated soil irrigation system to optimize water usage through real-time monitoring of soil moisture and surrounding temperature and pumping water to the cultivable land automatically. The planned scheme uses mainly a soil moisture sensor (FC-28), an Arduino Uno microcontroller, and a water pump to create a responsive irrigation mechanism. When soil is dry, that is, its moisture level falls below a certain threshold value and the ambient temperature rises above a certain threshold value, it is detected by the analog pins, A0 and A1 of the controller. The analog signal is converted by the ADC module of the microcontroller into a digital signal for further processing. If the detected signal is within the threshold level, then the system activates the water pump through a relay module to irrigate the soil. The sensors continuously check the moisture content and temperature, and sends the signal to the controller. The pump is stopped by the Arduino by sending a LOW signal to the relay module once the optimal moisture is restored. Simulation and experimental testing of the system at various environmental settings ensure the efficacy of the proposed system. It demonstrates significant water savings compared to manual irrigation. The cost of the designed system is only US\$15.26 (US Dollars fifteen point two six only). This closed-loop feedback system ensures crops receive adequate water based on actual needs. Thus, the system minimizes the energy consumption, reduces human involvement, and minimizes the water waste. The automated soil irrigation system offers a scalable, low-cost, and eco-friendly solution that can benefit small-scale farmers as well as large agricultural operations, making it a valuable innovation for smart farming and precision agriculture. In future, the system may be enhanced with wireless communication modules or solar power for remote or off-grid deployment.

Keywords: *Microcontroller-Based Automated Irrigation System, Soil Sensor, Arduino Uno, Smart Farming, Sustainable Agriculture, Irrigation, Energy-Efficiency, Water Conservation.*

Track 4: Electrical, Electronics, and Communication Engineering**Paper ID: 76****Presentation Mode: Oral****Techno Economic Analysis of a Solar Powered High Density Fish Farm in Bangladesh****Monjur Morshed, Nazmul Hasan*, Mehedi Hassan***Dept. of Electrical and Electronic Engineering, Jashore University of Science and Technology, Jashore, Bangladesh***Corresponding Author: nm.hasan@aol.com, na.hasan@just.edu.bd***Abstract**

Aquaculture is main source of protein for Bangladeshis and in many parts of south east Asia. Due to competitive nature of the market, and scarce water resources fish farming is carried out at increasingly higher density. In order to reduce fish mortality and ensure proper growth of the fish population, ample supply of oxygen is mandatory. Therefore, electricity driven instruments such as aerator and pump are used extensively in modern day fish farms. In addition, smart agriculture practices are leading to usage of more electricity driven internet of things (IoT) devices in fish farms. This paper proposes a solar power plant at the fish farm in order to reduce cost of electricity and increase profit. A detailed techno-economic analysis is carried out using PVSyst software in order to design a practical solar power plant and to evaluate economic benefits of the plant installation. A 6 kWp solar photovoltaic module is designed for an aeration system of 3kWp load. Net metering is found to be crucial in order to maximize profit and increase interest in farmers. When customer can sell excess electricity to the grid, payback period is only 4.0 years. On the other hand, only self-consumption savings results in a payback period of around 7.0 years. Considering the relatively long lifetime of a solar panel of around 20 years, and the positive cash inflow this investment is meaningful to the farmers. In fact, Return on Investment is 187.4 % and 400% for the two cases respectively. This research definitely points to the need of Government encouragement in order to increase green agriculture initiatives.

Keywords: *Aquaculture, Solar photovoltaic (PV) system, Techno-economic analysis, PVSyst, Net metering, Renewable energy, Sustainable agriculture, IoT-enabled fish farming, Return on Investment (ROI), Payback period*

Track 4: Electrical, Electronics, and Communication Engineering

Paper ID: 79

Presentation Mode: Oral

Performance Enhancement of p-n CsGeI₃-Based Solar Cell Through Device Architecture Optimization Using SCAPS

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Abstract

In recent years, there has been remarkable progress in the inorganic and organic perovskite solar cell performance. The inorganic perovskite solar cells are stable but suffer from lower power conversion efficiency compared to organic perovskite solar cells. To overcome this problem, we have proposed an inorganic germanium-based perovskite solar cell (FTO/Cd_{0.5}Zn_{0.5}S/IDL1/p-CsGeI₃/n-CsGeI₃/IDL2/CuI/Au), which enhances the performance rapidly. The SCAPS-1D simulation software has been used to optimize the performance, which works based on the continuity and Poisson equations. The proposed solar cell uses an n-type and p-type CsGeI₃ absorber layer, which has excellent electron properties and ionization characteristics. The band structure of Cd_{0.5}Zn_{0.5}S and CuI perfectly matches with the CsGeI₃ absorber layer, which is why Cd_{0.5}Zn_{0.5}S is used as an electron transport layer, and CuI is also used as a hole transport layer. Further, the IDL1 and IDL2 have asymmetry, and the IDL1 plays a crucial role in influencing the performance of the solar cell. In this work, we have optimized a set of parameters, including the defect density, thickness, bandgap, acceptor, and donor density of the absorber layer. The proposed structure is an optimized result of 28.12% power conversion efficiency, an open circuit voltage (VOC) of 1.09 V, a short circuit current (JSC) of 28.92 mA/cm², and a Fill Factor (FF) of 88.49% is obtained through the SCAPS simulation process and Compared to the previously reported work, the performance of our devices has improved significantly. These results are encouraging for the theoretical support for the development of novel all-inorganic germanium-based perovskite solar cells.

Keywords: Perovskite solar cells, Germanium-based perovskites, Power conversion efficiency, Electron transport layer (ETL), Hole transport layer (HTL), Band structure alignment, Defect density optimization, Photovoltaic performance.

Track 4: Electrical, Electronics, and Communication Engineering**Paper ID: 83****Presentation Mode: Oral****Next-Generation Electrical Systems for Efficient, Low-Emission Waste-to-Energy Conversion****Md. Ridwan Al Mustavy¹, Marowa Jahan¹, Md. Mobashir Tajuare Partho¹, Md. Helal Uddin¹, Somiya Afroz¹, Dr. Muhibul Haque Bhuyan²**¹ *Department of Computer Science & Engineering, Faculty of Science and Technology, American International University-Bangladesh, Dhaka 1229, Bangladesh*² *Department of Electrical & Electronic Engineering, Faculty of Engineering, American International University-Bangladesh, Dhaka 1229, Bangladesh***Abstract**

The constantly growing pace of urbanization and industrialization has considerably raised the levels of difficulty in efficient waste management, making the need for sustainable energy sources an essential need. The development and emergence of waste-to-energy technology has opened a fresh horizon in the field, making it possible to utilize municipal and industrial wastes in the production of electricity. Despite the inevitable advantages that come with the implementation and application of waste-to-energy technology, there has been a remarkable decrease in the electrical efficiencies and greenhouse gas and particulate matter emissions in the conventional systems. The paper proposes the development and application of advanced electrical systems in the field, ensuring efficient and environmentally safe waste-to-energy conversion. By the application and integration of advanced power electronics, optimal energy management systems, and advanced energy recovery technologies, the proposed systems shall be optimized to provide the highest possible electrical efficiencies and major decreases in the greenhouse gas and particulate matter emissions. The simulation and experimental investigations conducted in the paper have resulted in the revelation of remarkable improvements in the performance, stability, and levels of sustainability compared to the conventional waste-to-energy systems. Furthermore, the application and integration of real-time monitoring and control systems in the proposed systems shall ensure remarkable adaptability and responsiveness in the face of different compositions and demands. Hence, confirms that the application and implementation of the proposed advanced electrical systems in the field have the ability to make the waste-to-energy industry environmentally and technically stable power production units, hence supporting and promoting the concept and strategies related to the development and implementation of the circular economy.

Keywords: *Waste-to-Energy (WtE), Advanced Electrical Systems, Electrical Efficiencies, Power Electronics, Low-Emission Energy Conversion*

Track 4: Electrical, Electronics, and Communication Engineering**Paper ID: 174****Presentation Mode: Oral****Household Energy-Saving Behaviors And Effect On Campus Sustainability in AIUB****Md. Jahidun Muntaka Farabi ,Jamilatun Moshfirat Anika, Ayon Roy, Mutahhara Takia Jaman, Dr. Mohammad Saef Ullah Miah****Department of Computer Science, American International University – Bangladesh, Dhaka 1229, Bangladesh***Corresponding Author: saef@aiub.edu***Abstract**

This study investigates the relationship between household energy saving behaviors and support for campus sustainability initiatives among students of the American International University- Bangladesh. A total of 350 students participated in a structured survey, and the data were analyzed using SPSS through descriptive, correlation, and regression methods. The findings show that the average score for household energy saving behaviors was 3.12, while the average score for campus sustainability initiatives was 3.39, indicating that students are positive toward sustainability both at home and on campus. Correlation analysis revealed that the relationship between the two variables was statistically significant, suggesting that students who save energy at home are more likely to support sustainability initiatives at the university. Regression analysis further confirmed that household behaviors were a significant predictor of campus engagement, and the model explained about 8% of the variation. These findings highlight the importance of linking personal practices with institutional efforts, suggesting that universities can promote sustainability by encouraging household habits through awareness, incentives, and supportive policies.

Keywords: *Energy saving behaviors, Campus sustainability, Household practices, Student engagement, Behavioral correlation*

Track 4: Electrical, Electronics, and Communication Engineering**Paper ID: 178****Presentation Mode: Poster****Next-Generation Energy Dispatch Optimization Using an Automated Priority-Driven Control System for Bangladesh****Jahidul Islam Shanto***, Md. Ishtiaque Arefin, Md. Tazidul Islam, Md. Foysal Hossen Shohag, Md. Nayeem Uddin, Md. Amzad Hossain*Jashore University of Science & Technology (JUST), Jashore-7408, Bangladesh***Corresponding Author: ji.shanto.eee@gmail.com***Abstract**

Renewable Energy Sources (RES) transform free natural inputs of wind or sun into electricity without consuming fuel, but they are not steady. Sunlight, temperature and fuel/air conditions fluctuate constantly which causes the power of renewable sources to increase or decrease. This variability makes it difficult to maintain a smooth supply of load and also to protect the battery against undue strain. This project introduces a simple and efficient smart hybrid inverter system incorporating a PV module, battery and utility grid. A supervisory controller with forecasted PV output observes the DC-bus state which is an upper bound of renewable robustness and also monitors the battery charge level. On that basis, if the PV output power is less than the load power demand, the deficiency report will be sent to the grid a day before. As a result, the grid will be prepared for the power supply for next day and thus power generation shortage or load shedding can be reduced to minimal level. On the other hand, if the PV output is greater than our load demand, inverter will automatically prioritize renewable power for the load, store surplus energy in the battery and convert it into stored energy to cover brief shortages. It will reduce grid dependency by relying on the grid only during extended low-renewable periods. The setup maintains continuous power supply with variation, keeps the DC bus voltage constant, allows less energy to be drawn from grid power. .

Keywords: *Renewable Energy, Hybrid inverter, Photovoltaic, Battery management, PV power forecast*

Track 5: Environmental Science and Climate Change**Paper ID: 1****Presentation Mode: Oral****Spatiotemporal Monitoring of Shoreline Dynamics along the Bay of Bengal Coast of Bangladesh Using GIS and Remote Sensing****Md. Sharifuzzaman^{1*}, Md Rahedul Islam¹, Md. Alamgir Hossain²**¹ Department of Geography and Environment, Pabna University of Science and Technology (PUST), Pabna, Bangladesh² Environmental Oceanography and Climate Division, Bangladesh Oceanographic Research Institute, Cox's Bazar, Bangladesh

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Abstract

Bangladesh has a long coastline along with the Bay of Bengal. Every year, the shoreline is altered. For a number of reasons, including the country's coastline and coastal areas' susceptibility to various natural and man-made factors, accurate coastal shoreline detection and monitoring are crucial. This study's primary objectives were to extract the coastal region from satellite imagery, investigate changes in the type and nature of Land Use and Land Cover, and estimate the rate of shoreline change at 10-year intervals between 1980 and 2023. This study included both primary and secondary data from a variety of sources, including satellite imagery from the USGS, GPS locations, drone images, and data from several government agencies. After manually digitizing the shorelines, a transect-based study was executed using ArcGIS's Digital Shoreline Analysis System (DSAS) extension tools version 5.1. For the whole Bangladeshi coast, 14016 transects spaced with 50 meters have been used. Bangladesh's whole coastline was split into three zones in order to illustrate the rates of shoreline change and the net shoreline movement from the transects. The central zone had the greatest shoreline shift, with the highest accretion of 139.05 m/year and the maximum erosion of -533.88 m/year. At an average rate of 10.51 meters per year, the western zone has been steadily eroding over time, whereas the eastern zone is undergoing alterations in some specific areas. Net Shoreline Movement (NSM) was used to assess the net area of erosion and accretion, and the results indicate that the western zone has lost the most land—roughly 68 km²—while the eastern zone has gained 81 km². Additionally, the pattern of changes in land use and land cover along Bangladesh's coastline over time has been examined. This study will assist policymakers and vulnerable groups in managing Bangladesh's shoreline for various coastal problems.

Keywords: DSAS, Net Shoreline Movement, End Point Rate, Linear Regression Rate, LULC.

Track 5: Environmental Science and Climate Change**Paper ID: 22****Presentation Mode: Oral****Investigation of heavy metal levels in soil Hazaribagh tannery area near Dhaka-Narayanganj-Demra dam, Dhaka, Bangladesh****Md. Abul Hashem*, Jannatul Nime Tabassum, S. M. Shakil Islam, Mohammad Jony***Khulna University of Engineering & Technology (KUET), Faculty of Mechanical Engineering, Department of Leather Engineering, Khulna-9203, Bangladesh***Corresponding Author: hashem_518@yahoo.com***Abstract**

Globally, heavy metal pollution has become a major concern. In a tannery, the raw animal skin is converted into usable leather and leather products for our daily use. During leather processing, various tanning and retanning agents, dyestuffs, and synthetics chemicals are employed to obtain different effects on the final leather. A large amount of wastewater is generated through leather processing, and this wastewater contains a significant amount of heavy metals. These heavy metals contaminated the nearby soil or sediment and finally entered the human food chain. In this study, an investigation was conducted to assess the heavy metal levels in the soil of the western part of Hazaribagh near the Dhaka-Narayanganj-Demra (DND) dam, Dhaka, Bangladesh. The collected soil samples were sun-dried, oven-dried, ground, and homogenously mixed. The homogenously mixed soil samples were acid-digested following the standard methods. The heavy metals chromium (Cr), arsenic (As), lead (Pb), and cadmium (Cd) levels of the acid-digested aliquot were quantified through atomic absorption analysis with Atomic Absorption Spectroscopy (AAS). The levels of metals Cr, As, and Pb were found to be 556.5 ± 0.5 mg/kg, 1.5 ± 0.04 mg/kg, and 5.8 ± 0.6 mg/kg, respectively. The Cd was below the detection limit (BDL). The levels of As and Pb in the soil samples were below the recommended limits, but the Cr level was 11 times higher than the recommended limit. It seems that the surface soils in the investigated area are polluted by anthropogenic activities. This study delivers a scientific basis for taking proper measures and managing soil pollution in the areas of Hazaribagh tannery near the Buriganga River.

Keywords: *Pollution, AAS, Heavy metals, Standard limits.*

Track 5: Environmental Science and Climate Change**Paper ID: 61****Presentation Mode: Oral****Hydrochemical Assessment and Water Quality Index Evaluation of Industrial Canals in the National Special Economic Zone, NSEZ****Partha Pratim Nath^{1*}, Md Ali Akbar¹, Md Nizam Uddin², Barkat Ullah Bhuiyan²**¹*Department of Civil Engineering, Feni University, Feni.*²*Department of Chemistry, Shahjalal University of Science and Technology, Sylhet.***Corresponding Author: partha@feniuniversity.ac.bd***Abstract**

The National Special Economic Zone (NSEZ), the largest emerging economic zone in Bangladesh, is situated at the estuarine confluence of the Feni River and the Bay of Bengal and currently hosts approximately eleven operational industries. Bamonshundor Khal and Shusham Khal are the two principal canals traversing the zone, accommodating the majority of current industrial establishments at their banks and ultimately discharging into the Bay of Bengal. This study evaluates the water quality at selected upstream and downstream locations of both canals. A comprehensive study of hydrochemical parameters including turbidity, electrical conductivity (EC), pH, total solids (TS), total dissolved solids (TDS), total suspended solids (TSS), salinity, hardness, calcium, magnesium, total iron, and chemical oxygen demand (COD) was quantified using standard analytical instruments and titrimetric methods. Water Quality Index (WQI) values for each location based on these parameters were computed using the weight rating method to assess overall water quality status. The results reveal a marked deterioration in water quality along the downstream gradients of both canals. In Bamonshundor Khal, TDS increased by 64.18% from upstream to downstream, accompanied by an elevation in hardness from 61.17 to 105.09 mg/L and COD from 300 to 320 mg/L. In Shusham Khal, downstream water at the main sluice gate exhibited TDS of 792.62 mg/L, hardness of 210.19 mg/L, and COD of 450 mg/L, all exceeding established pollution guideline limits. WQI assessments classified all sampling points as “poor” to “highly polluted,” with the highest pollution loads detected at the terminal discharge points (WQI: 5187 for main sluiceway of Bamonshundor Khal and 6525 for main sluiceway of Shusham Khal). The study highlights the urgent need for integrated environmental monitoring, pollution mitigation, and regulatory enforcement to safeguard aquatic ecosystems in this rapidly industrializing region.

Keywords: *Hydrochemical Assessment, WQI, NSEZ, TDS, COD*

Track 5: Environmental Science and Climate Change**Paper ID: 182****Presentation Mode: Poster****Assessing The Dominance of Survival-Based Adaptation Strategies Among Urban Climate Migrants in Khulna City, Bangladesh****Khondoker Mahmud Parvez^{1*}, Md. Shahjahan Ali²**¹ Ph.D. student, Institute of Disaster Management, KUET, Bangladesh, e-mail: parudu@gmail.com² Professor, Department of Civil Engineering, KUET, Bangladesh, e-mail: msali@ce.kuet.ac.bd**Abstract**

This study analyzes whether the adaptation strategies adopted by climate migrants in Khulna City reflect long-term resilience or short-term survival behavior. Primary data were collected from 400 climate migrants using a simple random sampling method through structured household surveys. The analysis focuses on livelihood choices, housing arrangements, borrowing practices, skill development, and institutional support. Results show that informal employment dominates migrant livelihoods, with 28% engaged in rickshaw or van driving and 26% working in construction. Housing conditions remain highly insecure, as 52% of respondents live in shared accommodation and only 16% report self-arranged housing. Financial vulnerability is evident, with 45% of migrants depending heavily on community-based borrowing and 38% borrowing occasionally. Access to long-term adaptive mechanisms is limited, as only 26% of respondents have received skill training, while 36% are still planning to develop skills. Health-related adaptation remains weak, with 53% reporting no access to treatment facilities and relying on self-treatment practices. Factor analysis identifies three major adaptation dimensions: economic survival strategies, social-support arrangements, and skill-based activities. However, economic coping mechanisms dominate, indicating that most migrants remain focused on meeting immediate needs rather than building resilience. The findings suggest that urban adaptation among climate migrants is largely reactive and constrained by unstable income, insecure housing, and limited institutional engagement. Strengthening livelihood diversification, skill training, and inclusive urban support systems is essential to move migrants beyond survival-based adaptation.

Keywords: *Climate migration, survival strategies, urban adaptation, resilience assessment, Khulna City.*

Track 5: Environmental Science and Climate Change**Paper ID: 6****Presentation Mode: Poster****Assessing Groundwater Dynamics and Its Effective Use in Agriculture for Climate-Resilient Groundwater Management****Mahmuda Siddika^{*}, Rial Ahmed, Sakiba Akter Labonno, Sumaiya Khatun, Marjana Yesmen Toma, Khalid Mahmud, Deen Islam, M. G. Mostofa Amin***Department of Irrigation and Water Management, Bangladesh Agricultural University, Mymensingh-2202, Bangladesh**^{*}Corresponding author: m.siddika.61@gmail.com***Abstract**

Groundwater is a cornerstone of agricultural production and water security in Bangladesh; however, accelerated depletion and poor management undermine its long-term resilience in the face of climate change. This study aimed to develop a climate-resilient groundwater management strategy through a systematic assessment of aquifer dynamics and evaluation of groundwater contributions to crop water demands. Field monitoring, aquifer testing, and process-based modeling were applied. Nine observation wells recorded temporal groundwater levels. Transmissivity and storativity of the pumping aquifer were estimated from a time–drawdown pumping test interpreted using the Cooper–Jacob method. The UPFLOW model quantified upward capillary fluxes from shallow groundwater to the root zone and estimated groundwater contributions to crop evapotranspiration for representative crops. Three stratigraphic aquifers were identified: a shallow unconfined aquifer (water level of 0.1–2 m), a middle-confined aquifer (5–7 m), and a deep confined aquifer (9–15 m). The shallow aquifer responded rapidly to rainfall and irrigation, showing strong surface connectivity. The deep confined aquifer (transmissivity 1,791 m²/day; storativity 0.00035) responded steadily to rainfall but declined markedly at the start of the irrigation season, indicating seasonal pumping pressure. The middle aquifer remained mostly unresponsive due to limited recharge, low permeability and the absence of pumping. UPFLOW simulations indicated that upward capillary fluxes from shallow groundwater substantially supported maize, potato, and carrot for early November sowing dates, but for December sowings, groundwater contribution showed a 1.8 mm/day deficit, requiring one supplemental irrigation when the water table was <1.0–1.5 m. This also showed higher capillary rise in loam > silt loam > clay loam soil. Results highlight the need for integrated monitoring and crop-specific evaluation of groundwater use. Aligning irrigation with aquifer sustainability and optimizing shallow groundwater can boost water productivity and ensure long-term food and water security in Bangladesh.

Keywords: *Aquifer, Groundwater level, Aquifer properties, Evapotranspiration, Capillary fluxes*

Track 5: Environmental Science and Climate Change**Paper ID: 21****Presentation Mode: Poster****Estimating Soil Evaporation and Basal Crop Coefficients for Mulched vs. Non-mulched Maize: Experimental Measurements and Simulations****Md. Tasif Rayhan Sarker*, Farjana Afrin, M. G. Mostofa***Department of Irrigation and Water Management, Bangladesh Agricultural University, Mymensingh-2202, Bangladesh***Corresponding author: tasrifrayhan516@gmail.com***Abstract**

In Bangladesh, dry-season maize has emerged as a vital cereal crop, important for food security and achieving a record production of 4.87 million tonnes in 2024. With its cultivation largely dependent on rainfall, developing strategies for efficient water use is a key agricultural priority. This research adopted a process-based simulation approach, the SIMDualKc model, to gain a better understanding of mulching practice influencing soil water dynamics. The model is based on the well-established FAO-56 dual crop coefficient approach, was applied to assess soil water fluxes and determine stage-specific crop coefficients for rainfed maize. Crop coefficient is separated into two coefficients attributed to its fundamental components of plant transpiration and soil evaporation. The model was calibrated against the lysimeter measurements, including soil moisture content and evapotranspiration rates, collected throughout the 2023 growing season. The results showed that full plastic mulching significantly changed the field water balance, reducing soil evaporation from 58.3 mm to 22.5 mm. The total stage-wise evapotranspiration was lowered by up to 47% compared to the conventional bare soil practice. This water-conserving effect was reflected by lower soil evaporation coefficient (K_e) during the initial growth stages. The calibrated basal crop coefficients (K_{cb}) were 0.15 for the full mulch treatment compared to 0.30 for the no-mulch control. The soil water stress coefficient (K_s) was far more favorable under mulch, ranging from 0.9 to 1.0, versus a wider and often stressful range of 0.46 to 1.0 under no mulch. The stage-wise crop coefficients (K_c) derived from the model aligned well with lysimetric observations, confirming the model's reliability. SIMDualKc model proved to be a reliable tool for determining the dynamic soil-water balance and provides optimizing guideline for irrigation scheduling and improving water management strategies in rainfed maize production systems

Keywords: *SIMDualKc, dual crop coefficient, evaporation coefficient, basal crop coefficient, plastic mulching*

Track 5: Environmental Science and Climate Change**Paper ID: 24****Presentation Mode: Poster****Assessment of Noise Intensity and Its Relationship with Land Use at Major Intersections in Khulna City.****Mafrid Haydar*, Sumiya Tasnim, Akhi Hossain Anna, Tahsin Rahman Nijhum, Showmitra Kumar Sarkar***Department of Urban and Regional Planning, Khulna University of Engineering & Technology (KUET), Khulna-9203, Bangladesh.***Corresponding author: mafrid@urp.kuet.ac.bd***Abstract**

The issue of noise pollution is now being considered as a serious environmental health issue in the fast-developing cities of the world. The Khulna metropolitan, is one of the fastest developing urban centers in Bangladesh, facing the increasing traffic congestion that has increased exposure to noise in the congested urban areas. This study was conducted in the Khulna metropolitan city to determine the status of noise level at major traffic intersections and assess the possible effect of noise on surrounding land use. Noise levels (L_{eq} , L_{max} , L_{min}) were recorded during different time shifts at purposively selected intersections using a Sound Level Meter. The sampling points were mapped in ArcGIS with IDW used to interpolate noise intensity and buffers applied to identify surrounding land use. The relation of sound level with land use was identified through various statistical tests like ANOVA, Tukey's HSD, Correlation and Chi Square test. The result evidently specified that noise level in all major traffic circles of Khulna metropolitan city was much higher than the recommended threshold of acceptable noise level set by the Department of Environment of Bangladesh where residential areas averaging 71.32 dB, commercial areas 79.03 dB, industrial areas 78.90 dB and mixed areas 77.10 dB. The maximum noise level reached 99.5 dB in several busy traffic and commercial locations, while the lowest minimum value of 42.1 dB was recorded in sensitive zones. It was also found that noise pollution was positively correlated with population density, traffic volume and commercial and industrial land use and negatively correlated with green space and water bodies. These findings prioritize the need for policy interventions in urban noise management and the integration of land-use planning with environmental protection strategies. This study supports the achievement of Sustainable Development Goal 11 by promoting resilient, inclusive and environmentally sound urban development

Keywords: *Equivalent Continuous Sound Pressure Level (L_{eq}), Noise Pollution Mapping, Urban Land Use, Built Environment, GIS-Based Spatial Analysis.*

Track 5: Environmental Science and Climate Change**Paper ID: 35****Presentation Mode: Poster****RUSLE Model-Based Soil Erosion Estimation for Prioritization of Mitigation Strategies: A Case Study in Gaibandha District, Bangladesh****Akila Jamal*, Sourav Das, Debasish Roy Raja***Department of Urban and Regional Planning, Chittagong University of Engineering and Technology, Chattogram-4349, Bangladesh.***Corresponding Author: u2105035@student.cuet.ac.bd***Abstract**

Frequent soil erosion, driven by variations in rainfall, soil type, topography and land cover etc. threatens agricultural productivity and environmental stability across Bangladesh's floodplain regions. The Gaibandha district lies within the floodplains of Teesta, Jamuna and Brahmaputra rivers and experiences severe erosion driven by its geomorphological and hydrological characteristics. As the local communities are heavily reliant on land resources and agriculture, this intensifies the urgent need for effective erosion control interventions. This study aims to estimate and spatially analyze soil loss using the Revised Universal Soil Loss Equation (RUSLE) model integrated with Geographic Information System (GIS) and Remote Sensing (RS) techniques to identify erosion-prone zones and guide priority areas for erosion control interventions. Specifically, the RUSLE model incorporates five key factors: rainfall erosivity (R), soil erodibility (K), slope length and steepness (LS), cover management (C) and conservation practice (P). The datasets employed here were sourced from CHRS Data Portal, FAO Soil Database, DEM and Landsat-8 imagery from USGS Earth Explorer, which subsequently processed within GIS to derive the model parameters. Each factor was spatially analyzed and integrated in GIS to compute the annual soil loss rate. According to the analysis, the estimated annual soil loss in Gaibandha varied across five severity classes, ranging from 0 to 37 ton/ha/year. Specifically, the spatial distribution analysis revealed that approximately 53.07% of the area falls under very low soil loss, 33.22% under low, 10.95% under moderate, 2.46% under high and 0.30% under very high erosion risk. The most erosion-prone zones were predominantly concentrated in Fulchhari, Sundarganj and Gaibandha Sadar upazilas where steep slopes, sandy loam soils and sparse vegetation cover contribute significantly to higher soil loss. Overall, the study provides a reliable framework for soil loss assessment and a spatial basis for prioritizing soil conservation planning for implementing erosion control strategies in vulnerable floodplains.

Keywords: *Soil erosion; RUSLE model; GIS; Remote sensing; Erosion control; Gaibandha District.*

Track 5: Environmental Science and Climate Change**Paper ID: 42****Presentation Mode: Poster****Assessing Morphological Change and Future Dynamics of Three Major Transboundary Rivers in Sylhet District****Arpita Hoom Chawdhury¹, Nowrin Rahman Khanom²**¹Bangladesh University of Engineering and Technology, Dhaka-1000, Bangladesh²Khulna University of Engineering and Technology, Khulna-9203, Bangladesh**Abstract**

Transboundary rivers in Sylhet District are ever-changing natural systems that shape landscapes, support local communities, and sustain ecological balance across borders. Their shifting courses influence regional water flow, sediment movement, and land stability, making it crucial to understand how these rivers evolve over time. This study explored the morphological changes of the Surma, Kushiya, and Sari rivers from 2003 to 2023 and projected their likely conditions in 2033, focusing on patterns of stability, erosion, and accretion under both natural forces and human impact. River channels were extracted from Landsat 5 and Landsat 8 imagery using NDWI in ArcGIS 10.8, which highlighted shifting zones, eroding banks, newly formed land, and stable areas. Erosion and accretion were tracked through changes in river extent throughout the years, while future behavior was projected using the TerrSet Land Change Modeler alongside historical land use, vegetation, built-up areas, agricultural activity, and Markov Chain analysis. Findings show that the Surma River adjusted steadily, with channel shifting rising from 12.51 % to 14.18 %, stability declining from 10.73 % to 8.50 %, erosion increasing from 1.78 % to 4.01 %, and sediment deposition from 2.52 % to 5.68 %. The Kushiya River experienced the most instability, with transitions falling from 35.98 % to 14.38 %, stability dropping from 22.56 % to 12.03 %, and erosion reaching 23.95 %, though projections suggest recovery by 2033 with stability rising to 32.29 % and erosion falling to 3.69 %. The Sari River remained strongly deposition-driven, with accretion increasing from 1.68 % to 6.11 % and stability staying low around 3.43 %, with 2033 projections showing accretion at 4.14 % and stability at 3.51 %. These insights shed light on the rivers' evolving nature and provide a valuable basis for better land use planning, risk reduction, and sustainable river management in Sylhet.

Keywords: *Transboundary, Erosion, Accretion, Morphological Change, Sylhet.*

Track 5: Environmental Science and Climate Change**Paper ID: 44****Presentation Mode: Poster****Predicting Evapotranspiration for Vegetable Crops by Simulating Root-Zone Soil Water with HYDRUS-1D****Md. Farhan Galib*, Md. Tasrif Rayhan Sarker, Sadika Tanzil Mim, Alpona Khatun, Khalid Mahmud, M. G. Mostofa Amin***Department of Irrigation and Water Management, Bangladesh Agricultural University, Mymensingh-2202, Bangladesh***Corresponding Author: galib.1905015@bau.edu.bd***Abstract**

Accurate estimation of soil water dynamics is essential for predicting actual evapotranspiration (ET_c), optimizing irrigation scheduling, and improving water-use efficiency. This study assessed the HYDRUS-1D model performance in predicting ET_c and crop coefficients (K_c) for seven common vegetable crops through simulating observed soil water content within the root zone. Experiments were conducted in two growing periods in 2023–2024, i.e., November–February (carrot, radish, red amaranth, coriander, spinach) and February–April (okra, Malabar spinach) in Mymensingh. The HYDRUS-1D model was calibrated against measured soil water contents for each crop and two soils. Model performance was evaluated using the coefficient of determination (R^2) and Nash–Sutcliffe Efficiency (NSE). HYDRUS-1D reproduced observed soil water dynamics with satisfactory performance, with R^2 and NSE values of 0.52 and 0.49 for okra, 0.52 and 0.50 for Malabar spinach (loam), 0.53 and 0.53 for Malabar spinach (clay loam), 0.78 and 0.72 for carrot, 0.87 and 0.75 for radish, 0.89 and 0.85 for red amaranth, 0.68 and 0.58 for coriander, and 0.62 and 0.60 for spinach, respectively. Various performances for different vegetable crops suggest influences from root system distribution, evapotranspiration estimation, or soil heterogeneity. The ET_c values ranged 0.76–5.03 mm/day for okra, 1.74–6.76 mm/day Malabar spinach (loam), 1.56–9.07 mm/day Malabar spinach (clay loam), 1.01–1.93 mm/day carrot, 0.58–3.26 mm/day radish, 0.37–2.06 mm/day red amaranth, 0.86–2.01 mm/day coriander, and 0.88–1.76 mm/day spinach. The K_c values ranged for okra 0.29–0.75, Malabar spinach (loam) 0.60–1.00, Malabar spinach (clay loam) 0.54–1.34, carrot 0.57–0.72, radish 0.33–1.09, red amaranth 0.37–0.72, coriander 0.18–0.84, and spinach 0.38–0.72. This simulation technique can be an effective tool for predicting root-zone soil water content under the tested crop and soil conditions, reliably supporting irrigation planning for these vegetables.

Keywords: *Soil water dynamics; HYDRUS-1D; Evapotranspiration (ET_c); Crop coefficient (K_c); Irrigation scheduling*

Track 5: Environmental Science and Climate Change**Paper ID: 47****Presentation Mode: Poster****Assessment of Salinity Related Hazard, Vulnerability, and Risk in the South-West Coastal Region of Bangladesh Using Principal Component Analysis: A Case Study in Bagerhat District****Maria Akter Pospo, Debasish Roy Raja, Sourav Das***Department of Urban and Regional Planning, Chittagong University of Engineering and Technology***Abstract**

Salinity refers to the total amount of dissolved salt in the soil solution which adversely affect the vegetation, built up area, water quality. It mostly occurs in the South-Western coastal regions of Bangladesh leading to reduced crop yields, food insecurity, a scarcity of fresh drinking water and various health problems. The main aim of this study is to analyze hazard, vulnerability and risk of salinity and assess severity of salinity risks in Bagerhat district and propose major recommendations for environment, disaster and risk management. The study area is Bagerhat district which increasingly vulnerable to salinity intrusion due to its geographical position and upstream water management practices. Landsat 8 OLI/TIRS imagery were downloaded from USGS Earth Explorer and population data was collected from Worldpop website. ArcGIS software has been utilized to calculate various spectral indices such as SI, NDSI, BI, NDWI, NDVI, NDBI, NDMI, SAVI and VSSI which collectively facilitate a comprehensive analysis. The hazard and vulnerability indices were integrated using Principal Component Analysis method and the resulting composite was used to assess and map the overall risk. Approximately 922.33 sq km area of Bagerhat district are highly to very highly vulnerable to salinity, 1178.05 sq km area are moderately vulnerable and 1386.13 sq km area of the district are classified as low to very low vulnerable. Nearly 47.23% (1646.84 sq km) area of the district is under high to very high risk including Mollahat, Chitalmari, Bagerhat Sadar, Rampal. About 28.31% (987.06 sq km) area of the district is under moderate risk and 24.45% (852.61 sq km) area is under low to very low risk areas which is confined to Fakirhat, Sarakhula, Morrelganj and parts of Kachua. This study promotes salt tolerant crop variety and farming systems through government extension services by integrating salinity risk reduction into local government development plans..

Keywords: *Salinity, Soil salinity, Hazard and vulnerability assessment, Bagerhat district, Remote sensing, Spectral indices, GIS, Principal Component Analysis (PCA)*

Track 5: Environmental Science and Climate Change**Paper ID: 70****Presentation Mode: Poster****Alternate Wetting and Drying for Climate-Resilient Boro Rice: Impacts on Yield, Water Use Efficiency, and Water Quality****Md. Ashrafuzzaman^{1,2*}, M. G. Mostofa Amin¹**¹*Department of Irrigation and Water Management, Bangladesh Agricultural University Mymensingh 2202, Bangladesh*²*Bangladesh Agricultural Development Corporation, 49-51 Dilkusha C/A, Dhaka-1000***Corresponding Author: riponashraf.badc@gmail.com***Abstract**

In Bangladesh, rice is crucial to food security, and dry-season Boro rice accounts for over half of the national rice production. Nevertheless, it faces escalating threats from climate change and unsustainable water use, primarily through continuous flooding (CF) irrigation, which depletes groundwater and degrades water quality. Therefore, this study evaluates water conservation techniques to mitigate this issue, focusing on the trade-offs between water use, yield, and nutrient leaching. A lysimeter experiment was conducted during the 2025 Boro season at the Bangladesh Agricultural University, Mymensingh, with BRRI Dhan-88. Five treatments—alternate wetting and drying (AWD) at 10, 15, and 20 cm soil-water depletion, dry direct-seeded rice (DSR), and conventional continuous flooding (CF)—were arranged in a randomized complete block design with three replications. Additionally, the recommended AWD (15-cm depletion) was tested on two soil types: clay loam and loam. Comprehensive data on soil, water, nutrients, agronomic traits, and weather were collected and statistically analyzed. Results demonstrate that 15-cm AWD in clay loam soil achieved the highest grain yield (8.48 t ha⁻¹), while the lowest (4.43 t ha⁻¹) was recorded for DSR. Relative to CF, AWD saved 25–27% water with a 12–24% yield increase. This represents a significant improvement in water use efficiency, which is crucial for climate resilience. In contrast, DSR, despite reducing water use by 69%, incurred a 35% yield penalty, highlighting a critical trade-off. From a water quality perspective, 15-cm AWD showed the highest concentrations of nitrogen leaching (NH₄⁺-N: 1.61 mg/L; NO₃⁻-N: 9.87 mg/L), whereas CF had the highest phosphorus leaching (0.03 mg/L). The water-saving treatments reduced the total nutrient leaching load by reducing percolation, underscoring a key environmental co-benefit. The findings demonstrate AWD significantly improving water-use efficiency while minimizing environmental impacts, highlighting its potential as a sustainable, climate-resilient irrigation strategy for Boro rice production.

Keywords: *AWD; Climate-resilient irrigation; Nutrient leaching; Rice yield; Lysimeter experiment*

Track 5: Environmental Science and Climate Change**Paper ID: 71****Presentation Mode: Poster****Ecological health of the Kushiara River in the Meghna Basin as indicated by hydrological assessment****Md. Abdur Rouf*, M. G. Mostofa Amin***Department of Irrigation and Water Management, Bangladesh Agricultural University, Mymensingh-2202, Bangladesh***Corresponding Author: mrouf.1905072@bau.edu.bd,***Abstract**

Maintaining environmental flow is essential for sustaining the ecological health of a riverine ecosystem. This paper estimates the environmental flow requirements (EFR) of the Kushiara River in Bangladesh, and evaluates the decade-wise temporal variation of flows between two decades, 1998–2007 (G1) and 2008–2017 (G2). Daily discharge data of the Kushiara River (Station ID-73) were obtained from the Bangladesh Water Development Board. The EFR was estimated using three hydrological methods: Mean Annual Flow (MAF), Flow Duration Curve (FDC), and Constant Yield (CY). According to the average discharge of time series data, flow patterns were divided into low flow (February–May), medium flow (November–January/June), and high flow (July–October) seasons. The findings show that the mean annual flow has significantly decreased by 8.4% from 845.66 m³/s (G1) to 774.30 m³/s (G2). The Kolmogorov-Smirnov test ($p < 0.001$) also verified significant differences between the flow duration curves of each period. Drought severity has increased significantly, as indicated by a 35.2% decline in minimum flow, decreasing from 92.8 m³/s in G1 to 60.14 m³/s in G2. Between G1 and G2, the high flow reduced from 2220.0 m³/s to 2053.6 m³/s, while median flows fell from 517.5 m³/s to 483.5 m³/s. The EFR estimated by the FDC had an average of 92.8 m³/s. The observed flow was below the EFRs during the G2 period in the low-flow season (December–May). This might harm aquatic ecosystems and people who depend on water for their livelihoods. The analysis of hydrologic alteration shows that there is a similar decline in flow volumes for all seasons, and specifically, the most significant one happens during low flows. The study adds to the limited research on environmental flows in Bangladesh's transboundary rivers and provides valuable methodological insights applicable to hydrological assessments of river systems across South Asia.

Keywords: *River flow, Low flow season, Aquatic ecosystem, Hydrological drought, Flow duration curve*

Track 5: Environmental Science and Climate Change**Paper ID: 91****Presentation Mode: Poster****Household Waste Management Behavior Pattern Among Urban and Rural People of Jashore District****Ananna Krishna Roy*, Sadman Sadik, Shovic Ibtasam Shahir***Department of Leather Engineering, Khulna University of Engineering and Technology, Khulna-9203, Bangladesh.***Corresponding Author: anannakrishnaroy@gmail.com***Abstract**

Waste is the biggest problem in Bangladesh due to the urbanization. Municipality facing trouble to mitigate the effect. The objective of this study was to assess the current status of waste generation, segregation, disposal methods, awareness levels, and community participation in waste handling among households in Jashore district. A structured questionnaire survey was conducted to collect data from households across various upazila in Jashore Bangladesh. In Jashore almost half of the total waste come from households. More than 8% of urban people and 61% of rural people throw their waste in open area. And 33% of rural area, there is no proper dustbin area or proper authorized person to collect the waste. 63% of rural people dump their waste. And most surprisingly 39% of the people separate their waste as whether it is hazardous or non-hazardous, organic or inorganic. It reflects that they have the basic knowledge of aftereffect of throwing waste here and there. On the contrary 64% of rural and 60% of urban reuse and recycle many of their waste such as food waste by composting, plastic bottle in every 7 days. Jashore sadar upazila is more conscious about waste management comparative to other upazila in jashore. Rural people recycle their food waste mostly. Because they have enough area to utilize their waste as fertilizer in their agriculture or use that in fish farm. Many people are conscious and willing to help the authority to manage the household waste in a proper way. Near 67% of rural people agreed to help the authority. Based on the results, most of the people think that the lack of proper dustbin and no rule of collecting the wastes are the main reasons of mismanagement of waste. The paper recommends enhancing public education, strengthening collection systems, and introducing community-based waste management initiatives to improve the overall efficiency and sustainability of waste management in Jashore.

Keywords: *Household waste, Sustainable waste management, awareness, waste disposal, recycle and reuse.*

Track 5: Environmental Science and Climate Change**Paper ID: 105****Presentation Mode: Poster****Assessing Urban Density Transformation in Chattogram City Corporation, Bangladesh (2000–2024): A Remote Sensing and GIS Approach with LULC and NDBI Analysis****Md. Omar Faruk, Khondakar Arifuzzaman***Department of Geography and Environment, Pabna University of Science and Technology, Pabna, Bangladesh.***Abstract**

This study examines the spatiotemporal transformation of urban density in Chattogram City Corporation (CCC), Bangladesh, from 2000 to 2024 using integrated Remote Sensing (RS) and Geographic Information System (GIS) techniques. Multi-temporal Landsat (5, 8 & 9) were processed through supervised Land Use/Land Cover (LULC) classification and the Normalized Difference Built-up Index (NDBI) within the Google Earth Engine (GEE) environment to quantify changes across low, medium, and high density urban zones. The analysis reveals a substantial 38.5% increase in built-up area, expanding from 14.8 km² in 2000 to 71.7 km² in 2024, with an overall compound annual growth rate of 7.2%. Urban growth occurred in three major phases: low-density horizontal expansion (2000–2010), medium-density consolidation (2010–2015), and high-density vertical intensification (2015–2024). High-density areas experienced the fastest growth, reflecting increasing land scarcity and intensified development pressures in the urban core. Significant environmental impacts accompanied this expansion. The city lost 62% of its wetlands and 44.57% of forest cover between 2014 and 2024, contributing to rising urban heat island effects of 2–4°C. Bareland increased markedly in the later decade, indicating ongoing land conversion and construction activity. Despite policy frameworks such as the Chattogram Structure Plan 2035, unplanned and fragmented development persists, consuming 71.7% of the city's usable land area and elevating climate vulnerability in this coastal, hazard-prone region. Accuracy assessment of the LULC classification produced an 88.6% overall accuracy and a Kappa coefficient of 0.843, affirming the reliability of the adopted RS–GIS approach in delineating urban density gradients. The study offers a replicable analytical framework for monitoring urban dynamics in fast-growing cities and provides essential evidence for guiding compact urban development, ecological preservation, and climate-resilient planning in Chattogram.

Keywords: *Urban transformation, NDBI, LULC change, Sustainable development.*

Track 5: Environmental Science and Climate Change**Paper ID: 133****Presentation Mode: Poster****Vertical Distribution and Physico-Chemical Controls on Microplastic Deposition in Flood-Affected Soils.****Md. Nazimul Islam Methon****Abstract**

In terrestrial soils, MPs—often co-transported with flood—pose a growing ecological threat due to their mobility during flood-induced disturbances, highlighting strong interactions between hydrological processes and MP transport in vertical. However, the mechanisms governing vertical transport, depth-wise distribution of MPs in flood-affected “char” soils remain largely unexplored. To elucidate the vertical transport and depth-resolved distribution of MPs in flood-affected “char” soils, and assess the influence of MPs size, shape, and color on soil properties. Soil samples were carefully collected from six spots on three riverside agricultural sites at three different depth layers (0-5 cm), (5-10 cm), (10-20 cm) for a total of 54 samples to analyze. In addition, we treated our sample under phase-amenable before-flood-, during-flood-, and after-flood conditions, followed by ZnCl_2 density separation at 1.4 g/cm^3 , and then H_2O_2 digestion of the soil to remove its organic components. The filtered soil sample was observed by fluorescence microscopy for size, form, and color. FTIR Analysis: MPs are also characterized by FTIR spectroscopy. To identify the vertical relationship in transport, soil quality parameters, including pH, electrical conductivity (EC), and Soil Organic Carbon (SOC), were quantified. MP concentrations across the three study sites ranged from 15 to 53 MPs per sample. Analyzing all the samples, we finally observed that the most MPs accumulate in the 0-5cm layers, i.e., the surface layer. Fragments dominated in all layers, mostly being brown or red, and FTIR characterization showed the main polymer to be polythene, with other polymers having distinct site dependence. Soil pH, EC, and SOC were assumed to be the MPS transport variable, where MP has a 100-1000 μm size. In conclusion, these findings provide critical insights into MP fate in riverine agroecosystems and the long-term ecological risks due to global expansion.

Keywords: *Microplastics, Flood Induced Transport, FTIR, Vertical Migration.*

Track 5: Environmental Science and Climate Change**Paper ID: 151****Presentation Mode: Poster****Remote Sensing-Based Analysis of Lulc Impacts on PM2.5 And Gaseous Pollutants: A Study Of Dhaka, Bangladesh****Jubayad Bin Showkat***Department of Civil Engineering, Khulna University of Engineering and Technology (KUET), Khulna-9203, Bangladesh***Abstract**

Bangladesh has one of the highest levels of pollution in the world today, with an average PM2.5 level that is 15-18 times greater than the WHO-recommended level. Dhaka, Bangladesh's capital, is well known for its especially high humidity and low-quality air. Many studies focus on the source of emissions into our atmosphere, but the author of this research combined high-resolution mapping of Land Use/Land Cover (LULC) of Dhaka via satellites with the measured concentration of the pollutants to develop a spatial relationship between LULC and the pollutants. The goal of this study is to understand how land use, land cover, and air pollution change seasonally; thus, pays special attention to a high concentration of dust in Dhaka during dry months. The research in this paper uses (Landsat 8) land cover maps and (Sentinel-5P) measurements of (NO₂), (particulate matter), and (AOD) to establish the relationship between land use/land cover and the decline in air quality within Dhaka based on Land Use/Land Cover maps of all 3 indices. The high-resolution mapping of all these indices was generated via GIS-based database that linked and created a geographic representation of each region containing the indices within Dhaka and the surrounding areas. There is a positive relationship between new land cover and high concentrations of air pollutants, particularly during the dry season. In Dhaka, the World Bank examined two large construction sites and found that PM2.5 levels were 150% higher than the recommended standards of the World Health Organization (WHO). Consequently, using satellite imagery to keep track of the condition of the land and creating public awareness programs to fuel sustainable land-use, increasing vegetation-covering Land Area (i.e., through afforestation) and utilizing alternative, cleaner technologies for brick-making, will allow us to mitigate the increasing problem of air pollution in Bangladesh.

Keywords: *Air pollution, PM2.5, Land Use/Land Cover (LULC), Dhaka, Satellite monitoring, GIS, Aerosol Optical Depth (AOD), NO₂ concentration, Urban development, Sustainable land-use.*

Track 5: Environmental Science and Climate Change**Paper ID: 171****Presentation Mode: Poster****Smart Fog Harvesting for Urban Air Quality and Water Augmentation: An Integrated Green Technology Approach for Climate Resilience in Bangladesh.****Nawar Islam Sattik, Sadia Taz Oyshi***BGMEA University of Fashion & Technology***Abstract**

Urban environments in Bangladesh face severe, interlinked crises: acute water scarcity due to groundwater depletion and chronic air pollution from fugitive dust (PM_{2.5}). Traditional high-energy solutions fail to address both simultaneously. This proposal presents a sustainable, integrated approach leveraging atmospheric resources. To design and validate a low-cost, zero-energy Smart Fog Harvester (SFH) system that concurrently augments decentralized water supply through passive moisture capture and improves local air quality via wet-scrubbing filtration of airborne particulate matter. A prototype SFH unit (40 m² mesh area) will be constructed using low-cost local materials and bio-mimicry mesh. The system is equipped with a solar-powered ESP32 and a PM_{2.5} sensor suite for smart monitoring of meteorological data, water yield, and air quality (pre- and post-filtration). Performance analysis will track daily water output during transition seasons and quantify the efficiency of the mesh in passively trapping PM_{2.5}. The SFH is expected to yield 15,000 - 25,000 Liters annually per unit, providing non-potable water at a highly competitive long-term cost of approximately BDT 0.50 per Liter. Crucially, the system is projected to achieve a verifiable 15-25% reduction in near-mesh PM_{2.5} concentration, mitigating respiratory health risks. The SFH is an important green technology paradigm for urban climate resilience. This method provides a sustainable, low-carbon approach to meet SDG 6 (Clean Water) and SDG 11 (Sustainable Cities) in climate-vulnerable regions, demonstrating both economic feasibility and a scalable, dual-environmental benefit.

Keywords: *Smart Fog Harvester (SFH), Atmospheric water generation, PM_{2.5} filtration, Wet-scrubbing, Decentralized water supply, Biomimicry mesh, Urban air quality, SDG 6, SDG 11*

Track 6: Mechanical Engineering, Production Technology, and Management**Paper ID: 23****Presentation Mode: Oral****Assessment of Hardened Properties of Concrete Using VCCTL: A Simulation Based Approach****Md. Shahadat Hossain*, Prottoy Chanda, Md. Jihan Hasan, Korno Kumar Pall, Md. Ashikur Rahman***Department of Civil Engineering, Gopalganj Science and Technology University***Corresponding author: shahadat18ce018@gmail.com***Abstract**

Concrete is one of the most common materials in the world of civil engineering and plays a pivotal role in achieving successful construction outcomes. It takes a lot of time for performance evaluation of concrete which is based on several parameters such as compressive/tensile strength, transport properties as well as microstructural characteristics, shrinkage, porosity. An experienced engineer may select the correct aggregates and proportion based on experience and the familiar property variations of typically used concrete mixes. Yet, difficulties may arise when mineral admixture (fly ash, slag) is added to or there is a change in mix proportioning, cement type. This requires sample preparation with these aforementioned mix variations and conducting various laboratory test to determine changes in concrete properties which is quite time-consuming. Virtual Cement and Concrete Testing Laboratory (VCCTL) offers a virtual solution that can stipulate concrete properties through numerical analysis. This study assessed the variation in strength and shrinkage with change in different input parameters, such as fly ash content, W/C ratio, Cement type and tried to validate the results with the help of microstructure images. From the study, in case of strength, cement with 30% fly ash replacement yielded highest strength of about 43 MPa at the end of 28 days. The study also concludes that the relation between chemical shrinkage and fly ash replacement is approximately proportional; chemical shrinkage increases with increment of fly ash. Different cement types (typical Portland, pozzolanic, and rapid hardening cement) variedly impacted both mechanical and shrinkage results of concrete mixes. The obtained variation patterns of concrete's aforementioned properties agree with the typical observations from previously done studies. This further validates the suitability of this approach as an economic alternative and can help engineers with quick results to select the correct elements, proportions etc.

Keywords: *Virtual Cement and Concrete Testing Laboratory (VCCTL), Concrete mix design, Performance evaluation, Mineral admixtures, Fly ash, Compressive strength*

Track 6: Mechanical Engineering, Production Technology, and Management**Paper ID: 108****Presentation Mode: Oral****MECHANICAL SHEAR STRENGTHENING TECHNIQUES FOR RC BEAMS: A CRITICAL REVIEW****Md. Farden Ahosan Sakib^{*1}, Mahin Afroz Sinjan², Md. Fataur Rahman Sami³, Md. Naimul Haque²**¹*Student, Military Institute of Science and Technology, Bangladesh.*²*Student, East West University, Bangladesh.*³*Bangladesh University of Engineering and Technology.***Corresponding Author: 2020-2-22-030@std.ewubd.edu***Abstract**

Maximizing the efficiency of solar energy conversion is a path to a cleaner, more affordable, and energy-secure future, significantly reducing our carbon footprint while meeting the growing global demand for power. This research investigates the effectiveness of bio-based PCM assive cooling systems for improving the efficiency of photovoltaic panels. The experiments were conducted on a 20W monocrystalline Si panel in summer weather conditions in Sylhet during June and July. Three different setups were investigated: natural air cooling, PCM passive cooling, and PCM with fins hybrid passive cooling. Coconut oil and wax were chosen as PCM for their local availability, bio-based nature, and lower costs than commercial PCMs. A 50:50 blend of coconut oil and wax, with a melting temperature of 46°C, was filled into aluminum containers for PCM cooling. The containers were equipped with internal straight aluminium fins and copper strips for improved heat circulation of PCM. External aluminum fins were incorporated with the PCM container to investigate the hybrid cooling system's ability to enhance heat dissipation and accelerate thermal removal from the PV module. The experimental results showed that under normal conditions, the maximum PV module efficiency was 18.1%, and the efficiency dropped to 16.3%–16.9% when the panel temperature reached up to 62°C during midday without any cooling. For PCM-based passive cooling, the panel's peak efficiency increased to 18.5%, with efficiency not dropping by less than 17%, and the temperature profile improved, resulting in an average panel temperature of 49.32°C. For PCM with fin hybrid passive cooling, the system sustained stable efficiencies between 17.4–18.2%, with a peak efficiency of 18.5%, and the average panel temperature was 48.52°C. The bio-based PCM mix of coconut oil and wax, with or without external fins, was able to improve thermal regulation of the PV panel, thus sustaining increased efficiency.

Keywords: *Solar energy, Photovoltaics (PV) module, Module cooling, Phase change materials(PCM), Peak power.*

Track 6: Mechanical Engineering, Production Technology, and Management**Paper ID: 115****Presentation Mode: Oral****Experimental Investigation of Bio-Based PCM for Photovoltaic Panel Cooling****R. J. Tumu*, N. M. Robin, M. M. Hasan, A. B. M. A. Malek***Department of Industrial and Production Engineering, Shahjalal University of Science and Technology, Sylhet-3114, Bangladesh***Corresponding Author: tumuresmajerin@gmail.com***Abstract**

Maximizing the efficiency of solar energy conversion is a path to a cleaner, more affordable, and energy-secure future, significantly reducing our carbon footprint while meeting the growing global demand for power. This research investigates the effectiveness of bio-based PCM passive cooling systems for improving the efficiency of photovoltaic panels. The experiments were conducted on a 20W monocrystalline Si panel in summer weather conditions in Sylhet during June and July. Three different setups were investigated: natural air cooling, PCM passive cooling, and PCM with fins hybrid passive cooling. Coconut oil and wax were chosen as PCM for their local availability, bio-based nature, and lower costs than commercial PCMs. A 50:50 blend of coconut oil and wax, with a melting temperature of 46°C, was filled into aluminum containers for PCM cooling. The containers were equipped with internal straight aluminium fins and copper strips for improved heat circulation of PCM. External aluminum fins were incorporated with the PCM container to investigate the hybrid cooling system's ability to enhance heat dissipation and accelerate thermal removal from the PV module. The experimental results showed that under normal conditions, the maximum PV module efficiency was 18.1%, and the efficiency dropped to 16.3%–16.9% when the panel temperature reached up to 62°C during midday without any cooling. For PCM-based passive cooling, the panel's peak efficiency increased to 18.5%, with efficiency not dropping by less than 17%, and the temperature profile improved, resulting in an average panel temperature of 49.32°C. For PCM with fin hybrid passive cooling, the system sustained stable efficiencies between 17.4–18.2%, with a peak efficiency of 18.5%, and the average panel temperature was 48.52°C. The bio-based PCM mix of coconut oil and wax, with or without external fins, was able to improve thermal regulation of the PV panel, thus sustaining increased efficiency.

Keywords: *Solar energy, Photovoltaics (PV) module, Module cooling, Phase change materials (PCM), Peak power.*

Track 6: Mechanical Engineering, Production Technology, and Management

Paper ID: 143

Presentation Mode: Oral

Thermal influence of flower shaped inner geometry on MHD natural convection flow of hybrid nanofluid inside a square cavity.**Prianka Mallick*, Dr. Bijan Krishna Saha***Faculty of science and engineering, University of Barishal***Corresponding author: priankamallick144@gmail.com***Abstract**

This study investigates the characteristics of natural convection heat transfer and entropy generation in a square cavity with flower shaped inner geometry filled with Al_2O_3 - $\text{TiO}_2/\text{H}_2\text{O}$ hybrid nanofluid focusing on the effect of magnetic field strength. The insights of this study have practical implication in various electronic cooling and heating devices. Objectives: The aim of this study is to explore the effects of related key parameters such as Rayleigh number (Ra), Prandtl number (Pr), Hartmann number (Ha), Nusselt number (Nu), Bejan number (Be) and nanoparticle volume fraction (ϕ) on heat transfer, fluid flow and entropy generation. The main focus of this study is to enhance heat transfer rate and improve thermal management system. The governing equations and corresponding boundary conditions are solved using Galerkin weighted residual finite element method. To ensure the accuracy of the present study, the results are compared with different published works. This study is investigated varying Ra from 10^3 to 10^6 , ϕ from 0% to 5% and Ha from 0 to 60 while keeping Pr fixed at 6.9. The results indicate that with the increase of Ra and ϕ , the average Nusselt number (Nu_{avg}) increases but with the increase of Ha , Nu_{avg} decreases. At fixed $Ha = 0$ and $\phi = 0.05$, Nu_{avg} increases 65.66% as Ra increases from 10^3 to 10^6 . Again, at fixed $Ra = 10^6$ and $\phi = 0.05$, Nu_{avg} decreases 5.28% varying Ha from 0 to 60. For the case of average entropy generation (E_{avg}), with the increase of Ra , ϕ and Ha , E_{avg} become increased. At fixed $Ha = 60$ and $\phi = 0.05$, E_{avg} increases 98.94% varying Ra from 10^3 to 10^6 and at fixed $Ra = 10^6$ and $\phi = 0.05$, E_{avg} increases 1.99% varying Ha from 0 to 60. On the other hand, for the case of average Bejan number with the increase of Ra , ϕ and Ha , Be_{avg} decreases. At fixed $Ha = 60$ and $\phi = 0.05$, Be_{avg} decreases 96.63% varying Ra from 10^3 to 10^6 . Again, at fixed $Ra = 10^6$ and $\phi = 0.05$, Be_{avg} decreases 43.58% varying Ha from 0 to 60. The study reveals that the innovative design of the inner geometry and Al_2O_3 - $\text{TiO}_2/\text{H}_2\text{O}$ hybrid nanofluid play a significant role in enhancing convective heat transfer and improving thermal management efficiency of different electronic devices.

Keywords: *Natural convection, Magnetohydrodynamics, Nusselt number, Entropy generation, Hartmann number.*

Track 6: Mechanical Engineering, Production Technology, and Management**Paper ID: 149****Presentation Mode: Oral****Effect of SMAW and TIG welding parameters on defects and Heat-affected Zone (HAZ) hardness in various steels****Pranta Datta, M. S. Rabbi****Department of Mechanical Engineering, Chittagong University of Engineering & Technology, Chattogram-4349, Bangladesh***Corresponding Author: rabbi@cuet.ac.bd***Abstract**

In the era of the Fourth Industrial Revolution, welding remains critically important as it advances alongside cutting-edge manufacturing technologies. In this study, Shielded metal arc welding (SMAW) and Tungsten inert gas (TIG) welding techniques have been used to produce sound butt welds of two different metals-stainless steel AISI 304 and carbon steel ASTM A516. Welding current, and specimen thickness were considered as variable parameters and 36 samples underwent investigation. Dye penetration testing (DPT) was employed to investigate the defects of the samples. Various welding defects such as porosity, cracks, grinding defects, partial fusion, and joint faults were considered to be investigated. It was found that 10 mm thick stainless steel of SMAW welding has the highest porosity for a defined length when the welding current is 90 amps. Brinell hardness test of all specimens indicates that stainless steel has a higher hardness value in the heat-affected zone (HAZ) than carbon steel when it goes through TIG welding. Moreover the results revealed that, the sample of 8 mm stainless steel in TIG welding provided 181 HB average hardness, while it is 159 HB for the same sample underwent SMAW welding

Keywords: *SMAW, TIG, Welding, Heat-affected Zone, Heat-affected Zone (HAZ).*

Track 6: Mechanical Engineering, Production Technology, and Management**Paper ID: 118****Presentation Mode: Poster****Effect of non-Hermitian modulation for Topological acoustic transmission in a 2D periodic system based on phononic crystal****Md. Shuzon Ali***Department of Applied Mathematics, Rajshahi University, Rajshahi-6205, BANGLADESH.***Corresponding Author: sali_4222@ru.ac.bd***Abstract**

The topological acoustic phenomenon in the material showing a rich variety of unique characteristics in classical and quantum wave systems, possess parity-time (PT) symmetry in non-conservation Hamiltonian physics. In this letter, I investigated the topological acoustic transmission in a valley waveguide constructed using a two-dimensional phononic crystal (PnC) with C_{3v} symmetric arrangement of three rods filled by air and periodically immersed in polydimethylsiloxane (PDMS) through the unit cell. The numerical results clearly show the asymmetric eigenfrequency mode that occur the exceptional point (EP). The non-Hermitian Hamiltonian system with lossy air and gain air modulation (γ) have shown efficient transmission along the interface between two types of PnCs with differently oriented unit cells. All of these analyses are performed by using a general purpose software package (COMSOL Multiphysics) based on finite-element method. The present approach with an interface opens a new door that helps better understanding to stimulate research on acoustic topological properties beyond the Hermitian regime.

Keywords: *Phononic Band; Non-Hermitian System; Exceptional Point (EP); Super Cell Interface; Waveguide Transmission.*

Track 7: Petroleum, Mining, and Ocean Technology**Paper ID: 38****Presentation Mode: Oral****Land Subsidence Prediction Due to Coal Extraction in Barapukuria Coal Mine****Tonmoy Saha^{1,2*}, Istiak Ahmed^{1,2}, Md. Sohan Miah^{1,2}, Mohammad Forrukh Hossain Khan^{1,2}**¹*Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh*²*Mine Planning and Design Laboratory, Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh***Corresponding Author: tonmoy3632@gmail.com***Abstract**

Land subsidence prediction is a critical component of sustainable underground coal extraction, particularly in geologically sensitive and densely populated regions. The Barapukuria Coal Mine in northwest Bangladesh has experienced notable surface deformation associated with longwall mining activities, resulting in structural damage, land use disruption, and growing community concern. Field observations indicate that the maximum subsidence with significant areal extent has reached 5.8 m in the Northern Upper Panels (NUP) and 4.2 m in the Southern Lower Panels (SLP) following the extraction of a 10 m thick coal seam. This study evaluates land subsidence mechanisms in the Barapukuria coal basin by analyzing geological stratigraphy, mining geometry, overburden characteristics, and historical ground movement records. Both numerical and empirical prediction techniques are employed to estimate the magnitude, extent, and progression of subsidence under current and future extraction scenarios. Findings highlight the strong influence of weak overburden lithology, high extraction ratio, and panel orientation on surface deformation patterns. The study emphasizes the need for reliable forecasting approaches to support mine planning, mitigate environmental and socio-economic impacts, and guide the development of effective subsidence management strategies for Barapukuria and similar coal mines.

Keywords: *Subsidence prediction; Barapukuria coal mine; Ground deformation; Overburden response*

Track 7: Petroleum, Mining, and Ocean Technology**Paper ID: 84****Presentation Mode: Poster****Assessment of Land Deformation at Barapukuria Coal Mine Area and its Impacts with Geospatial Techniques****Md. Rubel Islam, Dr. Md. Rahedul Islam Rahe**¹ *Geospatial Lab. Of Environment and Disaster, Department of Geography and Environment, Pabna University of Science and Technology-PUST***Abstract**

Coal is a significant natural resource in Bangladesh and has been used to produce energy. The Barapukuria Coal Mine area is the first and largest Coal Mine area in Bangladesh. Due to the coal extraction, the mining area facing numerous environmental challenges and deformation is most concerned issue. This study aims to land deformation and its impacts at Barapukuria Coal Mine Area using SRTM and ASTER GDEM V3, LULC (Land Use Land Cover), Google Earth Pro, GIS (Geographical Information System) and RS (Remote Sensing). This study work used USGS (Earth Explorer) for SRTM DEM in 2014 and used Earth Data for 2004 and 2024 ASTER DEM for land deformation. Moreover, this study used time series satellite images from the Landsat 5 TM satellite 2004 and the Landsat 8 OLI/TIRS satellite between 2014 and 2024 for impacts of land deformation. The images were categorised into four classes like as water, agriculture, urban and vegetation. Analytical results clearly indicated High land subsided was 6m and low land subsided was 1m from year 2004 to 2014. Further, High land subsided was 9m and low land did not deform from year 2014 to 2024. Finally, we got results high land deformation was 15m and low land deformation was 1m from year 2004 to 2024. Moreover, also analysed LULC and got results indicated water bodies increasing and agriculture land decreasing trends over the past 20 year. Water bodies increased from 1.2555sq km to 3.537sq km between 2004 and 2024. However, agriculture land decreased from 25.7814sq km to 23.4027sq km in the same period. As a result, the local residents threatened, relocated and damaged agriculture land and building. The outcomes of the study, successfully identify amount of land deformation and its impacts of Barapukuria Coal mine area in Bangladesh.

Keywords: *ASTER, SRTM, DEM, LULC, USGS, Earth Data, GIS, RS and Barapukuria Coal Mine*

Track 7: Petroleum, Mining, and Ocean Technology**Paper ID: 93****Presentation Mode: Oral****Wave Energy Assessment and Optimal Site Selection in the Bay of Bengal: A Multi-Criteria Analysis for Wave Energy Converter Deployment****Animesh Roy*, Susmita Das Mumu, Dipta Akash Roy**¹*Department of Naval Architecture and Marine Engineering, Bangladesh University of Engineering and Technology, Dhaka, Bangladesh -1000*²*Department of Mechanical Engineering, Bangladesh University of Engineering and Technology, Dhaka, Bangladesh -1000***Corresponding Author: animeshbuet27@gmail.com***Abstract**

The transition to renewable energy is crucial for mitigating greenhouse gas emissions, thereby underscoring the importance of adopting green technologies. Among the available renewable sources, wave energy emerges as a particularly promising option. Owing to its extensive and uninterrupted coastline, Bangladesh possesses considerable potential for the effective utilization of wave energy. Besides the availability of energy, it is important to determine the most efficient place where temporal variability becomes steady and the appropriate choice of wave energy converter (WEC). In this study we use temporal variability index (COV, SV, and MV) and WEC location suitability (WLS) index to detect the best efficient place in Bay of Bengal. This study presents a comprehensive assessment of wave energy potential in the Bay of Bengal using 30 study points (P1,P2....P30) spanning latitudes 21.5oN to 15.00No and longitudes 88oE to 92oE. The research evaluates temporal variability indices including coefficient of variation (COV), seasonal variability (SV), and monthly variability (MV) to determine optimal locations for WEC deployment. Two WEC systems were analyzed: Pelamis and Wave Dragon converters. Results indicate that the highest wave power exceeds 21 kW/m at seven locations (P6, P11, P12, P17, P18, P23, P24), with maximum power of 23.82 kW/m at point P12. Multi-criteria analysis incorporating normalized wave power, capacity factors, temporal variability, distance, and water depth indices identified positions P3, P8, P15, and P21 as optimal sites for WEC deployment. The study reveals significant seasonal variation with summer months contributing approximately 53% of annual wave power, while winter months contribute only 8%. These findings provide crucial insights for renewable energy planning and maritime 20boundary utilization following Bangladesh's recent maritime boundary expansion.

Keywords: *Mining wastewater treatment, Barapukuria Coal Mine (BCMCL), Acid Mine Drainage (AMD), Dissolved Air Flotation (DAF), Ultrafiltration (UF), Reverse Osmosis (RO), Water recycling, Circular water economy, Resource recovery, Contaminant removal.*

Track 7: Petroleum, Mining, and Ocean Technology**Paper ID: 104****Presentation Mode: Oral****Integrated Investigation of Self-Heating and Combustion Risk in Coal Stockpiles at Barapukuria Coal Mine****Labiba Akter^{1,2}, Md. Manjurul Islam^{1,2}, Md. Sohan Miah^{1,2}, Mohammad Forrukh Hossain Khan^{1,2*}**¹ Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh² Mine Planning and Design Laboratory, Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh

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Abstract

Spontaneous combustion in coal stockpiles represents a significant safety and environmental hazard in tropical monsoon climates. This case study examines the susceptibility to spontaneous combustion within the coal storage yard at Barapukuria Coal Mine, Dinajpur, Bangladesh, which supplies coal to the adjacent power plant and regional industries. Employing an integrated methodological framework, the study incorporates assessment of coal stacking practices and storage conditions, collection of representative samples of raw coal, sized coal, and coal fines from various stockpiles, proximate and ultimate analyses to determine moisture, ash, volatile matter, fixed carbon, and sulphur content, as well as thermal analysis to estimate the onset temperature of self-heating. Field measurements of stockpile surface temperature and gas composition (CO, CO₂, CH₄, C₂H₄, and C₂H₆) near suspected hotspots are utilized to identify active low-temperature oxidation zones and related air quality impacts. Findings demonstrate that elevated volatile matter and sulphur content, a high proportion of fines, extended residence time in stockpiles, and insufficient control of pile height and ventilation collectively increase the vulnerability of Barapukuria coal to spontaneous heating, particularly during hot, dry pre-monsoon periods. The study proposes enhanced stockpile design and rotation, segregation and controlled handling of fines, routine monitoring of temperature and gases, and installation of an automated water spray system in the coal yard to provide controlled surface cooling with improved drainage and housekeeping as effective measures to mitigate spontaneous combustion risk and its environmental impact in Barapukuria's coal storage yard.

Keywords: *self-heating, spontaneous combustion, coal storage yard, coal handling.*

Track 7: Petroleum, Mining, and Ocean Technology**Paper ID: 110****Presentation Mode: Oral****Optimizing Mine Water Treatment Through Advanced Technologies: A Case Study of Barapukuria Coal Mine****Md. Sohan Miah^{1,2}, Md. Manjurul Islam^{1,2}, Istiak Ahmed^{1,2}, Mohammad Forrukh Hossain Khan^{1,2*}**¹ Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh² Mine Planning and Design Laboratory, Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh**Corresponding Author: dr.forrukh@just.edu.bd***Abstract**

Mining operations, including extraction and mineral processing, use large quantities of water and generate substantial contaminated wastewater. Process water becomes polluted with dissolved metals, suspended solids, and chemical reagents, while sulfide mineral oxidation produces Acid Mine Drainage (AMD), a highly acidic, metal-rich, and severe pollution source. Contaminated water infiltrates aquifers, degrading groundwater quality, and its discharge into surface water disrupts ecosystems, causes fish mortality, and reduces irrigation suitability, threatening agriculture and human health. These impacts highlight the need for effective wastewater treatment, as failure leads to long-term environmental damage, high remediation costs, and regulatory non-compliance that risks the industry's social license to operate. This study assesses, quantifies, and evaluates Dissolved Air Flotation (DAF), Ultrafiltration (UF), and Reverse Osmosis (RO) to identify the most viable water treatment method for Barapukuria Coal Mine Company Limited (BCMCL). It emphasizes the critical role of wastewater recycling and reuse in strengthening water security, reducing reliance on freshwater sources, and minimizing environmental impacts. Within a hierarchical wastewater management framework focused on responsible disposal, reuse-oriented treatment, and pollution prevention, the study demonstrates that advanced technologies outperform conventional methods, particularly in resource recovery and reuse. Key findings show that effective secondary treatment can remove about 85% of BOD and suspended solids, while RO membranes can eliminate up to 99% of contaminants to produce high-quality effluent. Overall, the study underscores how advanced treatment technologies enable a circular water economy and support long-term water sustainability amid growing global water scarcity.

Keywords: Mine wastewater; Dissolved Air Flotation (DAF); Ultrafiltration (UF); Reverse Osmosis (RO); BOD removal.

Track 7: Petroleum, Mining, and Ocean Technology**Paper ID: 132****Presentation Mode: Oral****Petrophysical Evaluation of Habiganj Gas Field using Well Log Interpretation****Md. Maharaj Mridul Soumic****Department of Petroleum and Mining Engineering, Jashore University of Science and Technology***Corresponding author: maharajmredul2002@gmail.com***Abstract**

The petrophysical study of Habiganj Gas Field wells HBJ-8 and HBJ-9 was carried out to quantify reservoir quality, hydrocarbon storage, and production potential using conventional wireline logs (GR, caliper, RHOB, NPHI, DPHI, deep & shallow resistivity, SP) and standard log-based workflows (shale-volume by Dresser-Atlas/Clavier correction, neutron-density porosity crossplots, Simandoux/Wyllie-Rose permeability estimation). The field-level GIIP is reported at ~3,684 Bscf, with the upper (Bokabil) and lower (Bhuban) reservoir architecture recurring across the field; upper reservoir porosity ranges 12–36% (mean ~28%) with average permeability on the order of ~500 mD, and an average water saturation in the upper zone of ~18% (implying very high gas saturation). Notably, HBJ-8 and HBJ-9 exhibit localized anomalous intervals: HBJ-8 contains a high-porosity (>40%) and high-permeability (reported locally at 5–16 D) interval (~1410–1450 m; ~40 m thick), and HBJ-9 shows similar laser-thickness features, interpreted as fracture-enhanced sands that substantially boost local transmissibility. Gas-water contacts cluster near 1415–1486 m (SSTVD) across wells, and log-derived gas saturation for gas-bearing sands commonly exceeds 70–80% in clean intervals. Volumetric outcomes indicate that wells 8 and 9 contribute materially to remaining field gas and that the fracture-enhanced intervals in these wells could sustain commercial deliverability despite field maturity. These petrophysical findings recommend targeted completion/ stimulation of the high-permeability intervals and re-estimation of well-level GIIP using core/pressure data to refine recoverable reserves and optimize late-life production.

Keywords: *Petrophysical Analysis, Reservoir Quality, Porosity, Gas Saturation (GIIP), Fracture-Enhanced Sands.*

Track 7: Petroleum, Mining, and Ocean Technology**Paper ID: 155****Presentation Mode: Oral****Integrated Desulfurization of Coal through Fenton Oxidation and Froth Flotation: A Hybrid Approach for Clean Coal Production****Asif Akbar Siyam^{1,2}, Sheikh Ayman Ahmad^{1,2}, Md. Raihan^{1,2}, Md. Riyad^{1,2}, Minhaj Uddin Monir^{1,2*}**¹*Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh*²*Energy Conversion Laboratory, Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh***Corresponding author: monir_pme@just.edu.bd***Abstract**

Coal continues to serve as a primary feedstock in power generation and in the conversion processes that yield liquid and gaseous fuels with high energy density. However, the large-scale use of coal for energy generation leads to severe environmental pollution and industrial problems. Coal's industrial applications are constrained by sulfur compounds, which lead to various technical difficulties and environmental concerns. In this work, imported coal was treated via an air oxidation aided froth flotation process, in which performic acid (a mixture of H_2O_2 and HCOOH) functioned as the co-oxidant and iron ions (Fe^{2+}) acted as catalytic agents within an aqueous medium. Under optimized conditions of ($50^\circ\text{--}70^\circ\text{C}$), a reaction time of 30–45 minutes, a 2:2 (v/v) ratio of H_2O_2 and HCOOH , and an Fe-ion concentration of 0.10%, the process achieved up to 75% total sulfur (TS) removal and 50% ash reduction. Desulfurization and deashing occurred through a Fenton-type reaction mechanism, in which Fe^{2+} catalyzed the decomposition of hydrogen peroxide (H_2O_2) to generate reactive hydroxyl radicals ($\bullet\text{OH}$). Formic acid (HCOOH) reacted with hydroxyl radicals ($\bullet\text{OH}$) to form performic acid and reactive peroxy radicals, which preferentially converted sulfur-containing compounds into sulfoxides and sulfones. Froth flotation was then conducted employing Triton X-100 or methyl isobutyl carbinol as surfactants and diesel oil (DO) or kerosene as collectors, resulting in the elimination of approximately 50% of total sulfur and (40–45)% of ash under the optimized process parameters. Under optimized parameters, the combined two-step batch treatment of oxidation followed by froth flotation resulted in total sulfur and ash removal of 87–90% and 80% in the froth fraction, and (60–65)% and 40% in the tailings. Characterization by SEM, FTIR, and XRD revealed significant structural, chemical, and mineralogical changes in the treated coal. This method shows promising prospects for industrial scale application in upgrading low-rank coal for electricity production and other industrial uses.

Keywords: *Fenton mechanism, Oxidative Desulfurization, Froth Flotation, Deashing.*

Track 7: Petroleum, Mining, and Ocean Technology**Paper ID: 156****Presentation Mode: Oral****Energy Perspective on Barapukuria coal: Functional and Morphological Characterization Using FTIR and SEM–EDX.****Md. Riyad^{1,2}, MD. RAIHAN^{1,2}, Sheikh Ayman Ahmad^{1,2}, Asif Akbar Siyam^{1,2}, Minhaj Uddin Monir^{1,2*}**¹*Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh*²*Energy Conversion Laboratory, Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh*

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Abstract

In a bid to achieve effective and sustainable utilization of the indigenous coal in Bangladesh, it is should be clear information on the effects of the properties of such coal on both in terms of chemical properties and microstructure on the combustion behavior and energy-use efficiency, on this note, the study is going to examine the functional properties of Barapukuria coal using Fourier Transform Infrared (FTIR) spectroscopy and morphological and elemental distribution shall be analyzed with the assistance of Scanning Electron Microscopy (SEM-EDX). The visual analysis of SEM-EDX samples shows that the coal particle under the study is predominantly consisting of carbon (63.6 wt%), a considerable portion of nitrogen (21.5 wt%), oxygen (13.1 wt%), sulfur (0.58 wt%) and the remnants of Al, Si, Ti, Ni and Sr. This fundamental geometry implies that it is an organic-rich matrix with heavily dispersed mineral and trace-element phases, in other words, that ash-forming species usually occur as fine inclusions, as opposed to homogeneous and individual mineral grains. The form of distribution can significantly affect the mineral transformation, ash formation and even combustion slagging and fouling since the reactive mineral phases can melt or react in coal particles as opposed to being separated out. The analysis of FTIR will involve data about the aromaticity of the coal, the cross-linkage of macromolecules, the presence of oxygenated structures that will regulate the ignition temperature, reaction at initiation and the burn out characteristics. With a mixture of SEM-EDX elemental patterns that were measured and an approximation of the expected FTIR-derived functional group data, the study stands to understand the possible combustion response, ash behavior, and applicability of Barapukuria coal in boilers. This finding is likely to give a spectral and microstructure-driven, energy-concentrated framework to provide a support to the decision-making with respect to the choice of fuels, coal mixing approaches and conditions in the power and industrial boiler rooms in Bangladesh.

Keywords: *FTIR, SEM-EDX, Combustion, Ash-forming.*

Track 7: Petroleum, Mining, and Ocean Technology**Paper ID: 101****Presentation Mode: Poster****Sustainable Underground Ventilation System: Selective Air-Distribution Design for Maddhapara Granite Mine****Farhana Yesmin^{1,2}, Md. Manjurul Islam^{1,2}, Labiba Akter^{1,2}, Mohammad Forrukh Hossain Khan^{1,2*}**¹Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh²Mine Planning and Design Laboratory, Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh

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Abstract

Maddhapara Granite Mine, Bangladesh's sole active underground hard rock mine, experiences persistent ventilation challenges, including blasting-induced dust, gas accumulation, heat, and substantial airflow losses through multiple drifts, roadways, and ventilation structures. Conventional air-distribution practices typically allocate airflow to all underground passages, including short crosscuts and auxiliary drifts, based on assumptions of unavoidable leakage across the network. This research reviews the existing ventilation air-distribution plan and develops a demand-based strategy for allocating airflow within the mine. Personnel demand factors, dust generation information, auxiliary fan requirements, and regulatory minimum airflow criteria were applied to determine the necessary air quantities for main stopes, haulage drifts, and other critical working areas. Ventilation coefficients were further used to estimate additional airflow needs arising from internal leakage and uneven distribution. Analysis showed that short, non-critical surface drifts- such as safety exits and minor crosscuts- do not require dedicated airflow, as minor leakage can be accommodated by surplus air from the main fan. In contrast, long or dust-prone drifts connected to active stopes require regulated airflow to ensure adequate dilution and permissible air velocities. The study also establishes a methodology for quantifying internal air leakage and outlines strategies for mitigating its influence on ventilation performance. Overall, the findings indicate that an optimized, selective air-distribution approach provides a more rational, energy-efficient, and operationally sustainable ventilation design for Maddhapara Granite Mine.

Keywords: *underground ventilation, air-distribution strategy, ventilation leakage, ventilation coefficients*

Track 7: Petroleum, Mining, and Ocean Technology**Paper ID: 125****Presentation Mode: Poster****A Probabilistic Prediction of Bangladesh's Earthquake Activity and Vulnerable Regions****Md.Abir hossain, Dr.Md. Rahedul Islam***Geospatial Lab Of Environment and Disaster, Department of Geography and Environment, Pabna University of Science and Technology, Pabna-6600, Bangladesh***Corresponding Author: abirhossainhossain172@gmail.com***Abstract**

Earthquakes are nowadays one of the most alarming environmental hazards across Bangladesh and globally. Although the happening of earthquake is a random phenomenon, Historical trend analysis, geo-spatial analysis and advanced methods of machine learning could be used to model them effectively. Regarding the background, the objectives are specified; to identify the active potential fault line, probabilistic prediction of earthquake occurrences, and vulnerable regions. From 1900 to 2024, 579 earthquakes within a range of 5.0 to 8.5 were included in the historical data which has been collected from the USGS, ISC, and BMD. The magnitude, depth, radius, and fault line previous earthquakes are analyzed to determine the relationships between and within them. Identifying potential active fault lines within and outside of the country is made easier with the implementation of multiple ring spatial clustering methods. In this investigation, the specific time historical earthquake catalog has been applied combined with several machine learning classifiers to perform probabilistic prediction of earthquake occurrence. Eight mathematically determined seismic parameters have been employed in constructing probabilistic prediction. These features are derived from the distribution of depth, magnitudes, frequency, and Gutenberg-Richter's inverse law. To predict future earthquake occurrences, four machine learning techniques are used autonomously: Pattern Recognition Neural Network, Recurrent Neural Network, Random Forest, and Linear Programming Boost Ensemble Classifier. Based on the prediction results, Random Forest scored 84.72%, PRNN scored 81.94%, and RNN scored 79.17%. Interestingly, the LP boost ensemble classifier, which integrates weak models, achieved the highest prediction accuracy of 86.11%. The Churachandapur Mao and Kabaw faults are the main sources of light shaking in Bangladesh, despite the Dauki fault representing the most active. According to our projection, Bangladesh's most vulnerable regions—Sylhet, Mymensingh, Chittagong and some portion of Dhaka—may sustain the most damage from a future catastrophic earthquake.

Keywords: *Earthquake, Probabilistic prediction, Geo-spatial analysis, Machine learning, Active potential fault line, Vulnerable region, Bangladesh.*

Track 7: Petroleum, Mining, and Ocean Technology**Paper ID: 158****Presentation Mode: Poster****Calculation of coalbed methane resource potential of Barapukuria coal mine using Meisner and Kim equation****Sheikh Ayman Ahmad^{1,2}, Asif Akbar Siyam^{1,2}, Md.Riyad^{1,2}, MD. Raihan^{1,2}, Minhaj Uddin Monir^{1,2 *}**¹*Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh*²*Energy Conversion Laboratory, Department of Petroleum and Mining Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh***Corresponding Author: monir_pme@just.edu.bd***Abstract**

Coalbed methane is a great source of energy. The prediction of coalbed methane resource is a critical step before the extraction process. This study outlines a simplified approach to the evaluation of the coalbed methane (CBM) resource potential of the Barapukuria Coal Mine attained through the combination of thermogravimetric analysis (TGA) with gas-content rate dependent correlations of the rank. The TGA data are used to measure the proximate parameters, which are moisture, volatile matter, fixed carbon, and ash content at carefully controlled heating value. The experimentally determined values would offer a laboratory-based estimate of the coal composition and devolatilization processes and thus the rank indicators required in CBM predictive models are improved. The Meisner and Kim equations, relating coal rank, depth, and proximate parameters, are then used to estimate gas content (m³/t) and the in-place CBM resources of the Barapukuria basin. This technique is an empirical alternative to desorption testing, specifically appropriate in those situations where field gas-content data is unavailable. This study showed that the methane content of Bituminous coal at Barapukuria coal mine varies from 15.88 m³/ton using the Meisner formula to 15.09 m³/ton using the Kim formula, representing a potential resource of approximately 4.22 billion cubic meters gas using the volumetric method. It is shown that incorporation of TGA properties of coal and depth dependent gas-content associations can improve preliminary estimates of CBM potential in Gondwana coalfields of Bangladesh and provide a more acceptable means of appraising unconventional power sources.

Keywords: Coalbed methane (CBM), Thermogravimetric analysis (TGA), Proximate analysis, Meisner-Kim correlation, Barapukuria coal mine.

Track 8: Textile Engineering and Sustainable Technology**Paper ID: 20****Presentation Mode: Oral****Sustainably Engineered PVA/CMC–Rice Husk Ash Bio-composites: Structure–Property Relationships and Barrier Enhancement toward Green Functional Materials****Md. Mebtahul Jannat, Mahfuzul Islam, Zannatun Naim, Md. Hafezur Rahaman***Dept. of Applied Chemistry and Chemical Engineering, Islamic University, Kushtia-7003***Abstract**

The urgent need to mitigate the environmental footprint of conventional plastics has accelerated research on sustainable, biodegradable materials with optimized structure–property balance. This study presents the eco-fabrication and comprehensive characterization of poly(vinyl alcohol) (PVA)/carboxymethyl cellulose (CMC) bio-composites reinforced with treated rice husk ash (TRHA)—a silica-rich agricultural residue valorized as a reinforcing nano-filler. The hybrid matrix leverages the mechanical integrity and film-forming properties of PVA, as well as the biodegradability and hydrophilic compatibility of CMC, to form a synergistic polymer network. TRHA was incorporated at varying loadings (0–20 wt%) using a solution casting method. HCl-treated RHA exhibited a highly amorphous, porous morphology and superior silica purity (98.82%) compared to H₂SO₄-treated samples, confirmed by SEM–EDX and compositional analyses. FTIR spectra revealed characteristic Si–O–Si, O–H, and C–H interactions, indicating effective interfacial bonding and hydrogen-bond-mediated compatibility between the TRHA particles and polymer chains. Enhanced crystallinity and phase uniformity, as detected by XRD, further substantiate the improved polymer–filler interfacial organization. Mechanical and thermal analyses demonstrated that silica incorporation up to 15 wt% significantly improved tensile strength, modulus, and thermal stability due to uniform filler dispersion and matrix crosslinking. Beyond this concentration, agglomeration effects slightly diminished performance. The water vapor transmission rate (WVTR) decreased markedly with moderate TRHA loading, confirming enhanced barrier properties, while soil burial degradation tests verified complete biodegradability with controlled degradation kinetics. These findings establish that optimized PVA/CMC–TRHA composites (10–15 wt%) exhibit an ideal balance of mechanical robustness, moisture resistance, and environmental degradability. The work highlights a sustainable pathway for transforming agro-waste into high-performance green functional materials, supporting circular economy strategies in polymer and materials engineering.

Keywords: *Rice husk ash, Sustainable materials, Poly(vinyl alcohol), Carboxymethyl cellulose, Waste valorization.*

Track 8: Textile Engineering and Sustainable Technology

Paper ID: 77

Presentation Mode: Oral

Environmentally Friendly Superabsorbent Bio-hydrogel for Agricultural Soil Conditioning and Plant Micronutrient Release

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Abstract

Water scarcity poses significant challenges to agricultural productivity, leading to substantial crop losses each year. In this study, we synthesized a superabsorbent, eco-friendly bio-hydrogel from carboxymethyl tamarind kernel gum (CMTKG), using epichlorohydrin as a crosslinking agent. The hydrogel is designed to act both as a water reservoir and a slow-release carrier for essential micronutrients. The prepared hydrogel was characterized by FTIR, SEM, TGA, UV visible, and rheological techniques to observe the hydrogel formation and stability. The bio-hydrogel synthesized under optimized conditions exhibited a swelling of 23,845% and a reswelling capacity of 6,217% after five cycles at room temperature (25°C). The swollen hydrogel retained water in air for up to 16 days. When incorporated into soil, this retention period was extended by about 48 days compared to soil without the hydrogel. The Maximum water-holding capacity of the soil was increased by 90% compared to the control soil sample. Biodegradation analysis showed a weight loss of the hydrogel by 98% after 60 days. In a blackberry plantation study, the germination rate and survival rate were 25% and 85% higher, respectively, than those planted in the control soil. As a proof of the concept, molybdenum (Mo) was loaded into the CMTKG-based bio-hydrogel to demonstrate controlled release of micronutrients. The hydrogel released approximately 86% of the loaded Mo into water over a period of 19 days. Kinetic modelling showed that both the first order and Peppas-Sahlin models provided the best fit, indicating that Mo release kinetics involve a combination of concentration-gradient-driven transport and CMTKG polymer chain relaxation dynamics. So, CMTKG hydrogel can be used for agricultural applications for a sustainable and technological solution to water scarcity in arid and semi-arid regions and sustained nutrient release for plant growth.

Keywords: Bio-hydrogel, Micronutrient, Molybdenum, Slow release, Soil amendment.

Track 8: Textile Engineering and Sustainable Technology**Paper ID: 78****Presentation Mode: Oral****Sustainable Fuel Production from Scrap Tires and Vegetable Waste via Electromagnetic Induction Heated Co-Pyrolysis****Md. Parvej Mosaraf Sohel¹, Maariz Rahman¹, Fabiha Nawshin Satta²**¹*Department of Glass & Ceramic Engineering, Rajshahi University of Engineering & Technology (RUET), Rajshahi-6204, Bangladesh*²*Department of Materials Science and Engineering, University of Rajshahi, Rajshahi-6205, Bangladesh***Abstract**

The sustainable conversion of solid wastes into value-added fuels is essential for addressing global energy shortages and environmental challenges. This study investigates the co-pyrolysis behavior of scrap tire waste and vegetable waste using a fixed-bed pyrolysis reactor heated by an electromagnetic induction system. Proximate, ultimate, and thermogravimetric analyses were conducted to determine the suitability of the wastes as feedstock. Blends containing varying proportions of scrap tire and vegetable waste (1:1, 1:2, and 2:1 by weight) were pyrolyzed at 380–430°C under inert N₂ conditions. The influence of feedstock ratio on product distribution, fuel properties, and energy conversion efficiency was systematically evaluated. The optimum liquid yield of 48 wt% was obtained for a blend of 67 wt% scrap tire and 33 wt% vegetable waste, producing pyrolytic oil with a calorific value of 32,541 kJ/kg, low viscosity (6.2 cP), and a density of 836 kg/m³. Increasing tire content enhanced liquid formation, whereas higher vegetable waste ratios favored char and gas formation due to elevated ash and oxygen content. FTIR analysis confirmed the presence of aliphatic and aromatic hydrocarbons along with minor oxygenated functional groups (ketones, aldehydes, ethers). GC-MS detected approximately 45 compounds, primarily within the C₇–C₂₄ range, indicating the production of fuel-like hydrocarbons comparable to petroleum fractions. The generated char, rich in carbon, shows potential for use as solid fuel or as a precursor for activated carbon. The use of electromagnetic induction heating demonstrated higher thermal efficiency and environmental compatibility than conventional gas-heated systems. Overall, the study validates co-pyrolysis as an effective route for waste management and renewable fuel production, offering a promising alternative to fossil-fuel-based energy resources.

Keywords: *Co-pyrolysis, scrap tire, vegetable waste, electromagnetic induction heating.*

Track 8: Textile Engineering and Sustainable Technology**Paper ID: 81****Presentation Mode: Oral****A Comprehensive Review of Soil Reinforcement Using Natural and Synthetic Fibers: Mechanisms, Performance Comparison, and Emerging Sustainable Trends****Sharifuzzaman Shafi^{1*}, Meherun Nessa Faruqui¹, Md. Sayed Ben Zobayer²**¹ Department of Civil Engineering, Bangladesh University of Engineering and Technology, Dhaka, Bangladesh² Department of Civil Engineering, Pabna University of Science and Technology, Pabna, Bangladesh

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Abstract

Fiber-reinforced soil has become an important area of geotechnical research due to its ability to enhance soil strength, deformation characteristics, and long-term stability. Both natural and synthetic fibers are increasingly used to improve engineering properties such as shear strength, ductility, load-bearing capacity, and resistance to cracking or erosion. Natural fibers, including jute, coir, sisal, and bamboo, offer advantages such as biodegradability, low cost, and environmental friendliness, making them suitable for short-term or temporary applications. Synthetic fibers, such as polypropylene, polyester, and polyethylene, exhibit high durability, uniform mechanical properties, and a long service life, making them suitable for permanent soil reinforcement. This review summarizes the fundamental mechanisms of fiber soil interaction, including frictional resistance, mechanical interlocking, and the contribution of fiber geometry and content to soil behavior. Existing findings on laboratory and field performance are compared to highlight the distinct benefits and limitations of each fiber type. Special focus is given to the influence of fiber length, aspect ratio, dosage, and dispersion on the improvement of shear strength and reduction of settlement. The key novelty of this review lies in its integrated comparative evaluation of natural, synthetic, and emerging hybrid fiber systems, with an emphasis on sustainability and long-term performance gaps. Unlike conventional reviews, the study synthesizes recent advancements involving bio-resin treatment, chemically coated natural fibers, recycled polymer fibers, and hybrid reinforcement approaches that aim to balance strength with environmental impact. The review also identifies critical research needs, such as improving the durability of natural fibers, optimizing fiber–soil interface bonding, and developing design guidelines for field applications. Overall, this work provides a comprehensive and up-to-date understanding of fiber-reinforced soils and offers insights for selecting appropriate fiber types to achieve both performance efficiency and sustainable geotechnical design.

Keywords: *Soil reinforcement, Natural fibers, Synthetic fibers, Fiber-reinforced soil, Geotechnical engineering.*

Track 8: Textile Engineering and Sustainable Technology**Paper ID: 85****Presentation Mode: Oral****Eco-Friendly Ceramic Tile Production Using Sugarcane Bagasse Ash and Eggshell Waste****Showvik Saha, Md. Parvej Mosaraf Sohel, Md. Nur Salam, Md. Bilal Hossain, Md. Khairul Islam***¹Department of Glass & Ceramic Engineering, Rajshahi University of Engineering & Technology (RUET), Rajshahi-6204, Bangladesh***Abstract**

This study investigates the incorporation of sugarcane bagasse ash (SCBA) and eggshell waste (ESW) as sustainable alternative raw materials in ceramic tile production, aiming to reduce environmental impact while improving material performance. SCBA, a silica-rich byproduct of the sugar industry, and ESW, a calcium-carbonate-based food processing waste, were introduced into standard ceramic tile formulations at varying proportions ranging from 3 wt.% to 15 wt.%. Raw materials including ball clay, china clay, quartz, feldspar, and limestone were prepared through milling, sieving, drying, and pressing, followed by sintering at 1150 °C. The physical properties—bulk density, water absorption, and firing shrinkage—and mechanical properties—compressive strength, tensile strength, impact resistance, and modulus of rupture—were evaluated to determine the influence of waste addition. Results reveal that incorporating 5–10 wt.% SCBA and ESW significantly enhances mechanical strength due to improved particle packing, reduced porosity, and beneficial phase formation such as anorthite and wollastonite. Optimal performance was observed at 10 wt.% substitution, where compressive strength, tensile strength, and modulus of rupture showed notable improvement compared to the control sample. However, waste addition beyond 10 wt.% led to increased closed porosity and microstructural discontinuities, causing a decline in both physical integrity and mechanical strength. The findings demonstrate that SCBA and ESW can partially replace conventional raw materials without compromising tile quality, offering a cost-effective and environmentally sustainable approach to ceramic manufacturing. This study highlights the potential of agro-industrial waste recycling in the ceramics sector and suggests further research on long-term durability, microstructural optimization, and scaling for industrial production.

Keywords: *Sugarcane Bagasse Ash (SCBA), Eggshell Waste, Ceramic Tiles, Sustainable Materials.*

Track 8: Textile Engineering and Sustainable Technology**Paper ID: 148****Presentation Mode: Oral****Study on the Rockwell hardness behavior of coal fly ash incorporated injection molded thermoplastic composites****Suman Bepari, M S Rabbi*, Shifat Hasan Naim***Chittagong University of Engineering & Technology, Pahartoli, Raozan, Chattogram - 4349, Bangladesh***Corresponding Author: rabbi@cuet.ac.bd***Abstract**

Filler materials could play a significant role to enhance the mechanical characteristics in composite materials. Coal Fly Ash (CFA) is an industrial by product for coal-based power plants. This study investigates the Rockwell Hardness behavior of CFA incorporated injection-molded thermoplastic composites. Three commonly used thermoplastics, such as: Polypropylene (PP), Polystyrene (PS) and High-density Polyethylene (HDPE) were chosen and blended with alkali-treated (with 5% NaOH) CFA. Influence of two different particle sizes (coarse and fine) were considered in the investigation. Three different filler loading were incorporated during the composite fabrication as 5, 10, and 15 wt.% to study the influence in characterization. Experimental data reveal that incorporation of FA affects the value of Rockwell hardness, with variation of fineness of FA and percentage of filler loading. For PP-FA composite, moderate amount of filler loading (up to 10%) exhibits better hardness value and fine FA shows comparatively better result in comparison to coarse FA. HDPE-FA composites shows modest enhancement of hardness values for fine and coarse both types of FA. PS-FA composites shows highest enhancement in hardness, especially with fine FA loading (up to 10%).

Keywords: *Thermoplastic Composites, filler, Fly Ash, Rockwell hardness, Injection molding.*

Track 8: Textile Engineering and Sustainable Technology

Paper ID: 107

Presentation Mode: Poster

Interface Engineering of PMMA/g-C₃N₄ Nanocomposites: Influence on Mechanical, Thermal and Efficient UV Shielding Properties

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Abstract

In this study, a transparent and flexible durable g-C₃N₄ incorporated PMMA film was developed by solvent casting method for UV shielding application. The mechanical, optical, thermal and structural properties of the synthesized PMMA/ g-C₃N₄ (PCN) films were investigated using a variety of characterization techniques, including Fourier transform infrared (FTIR), X-ray diffraction (XRD), field emission scanning electron microscope (FESEM), thermogravimetric analysis (TGA), differential thermogravimetric analysis (DTG), ultimate tensile strength (UTM) and UV-Visible spectra. The FESEM of the PCN film showed that g-C₃N₄ was evenly distributed throughout the PMMA film matrix and FTIR results indicated there was a strong intermolecular hydrogen bond, Van der Waals forces and electrostatic interactions between PMMA and g-C₃N₄. The composite PCN-0.01 films were found to have a tensile strength (TS) of 152.58 ± 1.86 MPa and a Young's modulus (YM) of 41.15 ± 0.397 MPa, respectively, which were approximately 14% and 39% greater than the PMMA film. Better UV and High Energy Blue Light (HEBL) blocking performance was also demonstrated by the PCN-0.01 composite films, which demonstrated 61.65% of UV-A radiation, 84.54% of UV-B radiation, 99.26% of UV-C radiation, and 68.88% of HEBL radiation. It was also discovered that the PCN film's UV and HEBL blocking capabilities rose dramatically with an increase in g-C₃N₄ content. Our research demonstrated that the straightforward and easy fabrication process is a viable approach to creating robust, transparent, and highly flexible PCN films with exceptional physicochemical qualities. These films also showed great promise for extending their practical applications in the blocking of UV/HEBL radiation.

Keywords: G-C₃N₄, PMMA, durable, mechanical properties, thermal properties, UV-shielding.

Track 8: Textile Engineering and Sustainable Technology**Paper ID: 172****Presentation Mode: Poster****Microwave-Assisted Extraction Optimization: Comparative Analysis of Gelatin Quality from Indigenous versus Crossbred Cattle Skins****Shahriar Hasan Shakib, Iqbal Hossain, Md. Abdullah Al Noman, Jannatul Ferdushi Jany, Md. Hafizur Rahman Bhuiyan, Afzal Rahman***Department of Food Engineering and Technology, Faculty of Agricultural Engineering and Technology, Bangladesh Agricultural University, Mymensingh 2202, Bangladesh***Abstract**

Optimizing gelatin extraction is critical for sustainable industrial production. In this study four extraction methods-Conventional Water Bath, Ultrasonic, Autoclave and Microwave Assisted Extraction (MAE) were systematically compared using indigenous and crossbred cattle skins to identify optimal protocols balancing yield and quality. MAE demonstrated superior performance, reducing extraction time by 83% (30 min vs 5h) compared to the conventional water bath method, while yielding premium gelatin from indigenous skin (gel strength: 182.81 g; viscosity: 16.70 mPa·s), a 27% functional improvement over conventional methods. Conversely, autoclave extraction produced the highest crude yield (19.66%) but caused severe thermal degradation, resulting in catastrophic functional loss (gel strength: 45.24 g) and confirming the critical yield-quality trade-off. The conventional water bath and ultrasonic methods yielded moderate (144.27 g) and lower (121.98 g) gel strengths, respectively. Indigenous cattle skin consistently outperformed crossbred skin, indicating breed-specific collagen differences. Correlation analysis confirmed processing intensity compromises molecular integrity via strong negative relationships between thermal indicators (color L*, a*, b*) and gel strength ($r < -0.94$). Furthermore, sensitivity analysis identified moisture as the most critical negative control parameter (10.66% strength reduction/ 1σ), while scenario modeling defined an "Optimal Balance" strategy for producing premium gelatin (gel strength: 134.79 g; transparency: 73.37%). These findings establish MAE as a rapid, energy-efficient technology for premium gelatin production, optimizing both processing efficiency and product quality.

Keywords: *Gelatin, Microwave-Assisted Extraction (MAE), Cattle Skin, Waste Valorization, Green Technology, Process Optimization.*

Track 9: Nuclear, Plasma, and Radiological Engineering**Paper ID: 103****Presentation Mode: Oral****Integrated DFT–MD–ML Multiscale Modelling of Hydrogen Isotope Trapping and Oxide Evolution in Tungsten under Fusion Relevant Plasma Exposure****Musavvir Haque, Md. Mushfiqur Rahman, Md. Shakil Hossain, Nabila Rahaman Ritu, Irin Akter Mim, Nure Modina Sathi, Abdullah Shadek Fahim, Saklain Mustak, Ragib Ishraq, Md. Amzad Hossain***Department of Electrical and Electronic Engineering, Jashore University of Science and Technology***Corresponding Author: musavvirhaquepial@gmail.com***Abstract**

Tungsten is widely regarded as the most promising plasma-facing material for next-generation fusion reactors such as ITER and DEMO. However, its operational lifetime is critically constrained by hydrogen isotope retention, oxide-layer formation, microstructural degradation, and plasma-driven cracking—phenomena that collectively influence fuel inventory, component integrity, and overall reactor safety. Understanding these coupled processes requires a modelling framework capable of resolving atomic-scale interactions while maintaining predictive capacity across operationally relevant conditions. In this work, we propose an integrated multiscale computational methodology that links Density Functional Theory (DFT), reactive Molecular Dynamics (MD), and Machine Learning (ML) to investigate hydrogen–tungsten interactions under diverse plasma exposure regimes. DFT simulations were employed to compute fundamental W–O–H energetics, which were subsequently used to parameterize and refine a Reactive Force Field (ReaxFF). This enabled high-throughput MD simulations covering more than 1000 physical scenarios involving three hydrogen isotopes—protium (^1H), deuterium (^2H), and tritium (^3H)—as well as systematically varied defect densities, oxide-layer thicknesses, and plasma flux–temperature ranges. These MD simulations capture key mechanisms such as hydrogen trapping, isotope-dependent diffusion, oxide restructuring, and the nucleation of crack-like defects. The resulting dataset was used to train two machine-learning models, Random Forest and XGBoost regressors, to generate fast and accurate predictions of binding energies, diffusion coefficients, and crack-density evolution. Among the models tested, XGBoost demonstrated superior performance, achieving a mean absolute error (MAE) of 0.017 eV with $R^2 = 0.88$ for predicting binding energies, and an MAE of 0.91% with $R^2 = 0.87$ for predicting crack density. The ML-driven predictions corroborate physical trends observed in the MD data, particularly enhanced hydrogen trapping in WO_3 (1.65–1.70 eV) relative to bulk tungsten (0.9–1.2 eV). Furthermore, the model identified diffusion coefficients reaching up to $1.0 \times 10^{-9} \text{ m}^2/\text{s}$ under high flux conditions and crack densities exceeding 20% in extreme plasma environments. Comparison with thermal desorption spectroscopy measurements and WEST tokamak datasets demonstrates strong agreement, validating the multiscale approach. Overall, this framework provides a transferable surrogate model that significantly accelerates predictive assessment of tungsten behavior, offering a powerful tool for the design and optimization of plasma-facing components in future fusion reactors.

Keywords: *Hydrogen isotope retention, DFT, Molecular dynamics, Machine learning, Fusion reactors.*

Track 9: Nuclear, Plasma, and Radiological Engineering**Paper ID: 117****Presentation Mode: Oral****Structural and Optical Properties of Ni-Doped CuO Thin Films Prepared by CBD Technique****Khadija Akter Akhi, Md Mosharof Hosen*, Apurbo Kumar Biswas, Rahima Nasrin.***Department of Physics, University of Barishal, Barishal-8200, Bangladesh***Corresponding Author: mmosharof20.phy@bu.ac.bd***Abstract**

Nowadays, cupric oxide (CuO) is receiving greater attention due to its special chemical and physical characteristics. Doping can have an impact on the optical and structural characteristics of CuO thin films (COTF) as well as their physical and chemical properties. This study aims to explore the influence of doping concentration (DC) on the structural and optical properties of pristine cupric oxide thin films (PCOTF) and Ni-doped cupric oxide thin films (NDCOTF). The PCOTF and NDCOTF were synthesized onto glass substrates through the CBD technique with various DC of Ni. From FTIR spectra it is noticed that the absorption peaks at 608 cm^{-1} and 444 cm^{-1} are assigned to CuO and Nickel oxide (NiO) stretching vibrations, confirming the formation of PCOTF and NDCOTF. Ultraviolet–visible spectroscopy analysis revealed that absorbance (A) increased with Ni content but decreased with wavelength. Whereas, the transmittance (T%) of NDCOTF exhibited an inverse trend with respect to A. The highest T% of the PCOTF, approximately 70% in the visible region (VR), was found to be reduced with increasing DC. Furthermore, the refractive index enhanced with DC, attaining a maximum value of 2.64, while the skin depth diminished with higher DC. Meanwhile, the dielectric constant showed a slight increase with increasing DC. The optical band gap energy (E_g) declined from 3.28 eV to 1.95 eV with rising Ni content. In addition, reflectance, absorption coefficient, and extinction coefficient showed notable alteration with Ni-incorporation. From all this analysis, it can be predicted that the fabricated NDCOTF becomes suitable for applications in optoelectronic devices.

Keywords: *Thin films, Chemical bath deposition (CBD), Ni-doped Cupric Oxide, Optical characteristics.*

Track 9: Nuclear, Plasma, and Radiological Engineering**Paper ID: 121****Presentation Mode: Oral****Optical And Structural Properties of Ni-doped Lead Sulfide Thin Films Synthesized via Chemical Bath Deposition Method****Khadija Akter Akhi, Md Mehedi Hasan shuvo*, Apurbo Kumar Biswas, Sudip Sital Gosh & Rahima Nasrin.***Department of Physics, University of Barishal, Barishal-8200, Bangladesh***Corresponding Author: shuvo1998m@gmail.com***Abstract**

In recent years, metal chalcogenide thin films have gained significant attention as promising materials for optoelectronic applications, owing to their tunable bandgap and strong absorption in the near-infrared region. Among different metal chalcogenides, PbS thin films (PSTF) have emerged as promising materials for thin-film solar cells, owing to the significant enhancement in their optical and structural properties achieved through Ni incorporation. The primary objective of the present research is to examine the effect of Ni-doping concentration (DC) on the structural, optical, and electrical properties of PSTF synthesized onto glass substrates using the chemical bath deposition (CBD) method. FTIR analysis confirmed the formation of PSTF, evidenced by the characteristic peak of PbS stretching vibration observed at 675 cm^{-1} . UV-Vis spectroscopic results showed that absorbance increased with higher DC in the UV region, and that it increased sharply. The highest absorbance was found for the 1% Ni-doped PbS thin film (NDPSTF). Transmittance (T%) exhibited an inverse relationship with absorbance, with the maximum value of approximately 80% at 750 nm. Furthermore, the refractive index of NDPSTF increased with DC, reaching a maximum value of approximately 2.6 at 675 nm wavelength. The Kubelka-Munk function showed a gradual rise with photon energy, indicating the onset of electronic transitions, and the corresponding optical band-gap energy values of NDPSTF lay within the range of 2.40-2.68 eV. Moreover, the variation of other optical parameters with DC, such as the extinction coefficient, absorption coefficient, skin depth, real dielectric constant, Urbach energy, and steepness parameter, was also computed. Based on these analyses, it can be inferred that the Ni-doped PSTF becomes more suitable for optoelectronic device applications as the DC is optimized.

Keywords: *Thin film, Chemical bath deposition, Lead Sulfide, Optical Properties, Doping.*

Track 9: Nuclear, Plasma, and Radiological Engineering**Paper ID: 124****Presentation Mode: Oral****Deposition Time Dependent Structural and Optical Properties of Cupric Sulfide Thin Films****Apurbo Kumar Biswas, Saifur Rahman Rakib*, Khadija Akter Akhi, & Rahima Nasrin.***Department of Physics, University of Barishal, Barishal-8200, Bangladesh***Corresponding Author: srahman19.phy@bu.ac.bd***Abstract**

In recent years thin films (TF) technology has shown remarkable progress in the discovery of new materials including nano and bio based materials, magnetic materials, etc. Among them, semiconductor TF are particularly significant due to their electrical and optical properties, which make them essential in modern optoelectronic device applications. The prime goal of the present research is to investigate the effect of deposition time (DT) on the optical properties of cupric sulfide (CuS) thin films. DT dependent CuS thin films (CSTF) were synthesized onto glass substrates via the chemical bath deposition (CBD) method. With the increase of DT, the thickness of the film increased ranging from 228 nm to 440 nm. The UV-Vis spectroscopic analysis revealed that the absorbance increased with longer DT in the UV region while it decreased with increasing wavelength in the visible region. The maximum absorbance was observed for 100 min deposited CSTF. Transmittance, T% inversely correlated with absorbance, exhibited a fast decline in the UV region and highest T% was found to be 23% for 100 min deposited CSTF. Moreover, the refractive index enriched with DT, attaining a maximum of approximately 2.2. The band gap energy values were found within the range 2.66-2.09 eV for CSTF at diverse DT. Urbach energy declined from 3.41 to 1.93 eV with higher DT. Additionally, the alteration of other optical parameters due to DT, such as extinction coefficient, absorption coefficient, skin depth, real dielectric constant, steepness parameter, etc., was also discussed for these CSTF. All of these observation indicates that DT significantly tunes the optical properties of CSTF, making them promising candidates for photovoltaics, and optoelectronic applications.

Keywords: *Thin film, Chemical bath deposition, Copper Sulphide (CuS), Optical Characteristics.*

Track 9: Nuclear, Plasma, and Radiological Engineering**Paper ID: 126****Presentation Mode: Oral****Preparation and Characterization of Lead Sulfide Thin Films Synthesized by Chemical Bath Deposition Technique****Khadija Akter Akhi, Sudip Sital Gosh*, Apurbo Kumar Biswas, Mehedi Hasan Shuvo & Rahima Nasrin***Department of Physics, University of Barishal, Barishal-8200, Bangladesh***Corresponding Author: sudip125598@gmail.com***Abstract**

In the past few years, lead sulfide thin films (PSTF) have attracted considerable attention due to their tunable bandgap and strong near-infrared absorption. The prime goal of the present research is to investigate the effect of precursor concentration (PC) on the structural and optical properties of PSTF. At different PC dependent PSTF were synthesized onto glass substrates through the chemical bath deposition method. From FTIR spectra, it is noticed that the absorption peak at 411 cm^{-1} is assigned to PbS bonding, confirming the formation of PSTF. The UV-Vis spectroscopic analysis revealed that the absorbance increased with higher PC in the UV region, while it decreased with increasing wavelength in the visible region. The maximum absorbance was observed for 0.4M PC of PSTF. Transmittance, T% inversely correlated with absorbance, of these thin films and exhibited their highest transmittance, reaching nearly 70% in the infrared region. Moreover, the refractive index is enriched with PC, attaining a maximum of approximately 2.2 at 700 nm. With increasing PC, the Kubelka-Munk function exhibited a gradual decrease with photon energy, and the corresponding optical band-gap values for PSTF were obtained in the range of 2.0-2.4 eV. Additionally, the alteration of other optical parameters due to PC, such as extinction coefficient, absorption coefficient, skin depth, real dielectric constant, Urbach energy, steepness parameter, etc., was also computed for these PSTF. From all this analysis, it can be predicted that the fabricated PSTF becomes suitable for applications in optoelectronic devices.

Keywords: *Thin film, Chemical bath deposition, Lead sulfide, Optical properties.*

Track 9: Nuclear, Plasma, and Radiological Engineering**Paper ID: 135****Presentation Mode: Oral****Structural, Optical and Electrical Properties of CuS Thin Films: Effect of Deposition Parameter****Apurbo Kumar Biswas, Parvez Mosarrof *, Khadija Akter Akhi & Rahima Nasrin***Department of Physics, University of Barishal, Barishal-8200, Bangladesh***Corresponding Author: pmosarrof21.phy@bu.ac.bd***Abstract**

Recently, transition metal chalcogenides have gained attention for their favorable physical and chemical properties. Among different metal chalcogenides, copper sulfide (CuS) is important and enormously used in large applications because of its suitable and excellent optical and electrical properties. The present study investigates the effect of precursor concentration (PC) on the structural, optical and electrical properties of copper sulfide thin films (CSTF) synthesized via chemical bath deposition method. The Films were deposited at different PC ranging from 0.2 M to 0.6 M. X-ray diffraction analysis confirmed the CSTF are polycrystalline in nature with hexagonal structure. From fourier transform infrared spectroscopic analysis it is noticed that the absorption peaks at 580 cm^{-1} is assigned to CuS stretching vibration, confirming the formation of CSTF. UV–Vis spectroscopic analysis revealed higher absorbance (A) of CSTF in the ultraviolet region (UVR), which gradually decreased in the visible region (VR). Transmittance (T%) spectra followed the inverse trend of A, with higher PC exhibiting comparatively greater T%. In the UVR, samples prepared at lower PC exhibited higher extinction coefficients, indicating stronger photon absorption. The estimated optical energy band gap values of the CSTF were found to be within the range of 1.75–2.32 eV, depending on PC. Additionally, the alteration of other optical parameters due to PC, such as skin depth, real dielectric constant, urbach energy, and steepness parameter, was also discussed for these CSTF. In electrical analysis, the obtained I–V curve for the CSTF exhibits a linear relationship of the current with voltage, confirming the ohmic nature of the film. All of these observation indicts that PC significantly tunes the structural, optical and electrical properties of CSTF, making them promising candidates for photovoltaics, and optoelectronic applications.

Keywords: *Thin film, Chemical bath deposition, Copper Sulphide (CuS), Optical Characteristics..*

Track 9: Nuclear, Plasma, and Radiological Engineering

Paper ID: 136

Presentation Mode: Oral

A DFT Study on Penta-Boron Phosphorus (P-BP₅) Nanosheet for Oxygen-Based Toxic Gases Sensing.

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Abstract

In this study, we designed Penta-boron phosphorus (P-BP₅) nanostructure and investigated its optical, electrical, and adsorption energy characteristics for toxic gas molecules (NO, SO₂, NO₂, and O₃) using density functional theory calculations. These gas molecules are successfully absorbed on the P-BP₅ monolayer. The result showed that NO₂ gas had a significant adsorption energy and a small adsorption distance of -3.5832 eV, and 2.754 Å, as well as being better at gas sensing and chemisorption ability. P-BP₅ structure found recovery times of 0.792 ns to 3.99×10^{48} s, indicating a strong attractive energy for the NO₂, and O₃ gas molecules. On the P-BP₅ monolayer, -5.692 eV cohesive energy reveals structural stability and shows a direct band gap of 1.85 eV. Absorbent exhibited a high adsorption coefficient of over 10^{12} cm⁻¹ orders. The electron localization function and Electron density difference study indicated that gas molecules are physically adsorbed onto the adsorbent. The research demonstrates that the P-BP₅ can be a good choice for sensing NO, SO₂, NO₂, and O₃ gases.

Keywords: Penta-boron phosphorus (P-BP₅) nanostructure, Density functional theory (DFT), Gas sensing, Adsorption energy, Toxic gas molecules.

Track 9: Nuclear, Plasma, and Radiological Engineering

Paper ID: 140

Presentation Mode: Oral

A DFT Study on Pristine and Co-Doped Graphene Boron Nitride Nanosheets for PH₃ and H₂Se toxic gas Detection

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Abstract

In order to reduce the effect of several toxic gases produced during modern technology activities, it is essential to develop promising gas adsorbents in the present day. In this study, we investigated the adsorption of PH₃ and H₂Se gas in pristine and Co-doped graphene boron nitride (G-BN) using density functional theory calculations. The adsorption of PH₃ and H₂Se gases was investigated to determine how it affects the geometrical, electronic, and optical characteristics of pristine and Co-doped G-BN nanosheets. The negative cohesive energies of -8.882 eV/atom for G-BN and -8.795 eV/atom for Co-doped G-BN indicate that both sheets are energetically stable. The investigated G-BN adsorbent exhibited physisorption toward PH₃ gas, while both sheets showed chemisorption toward H₂Se. The highest adsorption energy, -2.154 eV, was observed for H₂Se on the Co-doped G-BN nanosheet. The G-BN and Co-doped G-BN nanosheets exhibited direct band gaps of 1.099 eV and 0.159 eV, respectively. However, a noticeable change in the band gap of both adsorbents is seen after the interaction with the gases. The Co-doped G-BN nanosheet demonstrated a semiconductor-to-conductor transition by the adsorption of PH₃ and H₂Se. Both sheets have a very high recovery time, suggesting that they interact most strongly with the H₂Se gas molecule. Both the geometry structures exhibit absorption coefficients greater than 10³ cm⁻¹ order in the UV region. However, gas adsorption causes a slight peak shifting in this region. According to this study, both nanosheets are suitable as sensors of harmful gases.

Keywords: Graphene-boron nitride (G-BN), Co-doped G-BN, PH₃ adsorption, H₂Se adsorption, Density functional theory (DFT), Gas sensing

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